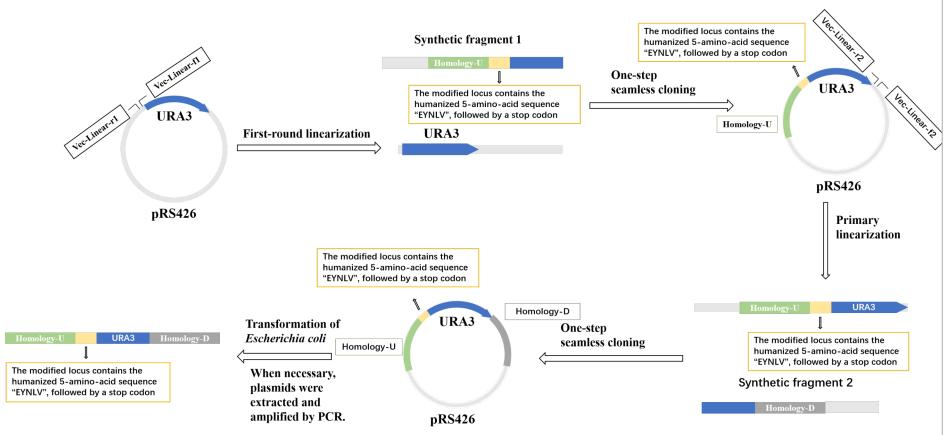
Notebook： Humanization Engineering of *GPA1* in *Saccharomyces cerevisiae*

No 1 Date：2026.04.12-04.15 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	C-terminal humanization engineering of the *GPA1* gene (obtaining the target fragment via engineering of the *pRS426* plasmid): design of primers, homology arms, and the Gibson assembly strategy
Note	YeastGenome（https://www.yeastgenome.org/） NCBI（https://www.ncbi.nlm.nih.gov/） SnapGene（https://www.snapgene.com/） Addgene（https://www.addgene.org/）
Result	 **Basic workflow**

Forward primer	Reverse primer	Objective
Vec-Linear-f1:AGAGTGCACCATACCACAGC	Vec-Linear-r1:GGAGCTGCATGTGTCAGAGG	Initial linearization upstream of *URA3*
Val-Hom-U-f:CCCAGGCATTACAGAAACCG	Val-Hom-U-r:TCGTTCTTCCTTCTGTTCGG	Verification of whether the upstream homology arm has been introduced into the plasmid
Vec-Linear-f2:GACCGAGATAGGGTTGAGTGT	Vec-Linear-r2:CCGATTTCGGCCTATTGGTT	Secondary linearization downstream of *URA3*
Val-Hom-D-f:AACCAATAGGCCGAAATCGG	Val-Hom-D-r:CTCGGTCGCCTAGTAGATCT	Verification of whether the downstream homology arm has been introduced into the plasmid
Donor-f:CCCAGGCA	Donor-r:CTCGGTCG	Final linearization,

	TTACAGAAACCG	CCTAGTAGATCT	followed by subsequent transformation into *BY4741*
	Val- Yeast- f:AGAGTGCACCATACCACAGC	Val- Yeast- r:CCGATTTCGGCCTATTGGTT	Verification of successful transformation and homologous recombination within yeast

Primer design

TTCGGTGATGACGGTGAAAACCTCTGACACATGCAGCTCCCAGGCATTAC
AGAAACCGAATTTAACATCGGCTCGTCCAAATTCAAGGTTCTCGACGCTG
GTGGGCAGCGTTCTGAACGTAAGAAGTGGATTCATTGTTTCGAAGGAATT
ACAGCAGTTTTATTTGTTTTAGCAATGAGTGAATACGACCAGATGTTGTTT
GAGGATGAAAGAGTGAACAGAATGCATGAATCAATAATGCTATTTGACAC
GTTATTGAACTCTAAGTGGTTCAAAGATACACCGTTTATTTTGTTTTTAAA
TAAAATTGATTTGTTCGAGGAAAAGGTAAAAAGCATGCCCATAAGAAAGT
ACTTTCCTGATTACCAGGGACGTGTCGGCGATGCAGAAGCGGGTCTAAAA
TATTTTGAGAAGATATTTTTGAGCTTGAATAAGACAAACAAACCAATCTA
CGTGAAACGAACCTGCGCTACCGATACCCAAACTATGAAGTTCGTATTGA
GTGCAGTCACCGATCTAATCATCCAGCAAAACCTTAAAGAATATAATTTG
GTTTGA（EYNLV）
AGAGTGCACCATACCACAGCTTTTCAATTCAATTCATCAT

Synthetic upstream homology arm carrying homology sequences to the linearized plasmid

GTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGAGGAACTGTA
TAATTAAAGTAGTGTTTAGATACGTAAATTCTGTTTCCGAAGATGCAAGA
AGGAGCAGCAGCACCAGAAAAAATTACTATTTTTCTTCTCCATTAGAGTCT
ATGATGGAATGCCAAATGAAAAAGCCATTTTGTTCAACAGTTCTTGATCTC
GTTAAATCGTTCCGGGTTTTCAATTGAAAAACAAGGGTAATAAAATCGCA
TGAGAAAAAAAAGGTCCAGACACTTTTTTAGAAGAAAAACTCTTGAAAAC
AATTGATTCTTCTACGTAAGGACAGGGTATTATCTCGAATTCTTCTCCCTG
GTAAATACTCGGGATATATAGAACCTATTTTGCATCCTATTCTCACTTTTC
TTTATTGTTGTTAATTTCTGGACATTGTTTCATTAATTGAATGAATGCCCTA
AAATATTTCTCAAATCATTTAATAACTACAAAGAAACAAAAAAAAATCAA
TGTTGAGGTGACAAAGAATCAAGATCTACTAGGCGACCGAGATAGGGTTG
AGTGTTGTTCCAGTTTGGAACAAG

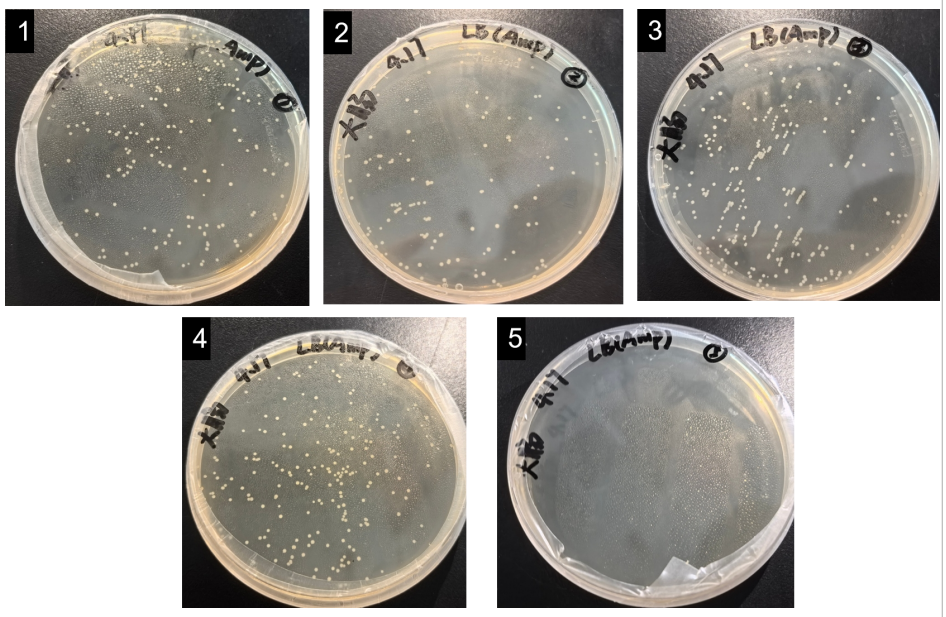
Synthetic downstream homology arm carrying homology sequences to the linearized plasmid

Explanation	This experiment completed the design of the full set of primers and homology arm sequences for the C-terminal humanization engineering of the *GPA1* gene.

	The five C-terminal amino acids of yeast *GPA1*, KIGII, were replaced with EYNLV of human Gαq, and a TGA stop codon was added, with *URA3* selected as the selection marker. Because the GC content of the homology arm regions was excessively low, direct PCR amplification was difficult; therefore, artificially synthesized upstream and downstream fragments carrying plasmid homology overlapping sequences were employed, and vector construction was accomplished through two rounds of Gibson Assembly. Plasmid verification primers for each step and genome identification primers for post-yeast-transformation analysis were designed, thereby laying the foundation for subsequent vector construction and yeast transformation experiments.	
Next plan	Synthesize primers and upstream/downstream homology arm fragments, and conduct wet-lab experiments for vector construction and yeast transformation.	

No 2 Date：2026.04.17- 04.25 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Transformation of the *pRS426* plasmid into *Top10* competent cells, bacterial culture expansion, plasmid extraction, and double restriction enzyme digestion verification.
Note	Protocol 1(E. coli Transformation and Culture Expansion)、Protocol 2(Glycerol Stock Preparation and Cryopreservation)、Protocol 3(Plasmid Extraction from E. coli)、Protocol 4(Restriction Enzyme Digestion)、Protocol 5(Agarose Gel Electrophoresis)
Result	4.19



Transformation results

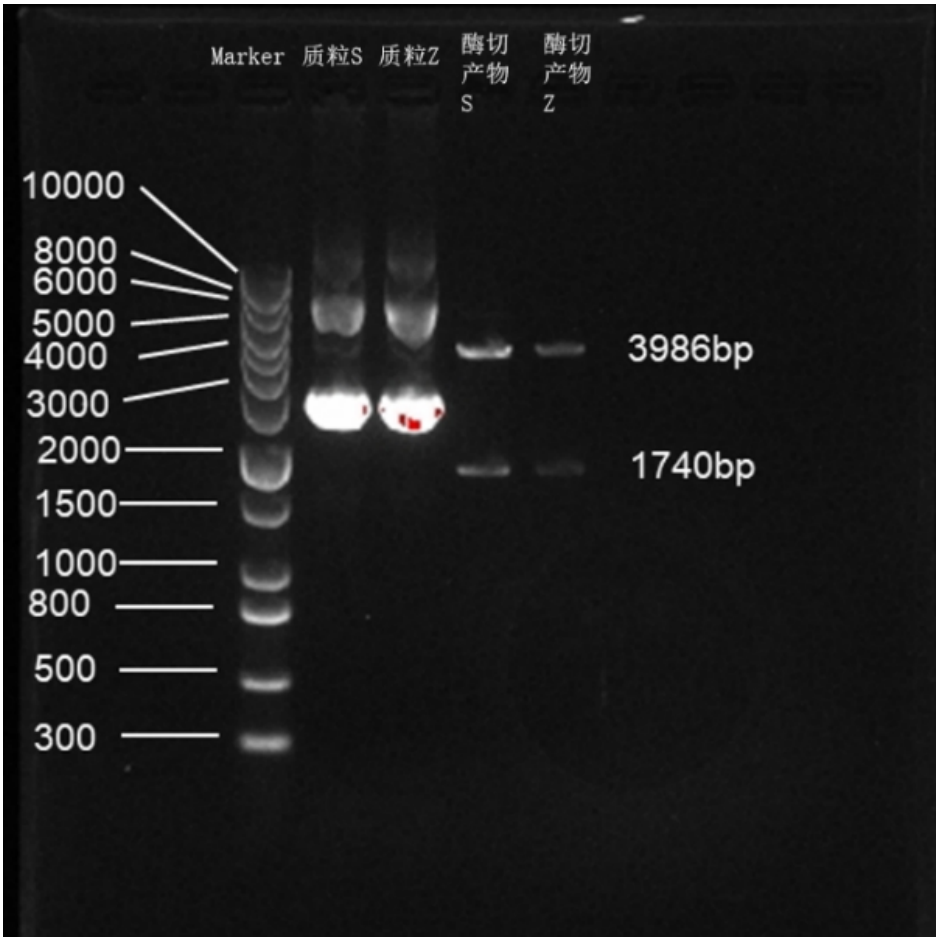
1–4 transformed with the *pRS426* plasmid, 5 blank control

Sample	Concentration (ng/μL)	A260/A280
1	229.30	1.802

2	287.60	1.905
3	294.10	1.907
4	259.75	1.899
5	282.50	1.903
6	285.60	1.901

Concentration and purity of the extracted *pRS426* plasmid

4.25



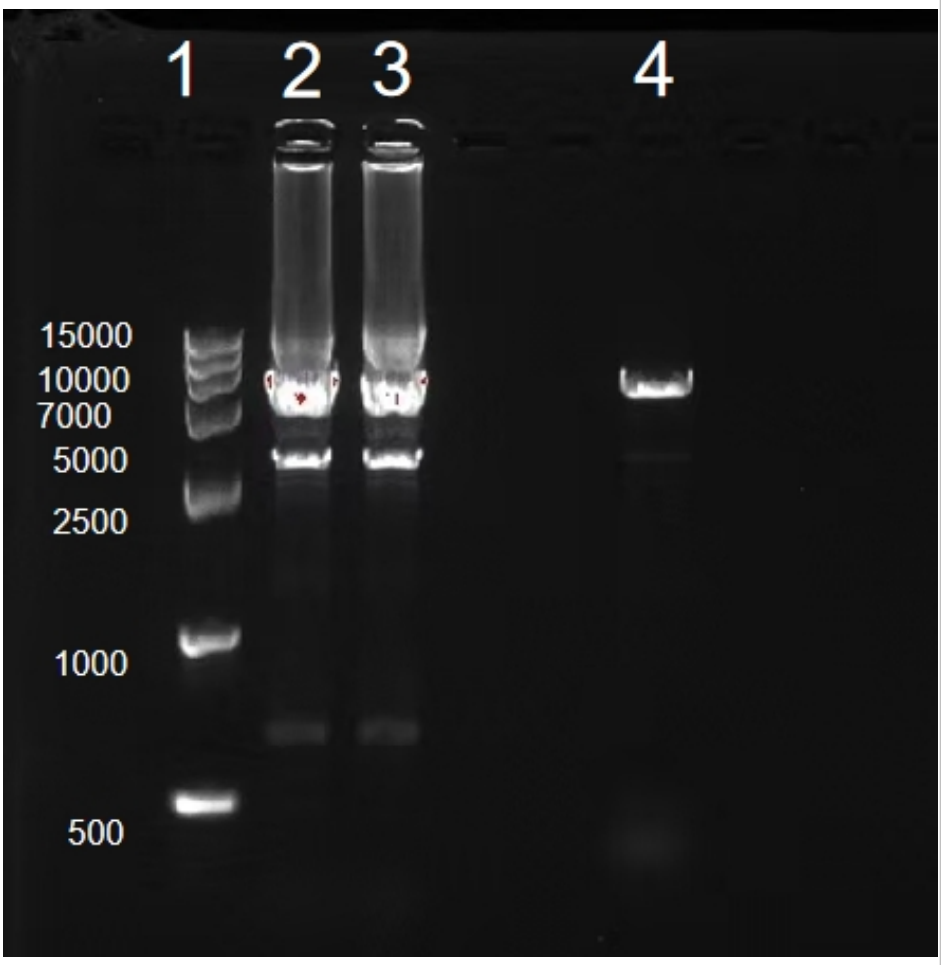
Double restriction enzyme digestion verification (the target bands are 1740 bp and 3986 bp, respectively)

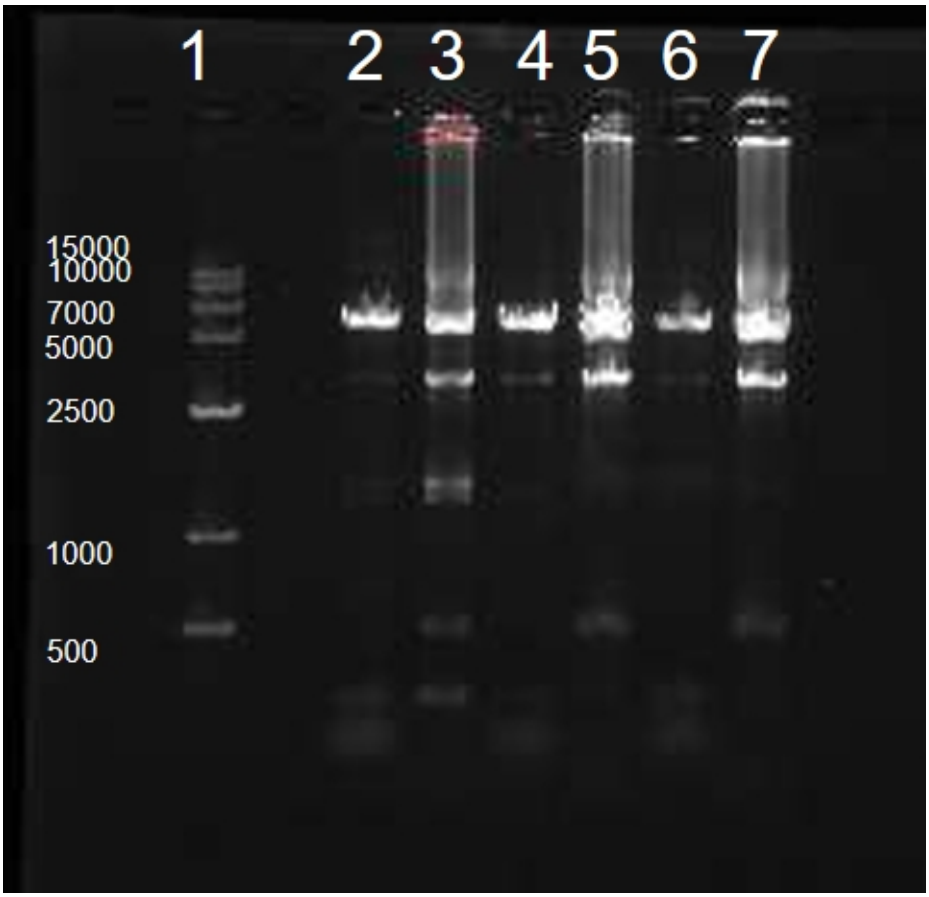
Explanation

This experiment completed the *Escherichia coli* transformation of the *pRS426* plasmid, expansion culture of positive clones, plasmid extraction, and double restriction enzyme digestion identification. During the initial plasmid extraction, residual ethanol on the column membrane caused inhibition of restriction enzyme digestion, and the plasmid yield from small-volume shaking culture in test tubes was relatively low. Subsequent procedural optimization was performed: the empty column was sufficiently centrifuged and air-dried to remove ethanol, culture expansion was switched to 100 mL Erlenmeyer flasks, and the restriction digestion system was prepared sequentially on ice, thereby successfully obtaining high-purity *pRS426* plasmid with correct structure, which can serve as the backbone vector for subsequent Gibson Assembly construction of the *GPA1*

	humanization-engineered plasmid.
Next plan	Upon arrival of the DNA fragments to be synthesized, PCR linearization of the vector and Gibson Assembly experiments were carried out.

No 3 Date：2026.05.10- 05.25 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

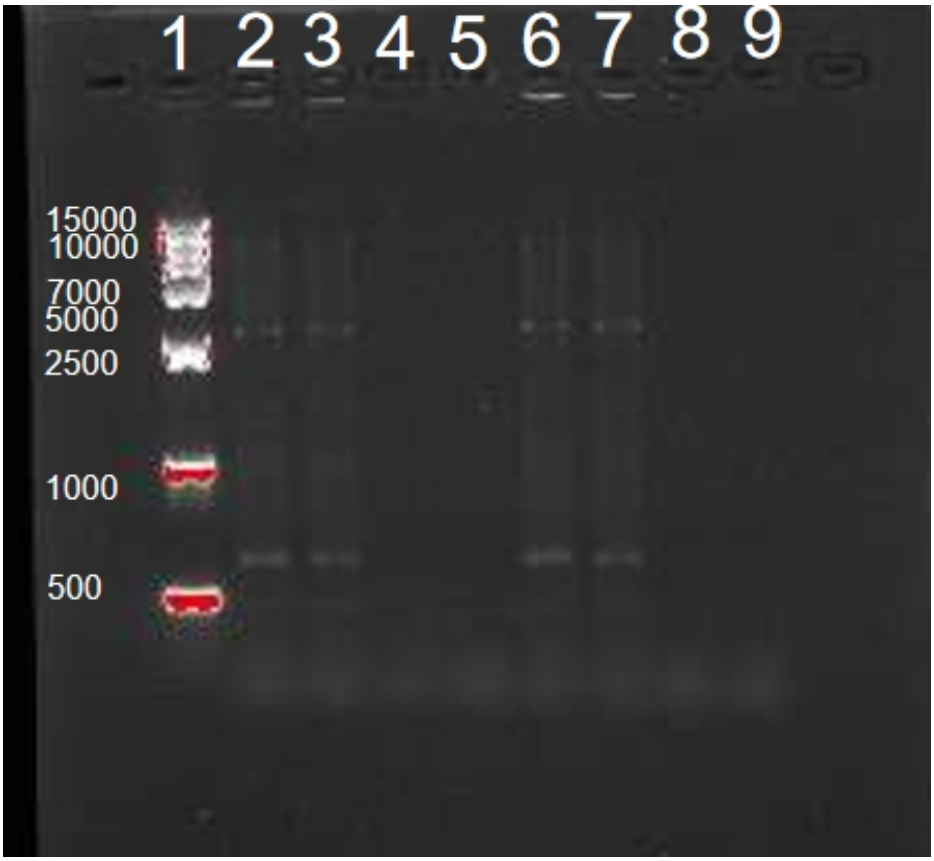
Experiment	Initial PCR linearization of the *pRS426* plasmid, electrophoresis verification, and gel extraction recovery.
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)
Result	5.10  **First preliminary experiment for initial plasmid linearization (the target band is 6092 bp)** 1: 15000 bp marker; 2–3: experimental group; 4: blank control 5.10



Second preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control (annealing temperature 58°C, template amount 50 ng); 3: experimental group (annealing temperature 58°C, template amount 50 ng); 4: blank control (annealing temperature 60°C, template amount 50 ng); 5: experimental group (annealing temperature 60°C, template amount 50 ng); 6: blank control (annealing temperature 58°C, template amount 100 ng); 7: experimental group (annealing temperature 58°C, template amount 100 ng)

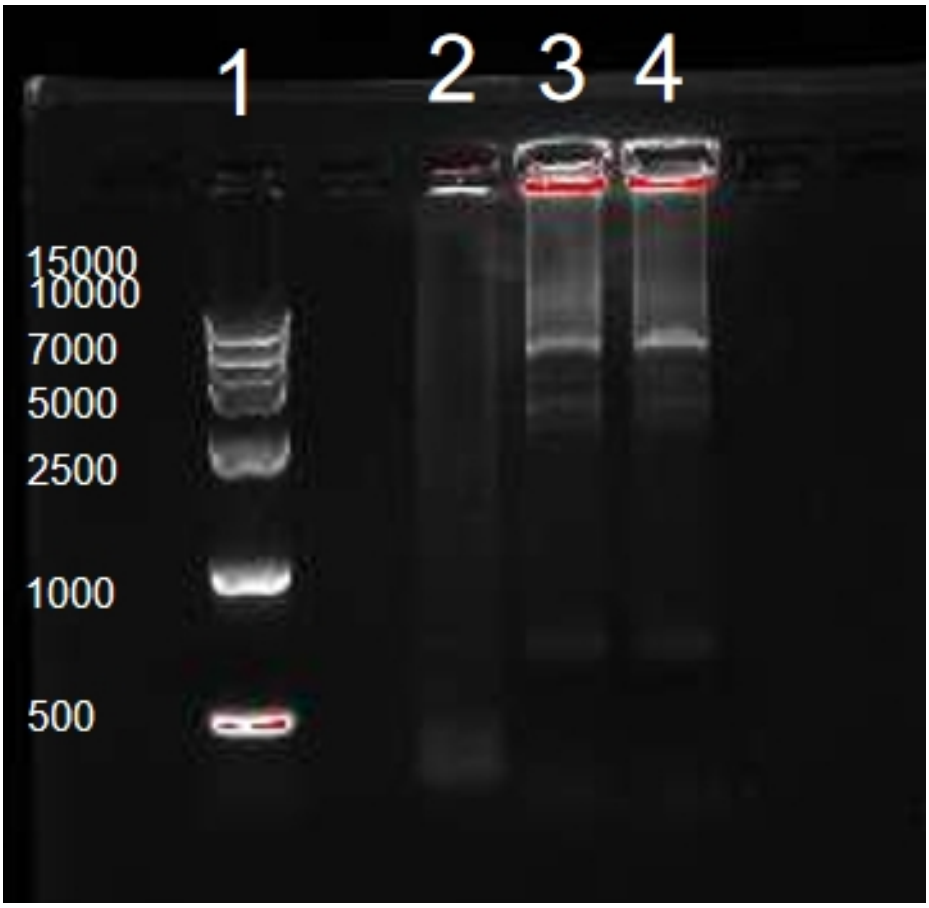
5.11



Third preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2–3: experimental group (annealing temperature 60°C); 4–5: blank control (annealing temperature 60°C); 6–7: experimental group (annealing temperature 62°C); 8–9: blank control (annealing temperature 62°C)

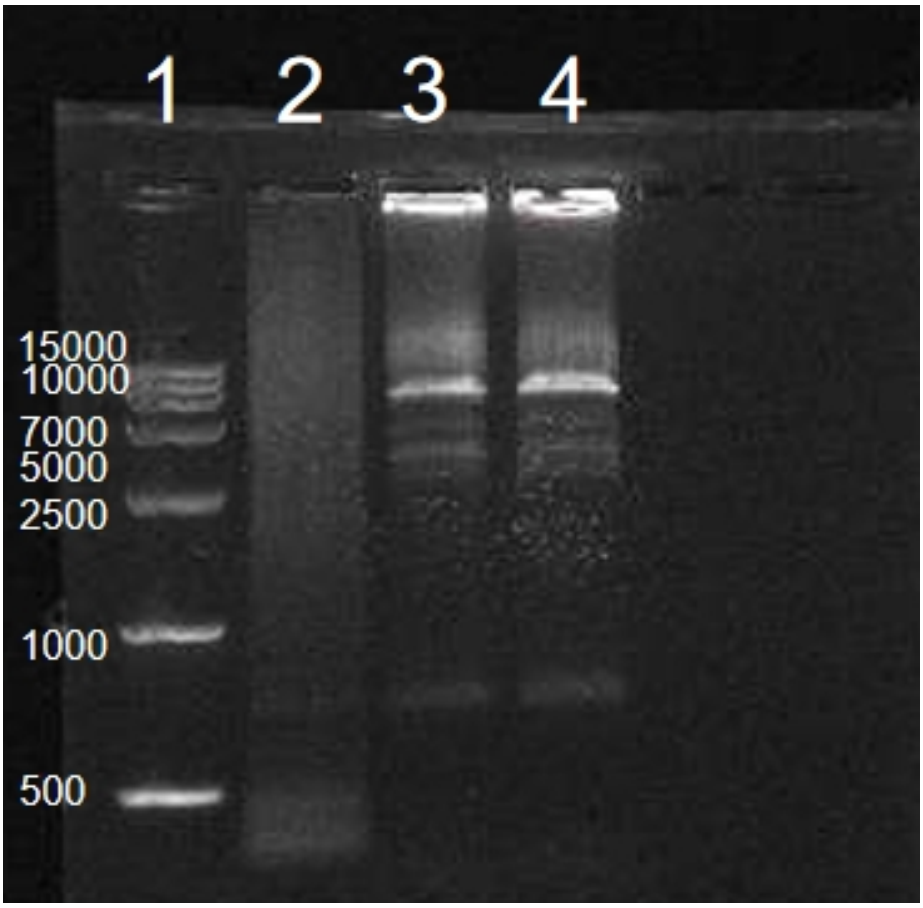
5.12



Fourth preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control; 3–4: experimental group

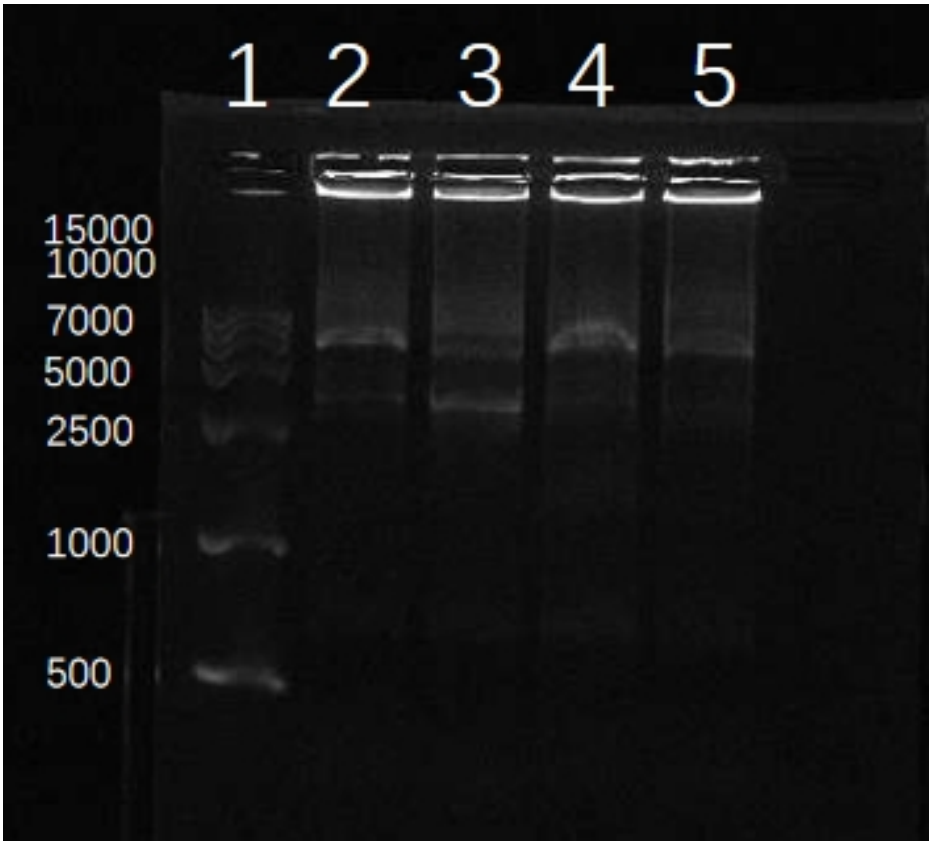
5.12



Fifth preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control; 3–4: experimental group

5.19



Sixth preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

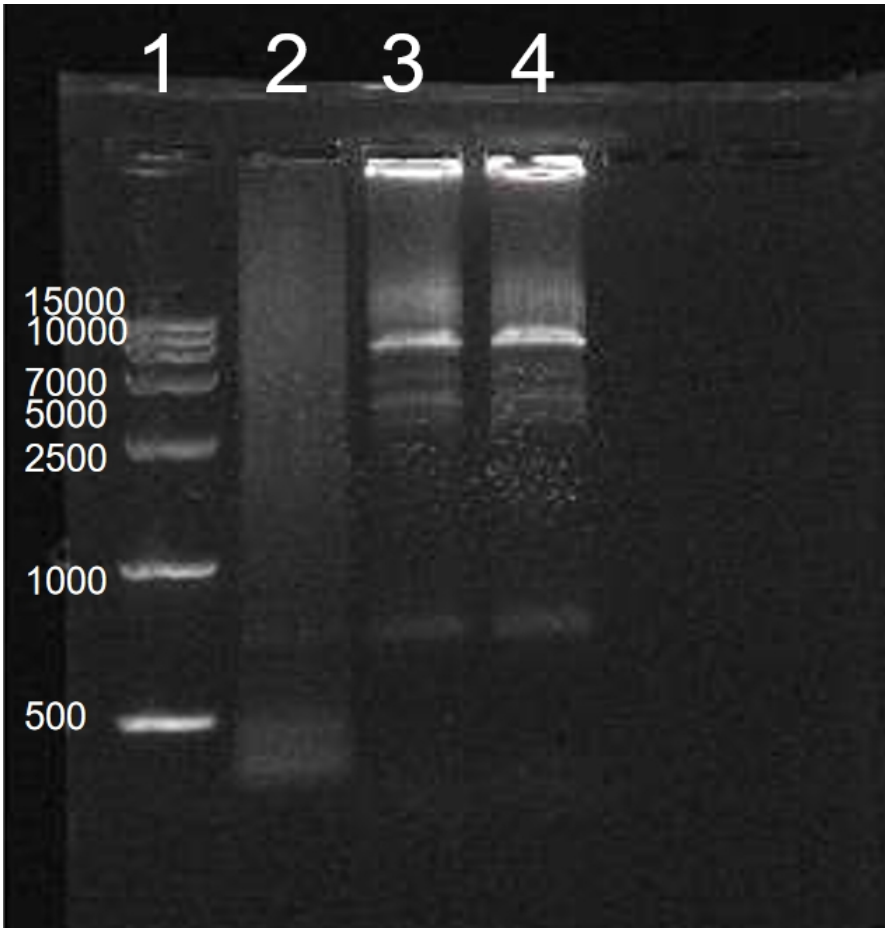
1: 15000 bp marker; 2–5: experimental group

5.23

Seventh preliminary experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2–3: experimental group (annealing temperature 60°C); 4–5: blank control (annealing temperature 60°C); 6–7: experimental group (annealing temperature 62°C); 8–9: blank control (annealing temperature 60°C)

5.25



Formal experiment for initial plasmid linearization (the target band is 6092 bp)

1: 15000 bp marker; 2: blank control; 3–4: experimental group

Sample	Concentration (ng/μL)	A260/A280
1	14.5	2.042
2	12.4	2.756

Concentration and purity of the linearized product recovered by gel extraction

Explanation	In this experiment, the *pRS426* plasmid previously verified as correct by double restriction enzyme digestion was used as the template, and PCR was performed with the primers VecLinear_f1/VecLinear_r1 to accomplish the initial linearization of the plasmid (the blank control had consistently shown a band in the earlier stage, and a considerable amount of time was spent excluding contamination sources, ultimately confirming that it was likely caused by operational issues). Agarose gel electrophoresis confirmed that the size of the PCR product band matched the expectation, with no obvious non-specific bands; subsequently, gel extraction recovery was carried out to remove residual circular plasmid template and impurities, yielding the purified linear vector *pRS426_L*, which provides a qualified linear backbone for the Gibson Assembly of the

	upstream homology arm fragment.
Next plan	Gibson Assembly of the upstream homology arm fragment with *pRS426_L* and *Escherichia coli* transformation were carried out.

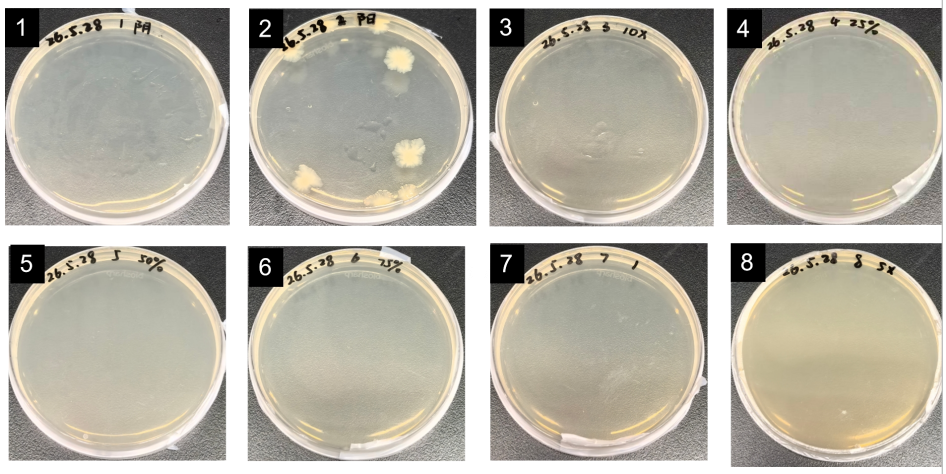
No 4 Date：2026.05.25- 05.30 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	One-step cloning (Gibson Assembly) of the upstream homology arm fragment with *pRS426_L* and *Escherichia coli* transformation.
Note	Protocol 11(Gibson Assembly)、Protocol 1(E. coli Transformation and Culture Expansion)
Result	

Component	Volume
Volume	2.5 μL（50 ng）
Linearized vector	1.5 μL（Approximately 20 ng）
2×CE Mix V3	5 μL
DdH_2O	1 μL

One-step cloning system

5.30



Transformation plate images

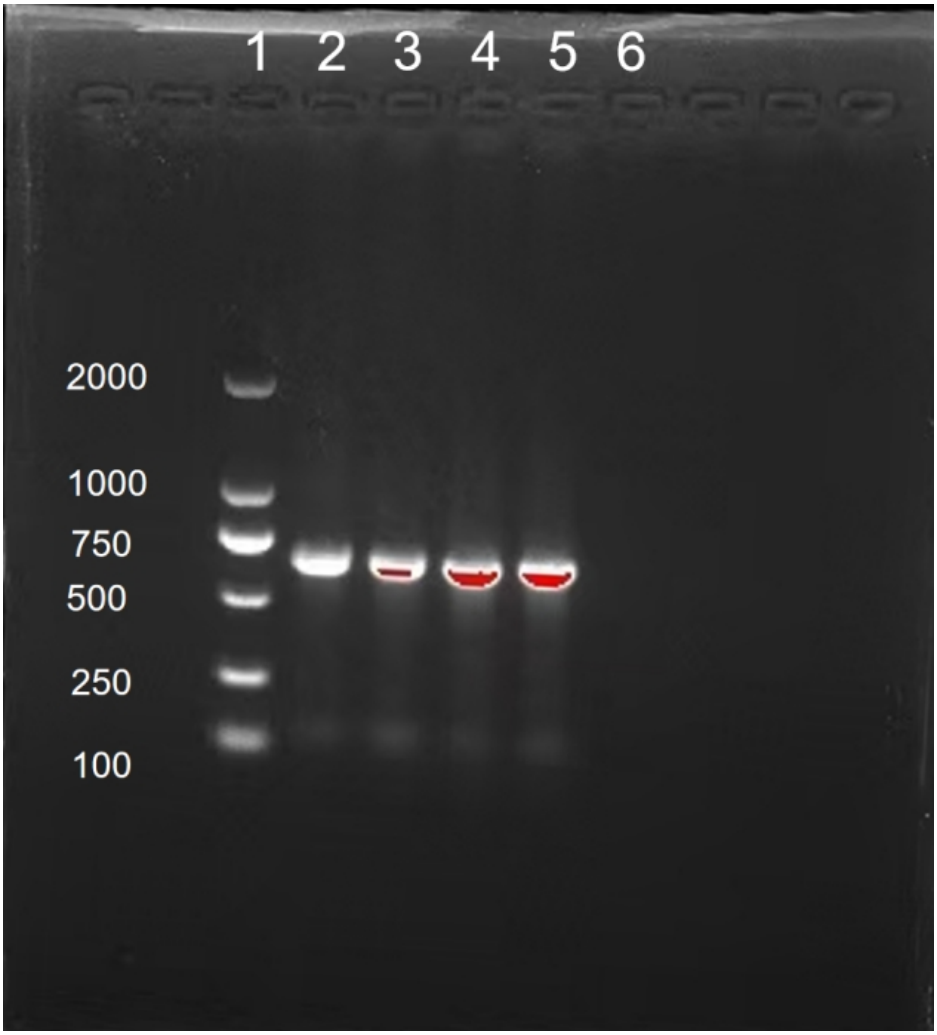
1: blank control; 2: transformed with *pRS426*; 3: one-step cloning product, plated after 10-fold concentration; 4: one-step cloning product, plated after dilution to 25% of the original solution; 5: one-step cloning product, plated after dilution to 50% of the original solution; 6: one-step cloning product, plated after dilution to 75% of the original solution; 7: one-step cloning product, plated as the original solution; 8: one-step cloning product, plated after 5-fold concentration

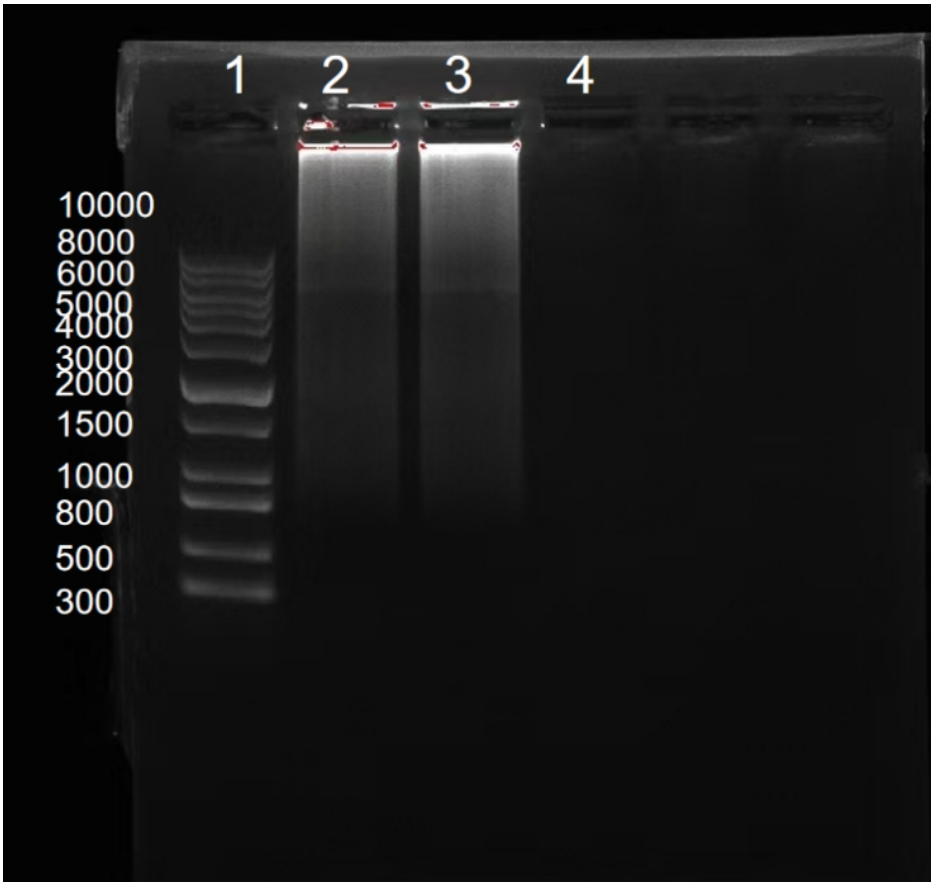
Explanation	In this experiment, the purified linear vector *pRS426_L* was subjected to Gibson one-step cloning assembly with the artificially synthesized upstream homology arm fragment. The blank control excluded background contamination from the plates and competent cells; the positive control exhibited bacterial contamination and thus lost its reference significance; multiple concentration gradient plating was set up, yet *Escherichia coli* positive transformants could not be obtained throughout. PCR verification demonstrated that the *in vitro* system had completed

	correct fragment assembly, but the assembled product was unable to undergo transformation screening *in vivo* in *Top10*. To avoid repeatedly consuming reagents and time, the experimental strategy was adjusted: obtaining the circular recombinant plasmid *pRS426_circ1 in vivo* in *Escherichia coli* was abandoned, and the one-step cloning reaction mixture was directly used as the template for subsequent secondary PCR linearization of the vector, thereby continuing to advance the gene editing experiment.
Next plan	PCR verification was performed on the one-step cloning reaction stock solution, and the stock solution was used as the template for the secondary PCR linearization experiment of the vector.

No 5 Date：2026.06.01-06.07 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	PCR verification of the one-step cloning reaction stock solution and secondary linearization of the vector
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)

Result	6.1  **Initial one-step cloning electrophoresis verification (the target band is 640 bp)** 1: 2000 bp marker; 2–5: experimental group; 6: blank control 6.7



Electrophoretogram of secondary linearization of the vector (the target band is 6092 bp)

1: 10000 bp marker; 2–3: experimental group; 4: blank control

Sample	Concentration (ng/μL)	A260/A280
1	4.9500	1.623

Concentration and purity of the linearized product recovered by gel extraction

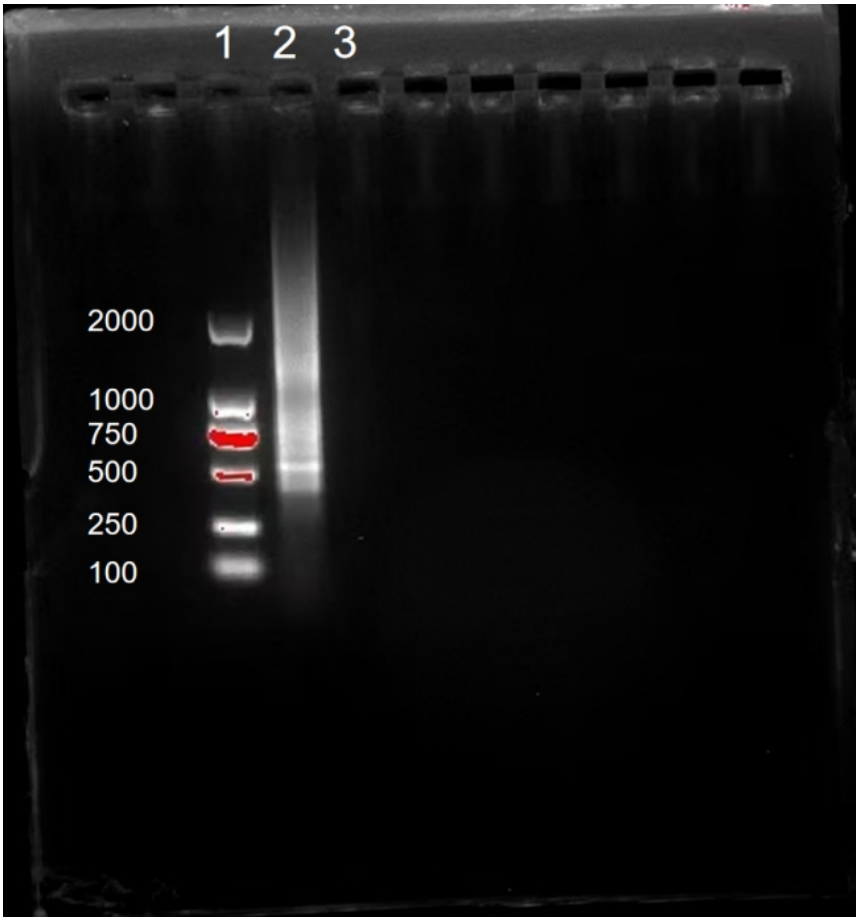
Explanation	Because the circular *pRS426_circ1* plasmid could not be obtained through *Escherichia coli* transformation, the one-step cloning reaction stock solution was directly used as the PCR template, and the expected 640 bp band was amplified, further confirming that the *in vitro* Gibson Assembly had achieved correct splicing of the upstream fragment. The one-step cloning reaction stock solution was used to accomplish the secondary PCR linearization of the vector, and gel extraction recovery yielded the purified *pRS426_L2* linear backbone, thereby circumventing the experimental obstacle caused by the failure of *Escherichia coli* transformation screening and rendering it suitable for the Gibson Assembly of the downstream homology arm fragment.
Next plan	The artificially synthesized downstream homology arm fragment was subjected to Gibson Assembly with the *pRS426_L2* backbone, followed by PCR verification.

No 6 Date：2026.06.07-06.08 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Gibson Assembly of the downstream homology arm fragment with the *pRS426_L2* backbone and PCR verification
Note	Protocol 11(Gibson Assembly)、Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)
Result	6.8

Component	Volume
Linearized fragment(4.9500 ng/μL)	2.5
Downstream homology arm(20 ng/μL)	1.25
2×CE Mix V3	5
ddH_2O	1.25

Secondary one-step cloning system

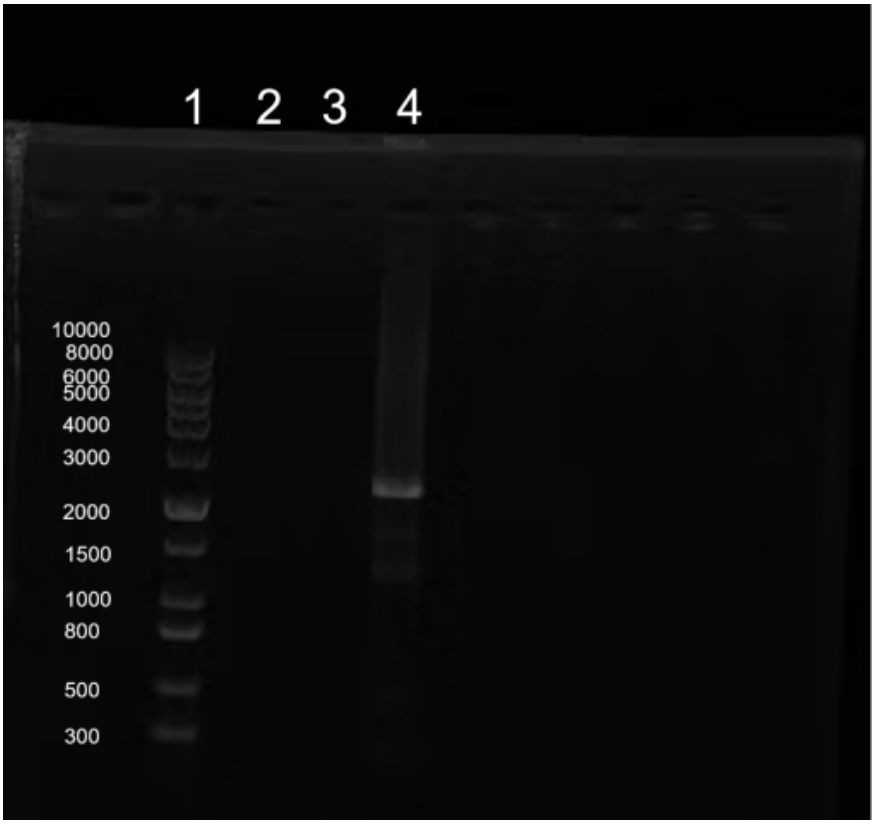


Secondary one-step cloning electrophoresis verification (the target band is 640 bp)

1: 2000 bp marker; 2: experimental group; 3: blank control

Explanation	In this experiment, the *pRS426_L2* linear backbone was subjected to *in vitro* Gibson Assembly with the artificially synthesized downstream homology arm fragment. The assembly stock solution was directly used as the template for PCR verification, and the expected target band was amplified, demonstrating that the downstream homology arm had been successfully spliced with the vector backbone *in vitro*. Combined with the verification results of the upstream fragment from the previous step, this indicates that the complete recombinant sequence comprising the upstream homology arm, the humanized sequence, the *URA3* expression cassette, and the downstream homology arm had already been obtained *in vitro*.
Next plan	Using the Gibson Assembly stock solution from this round as the template, PCR amplification was performed to obtain the complete linearized DNA fragment for yeast transformation.

No 7 Date：2026.06.11-06.15 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	The complete linear target DNA fragment for yeast transformation was amplified and sent for sequencing identification.
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)
Result	6.11  **Obtainment of the target fragment (the target band is 2289 bp)** 1: 12000 bp marker; 2–3: blank control; 3: experimental group 6.15

	 Comparison of sequencing results 1: 2000 bp marker; 2: experimental group; 3: blank control
Explanation	Although electrophoresis yielded a linear recombinant DNA fragment of 2289 bp with the expected size, sequencing exposed two critical defects: insufficient primer design for long-fragment sequencing resulted in incomplete sequence coverage; and the accumulation of multiple *in vitro* PCR cycles introduced numerous base mutations, making it impossible to obtain an editing donor fragment with a completely correct sequence. Therefore, the current experimental route combining Gibson Assembly with multiple rounds of PCR amplification was terminated to circumvent the risk of mutations, and the experimental scheme was changed to reconstruct the *GPA1* humanization editing donor fragment using Overlap Extension PCR (fusion PCR).
Next plan	The fusion PCR scheme was adopted to redesign the primers and construct the editing donor fragment.

No 8 Date：2026.06.15-06.18 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

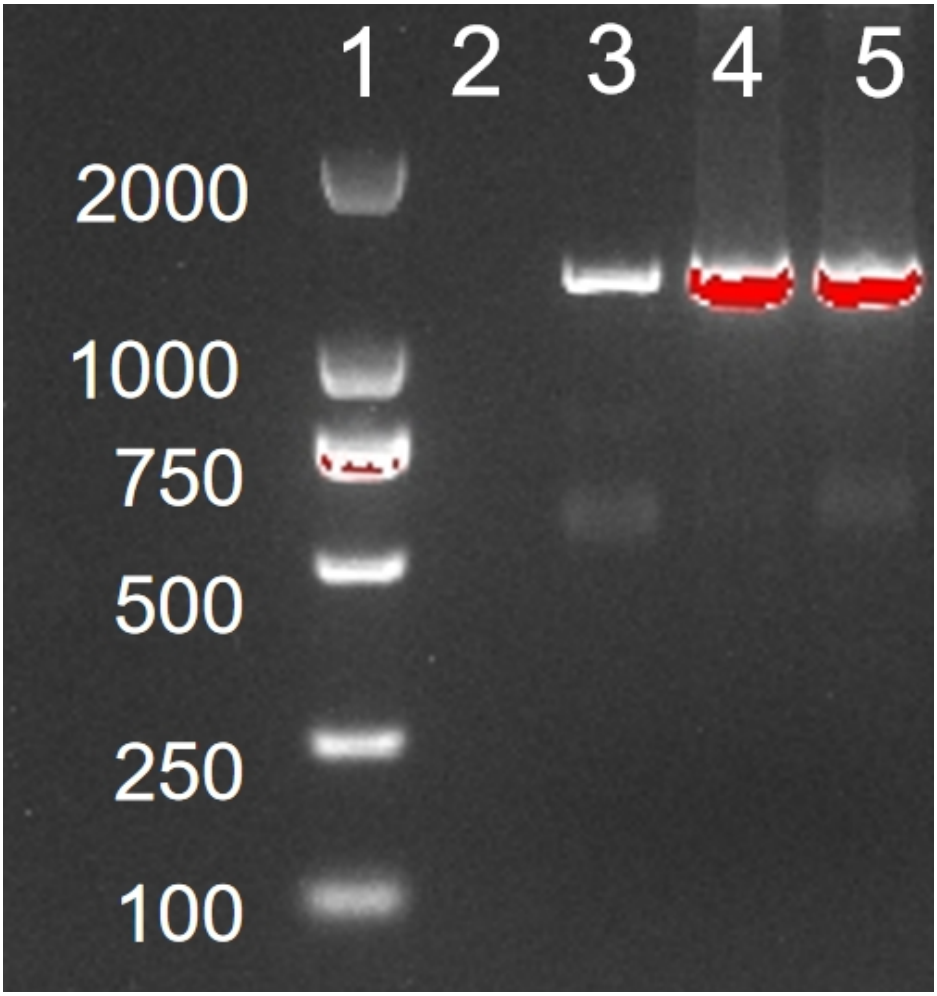
Experiment	Primer design for constructing the *GPA1* humanization donor fragment by fusion PCR (Overlap Extension PCR)
Note	YeastGenome（https://www.yeastgenome.org/） NCBI（https://www.ncbi.nlm.nih.gov/） SnapGene（https://www.snapgene.com/） Addgene（https://www.addgene.org/）
Result	

Forward primer	Reverse primer	Objective
*URA3*_f:AGAGTGCACCATACCACAGC	*URA3*_r:CCGATTTCGGCCTATTGGTT	Obtain *URA3*
U_f:CAGGCATTACAGAAACCGAATTTAACATCG	U_r:ATGATGAATTGAATTGAAAAGCTGTGG	Obtain the upstream homology arm
D_f:GTTAAATCAGCTCATTTTTTAACCAATAGGC	D_r:GCCTAGTAGATCTTGATTCTTTGTCAC	Obtain the downstream homology arm
T1_f:CGGATGAAGACATTTTGAAGGGCC	T1_r:CCGATTTCGGCCTATTGGTT	Verification after
T2_f:CGGATGAAGACATTTTGA	T2_r:CAACAGGACTAGG	transformat

	AGGGCC ATGAGTAGCAGC ion *URA3*_f:AGAGTGCACCATACCA *URA3*_r:CCGATTTCGGCC 获得 *URA3* CAGC TATTGGTT Primer design
Explanation	The earlier Gibson multi-round PCR route was declared a failure due to accumulated mutations from multiple rounds of amplification. In this experiment, based on the fusion PCR scheme in Protocol 8, primer splitting design was carried out for the *GPA1* humanization donor fragment. The complete donor was divided into three segments, A, B, and C, with homologous overlapping sequences added between the fragments, so that the full-length donor could be directly spliced through overlap extension PCR, thereby reducing the number of PCR cycles and lowering the risk of base mutations. The outer amplification primers continued to use the previously verified target DNA_f/r to ensure the correctness of the homology arm sequences at both ends. In this stage, the design of the entire primer strategy was completed, laying the groundwork for the subsequent wet-lab fusion PCR experiments.
Next plan	The complete set of fusion PCR primers was synthesized, and segmented PCR amplification of each DNA fragment was carried out.

No 9 Date：2026.07.06- 07.12 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

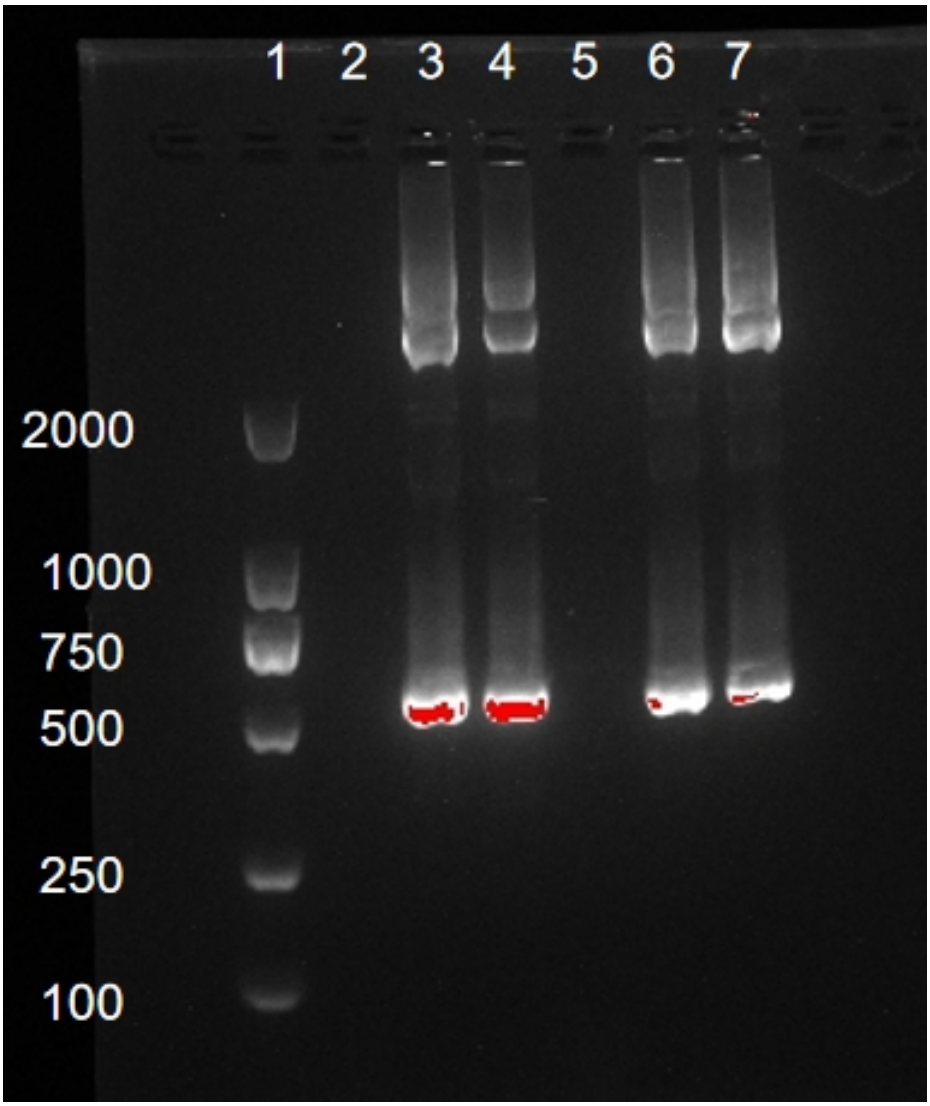
Experiment	Fusion PCR: separate amplification of the upstream homology arm, the *URA3* expression cassette, and the downstream homology arm fragment
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)、Protocol 8（Overlap Extension PCR）
Result	7.7



Preliminary experiment: obtainment of the target fragment (the target band is 1262 bp)

1: 2000 bp marker; 2: blank control; 3–5: experimental group

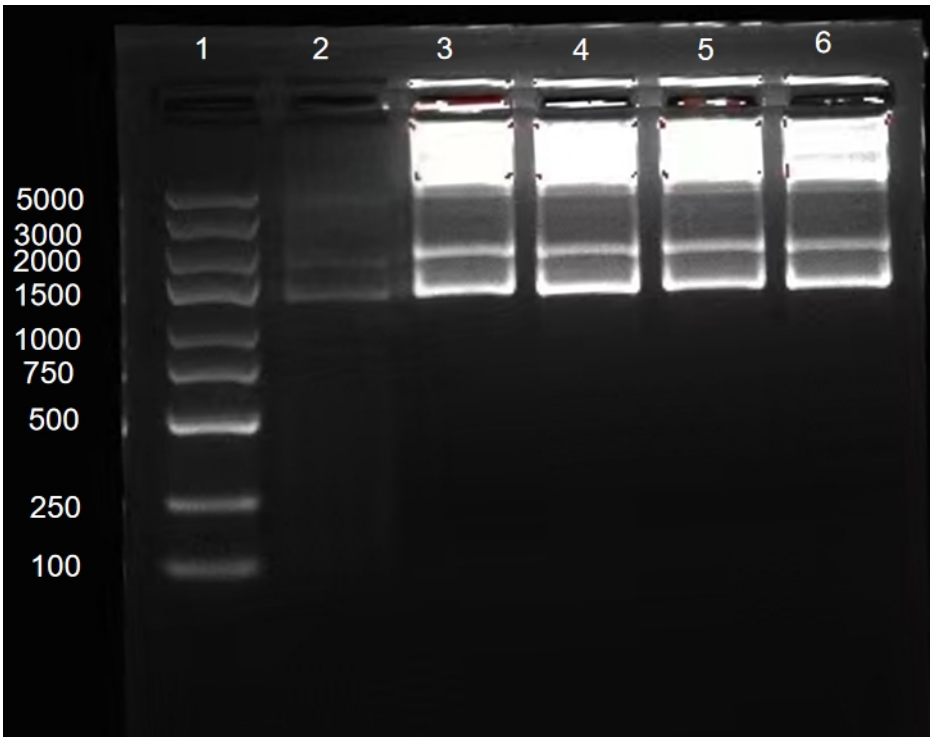
7.9



Preliminary experiment: obtainment of the upstream and downstream homology arms (the target bands are 558 bp and 540 bp, respectively)

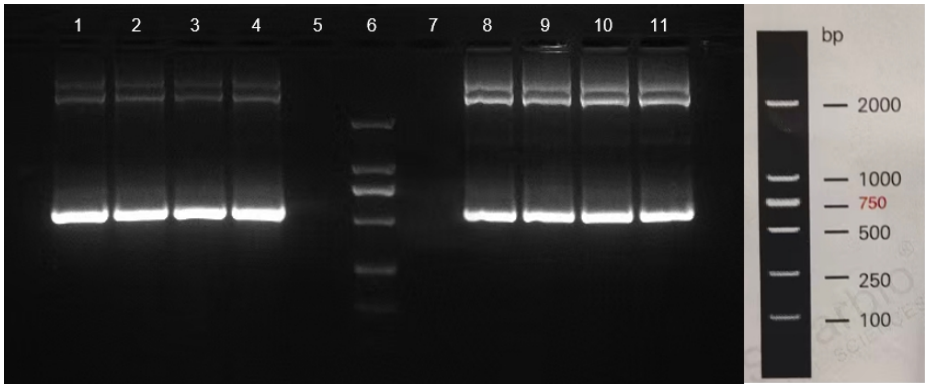
1: 2000 bp marker; 2: blank control for upstream homology arm amplification; 3–4: experimental group for upstream homology arm amplification; 5: blank control for downstream homology arm amplification; 6–7: experimental group for downstream homology arm amplification

7.11



Formal experiment: obtainment of the target fragment (the target band is 1262 bp)

1: 2000 bp marker; 2: blank control; 3–6: experimental group



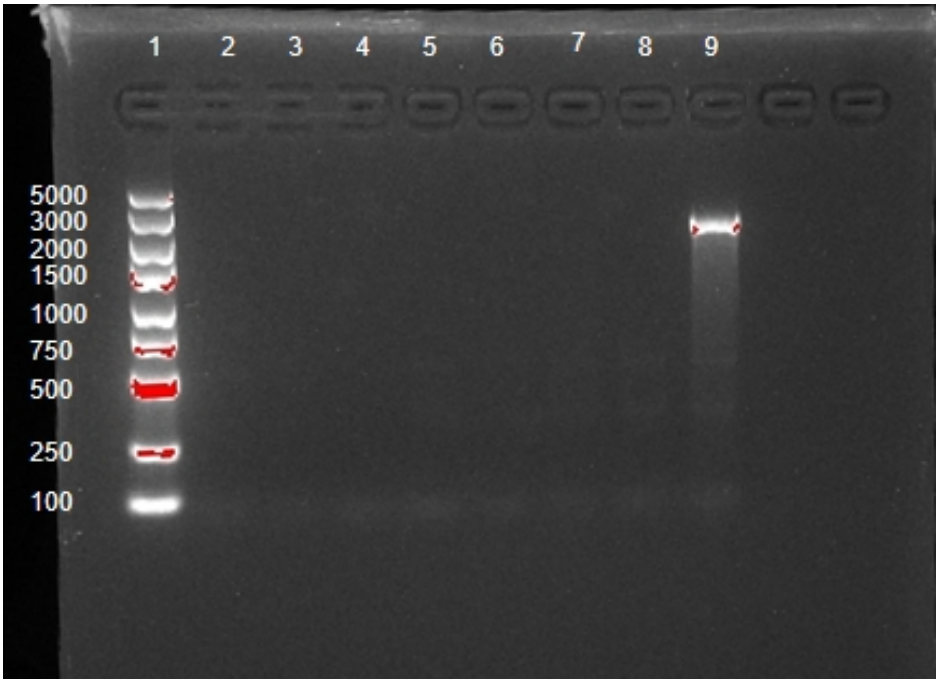
Formal experiment: obtainment of the upstream and downstream homology arms (the target bands are 558 bp and 540 bp, respectively)

1–4: experimental group for upstream homology arm amplification; 5: blank control for upstream homology arm amplification; 6: 2000 bp marker; 7: blank control for downstream homology arm amplification; 8–10: experimental group for downstream homology arm amplification

Sample	Concentration (ng/μL)	A260/A280
Upstream homology arm 1	121	1.814
Upstream homology arm 2	111	1.861

URA3 1	72.95	1.812
URA3 2	140.60	1.668
Downstream homology arm 1	75.45	1.711
Downstream homology arm 2	82.000	1.739

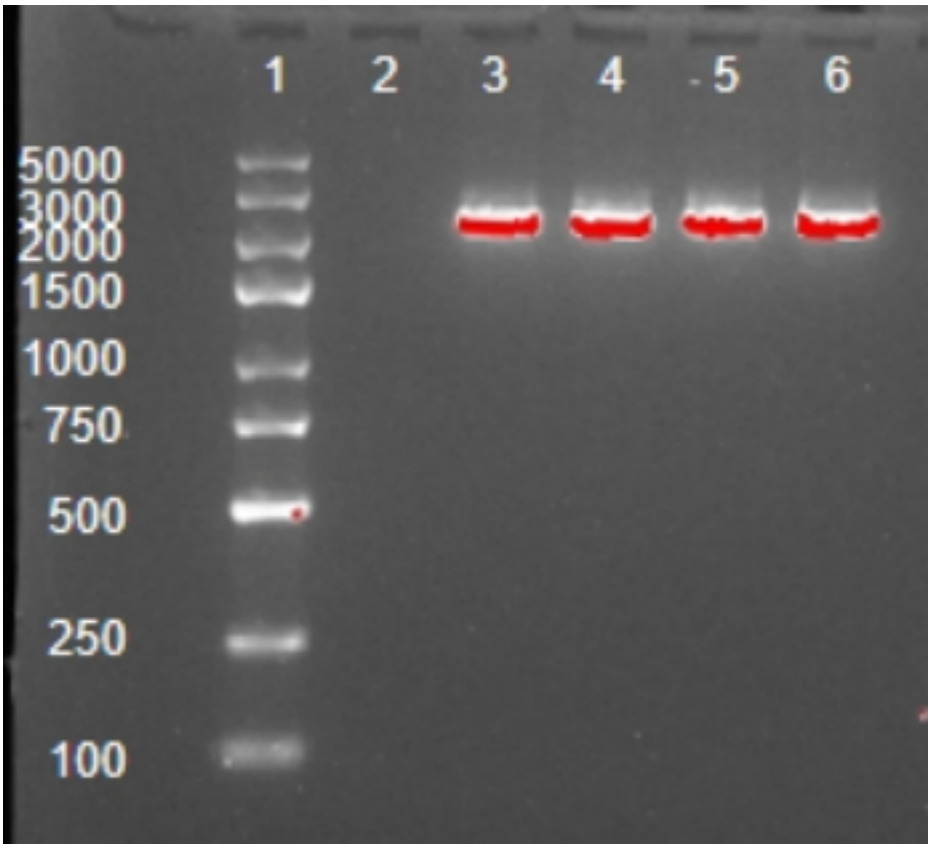
Concentration and purity of the linearized product recovered by gel extraction



Preliminary experiment: fusion PCR (the target band is 2280 bp)

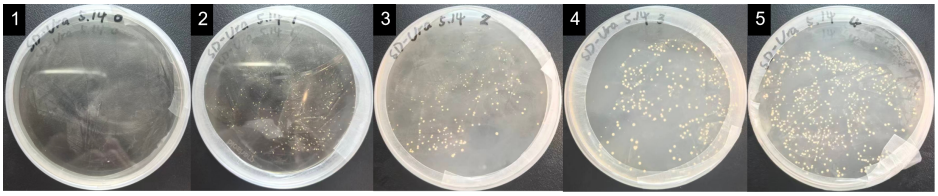
1: 5000 bp marker; 2: blank; 3: upstream homology arm; 3: *URA3*; 4: downstream homology arm; 5: upstream homology arm + *URA3*; 6: *URA3* + downstream homology arm; 7: upstream homology arm + downstream homology arm; 8: upstream homology arm + *URA3* + downstream homology arm

7.12

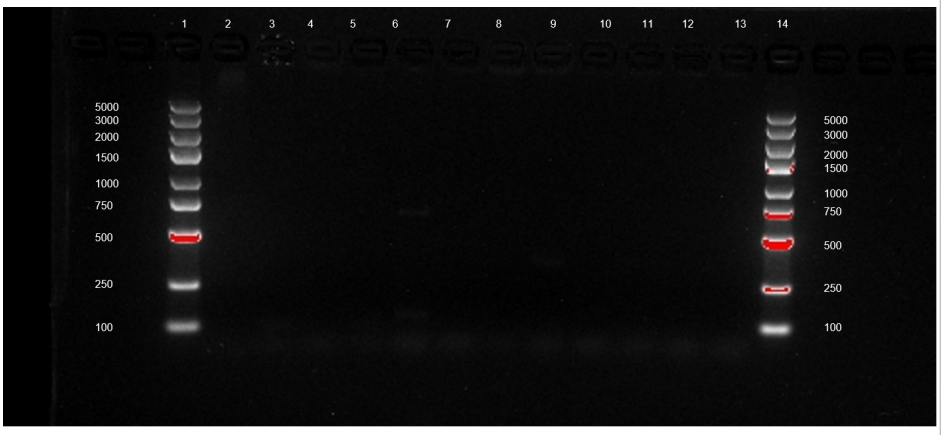
	 **Formal experiment: fusion PCR (the target band is 2280 bp)** 1: 5000 bp marker; 2: blank; 3–6: experimental group
Explanation	This experiment implemented Protocol 8 overlap extension fusion PCR, completing the construction of the full-length donor fragment in two steps. In the first round, the primer-free reaction relied on the homologous overlapping regions between fragments to achieve bridging and splicing of fragments A, B, and C; in the second round, outer primers were added for amplification to obtain the complete 2280 bp editing donor fragment. Compared with the previous multi-round PCR–Gibson route, fusion PCR reduces the number of cycles and lowers the risk of mutations introduced by amplification; the full-length product was recovered and purified.
Next plan	The fusion PCR band was single, and it was prepared for transformation.

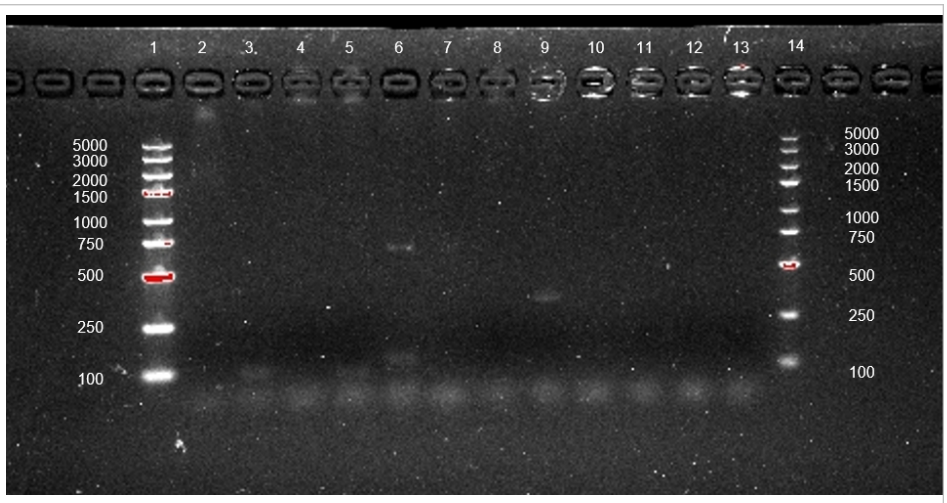
No 10 Date：2026.07.15- 07.18 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	The fusion PCR product was transformed into *Saccharomyces cerevisiae*.
Note	Protocol 12(Yeast Transformation)
Result	7.18

	 **Yeast transformation plate images** 1: blank control; 2–5: experimental group (the date marked on the plates in this figure is the plate-pouring date)
Explanation	The full-length donor fragment obtained by fusion PCR splicing largely circumvented the base mutation problem of the previous Gibson–multi-round PCR scheme. This linear editing donor fragment was introduced into *BY4741* yeast cells, and potential yeast transformants with C-terminal humanization of *GPA1* were obtained on SD-Ura plates by relying on the *URA3* selection marker.
Next plan	Colony PCR verification was performed and the samples were sent for sequencing.

No 11 Date：2026.07.19- 07.23 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

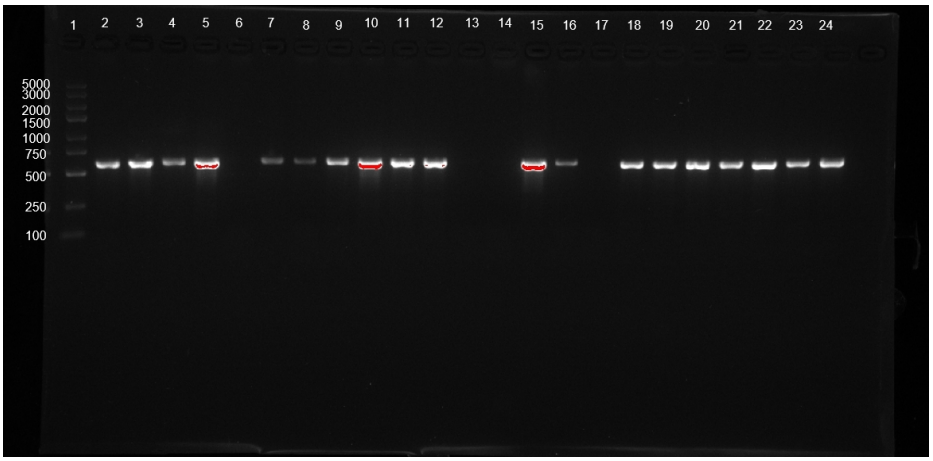
Experiment	Colony PCR verification and analysis of the sequencing results.
Note	Protocol 13(Colony PCR for *Saccharomyces cerevisiae*)、Protocol 5(Agarose Gel Electrophoresis)
Result	7.19  **Electrophoretogram of the first colony PCR (the target band is 2280 bp)** 1: 5000 bp marker; 14: 5000 bp marker; 2–13: corresponding to picked colonies 1–12, respectively



Electrophoretogram of the second colony PCR (the target band is 2280 bp)

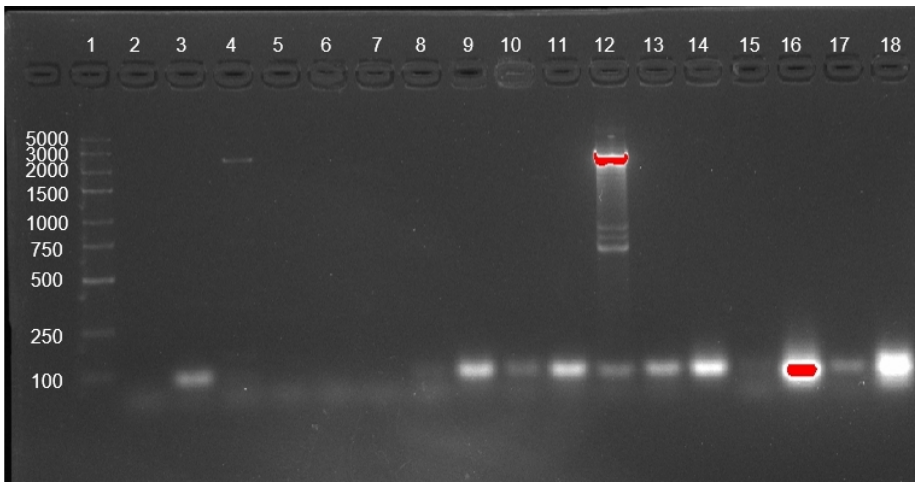
1: 5000 bp marker; 14: 5000 bp marker; 2–13: corresponding to picked colonies 1–12, respectively

7.20



Electrophoretogram of the third colony PCR (verification of the upstream homology arm) (the target band is 558 bp)

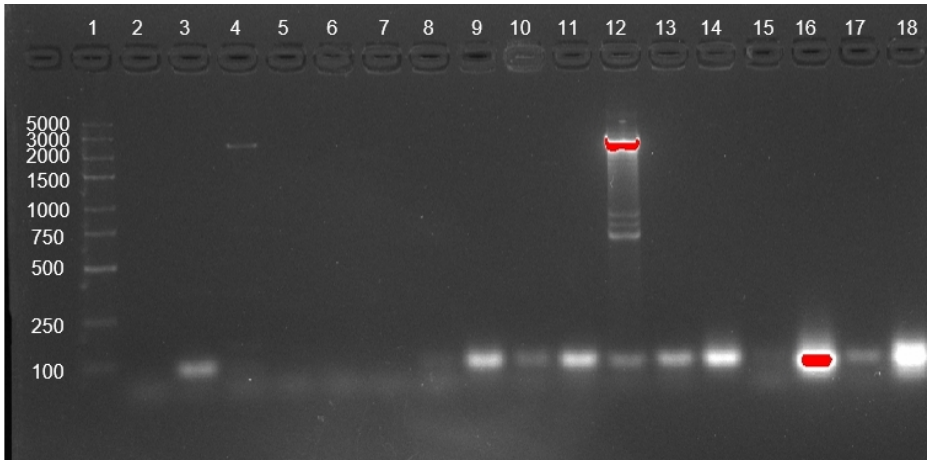
1: 5000 bp marker; 2–24: corresponding to the picked colonies, respectively (1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 8, 5) (because it was felt that 8 and 5 had not been loaded well during sample loading, they were re-loaded)



Electrophoretogram of the fourth colony PCR (verification of the full length) (the target band is 2280 bp)

1: 5000 bp marker; 2–18: corresponding to the picked colonies, respectively (1, 3, 5, 6, 8, 10, 11, 12, 13, 14, 17, 18, 20, 21, 22, 23, 24)

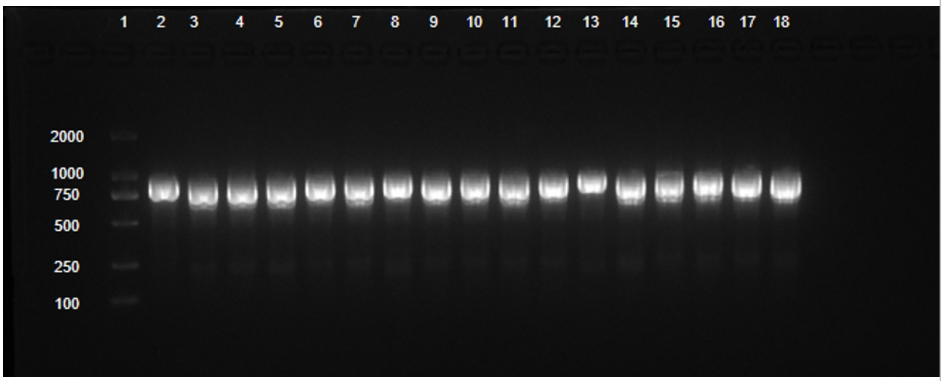
7.21



Electrophoretogram of the fifth colony PCR (verification of the full length) (the target band is 2280 bp)

1: 5000 bp marker; 2–18: corresponding to the picked colonies, respectively (1, 3, 5, 6, 8, 10, 11, 12, 13, 14, 17, 18, 20, 21, 22, 23, 24)

7.22

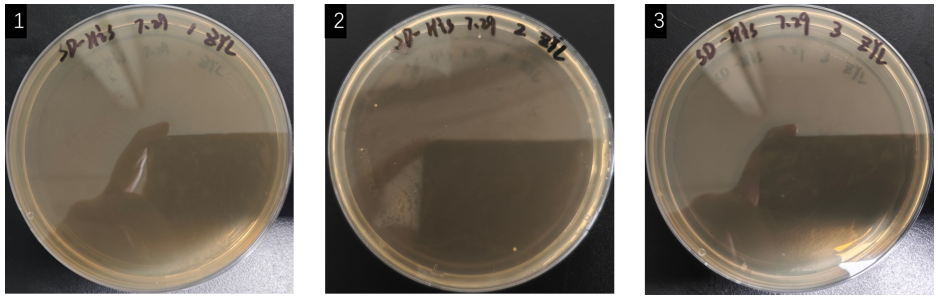
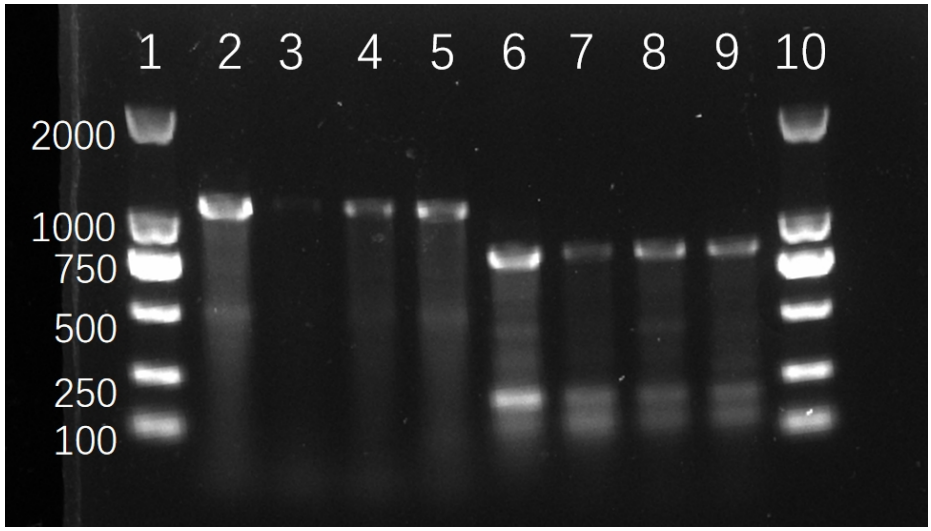


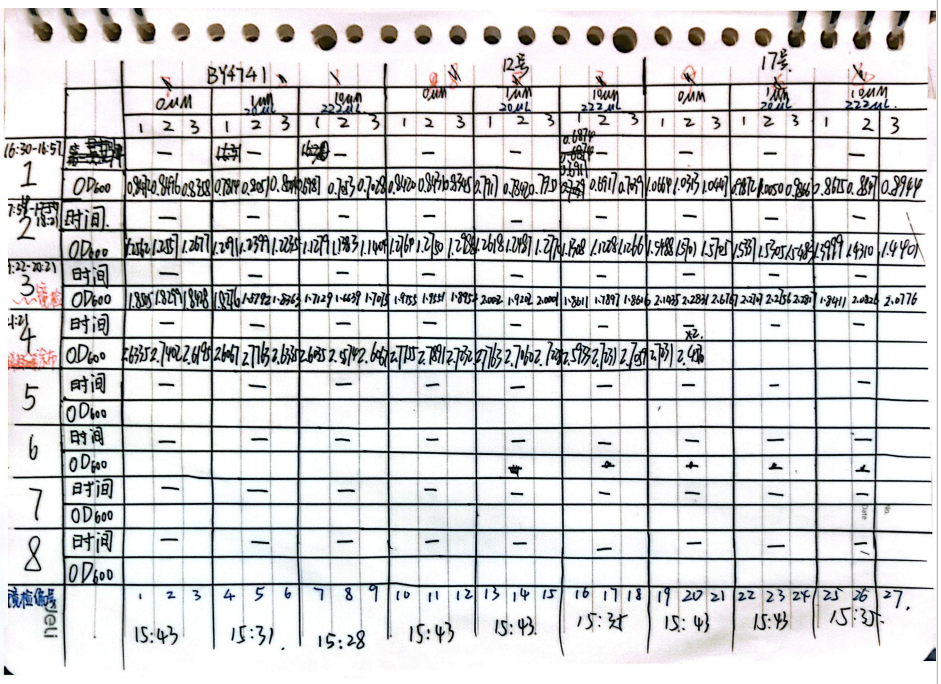
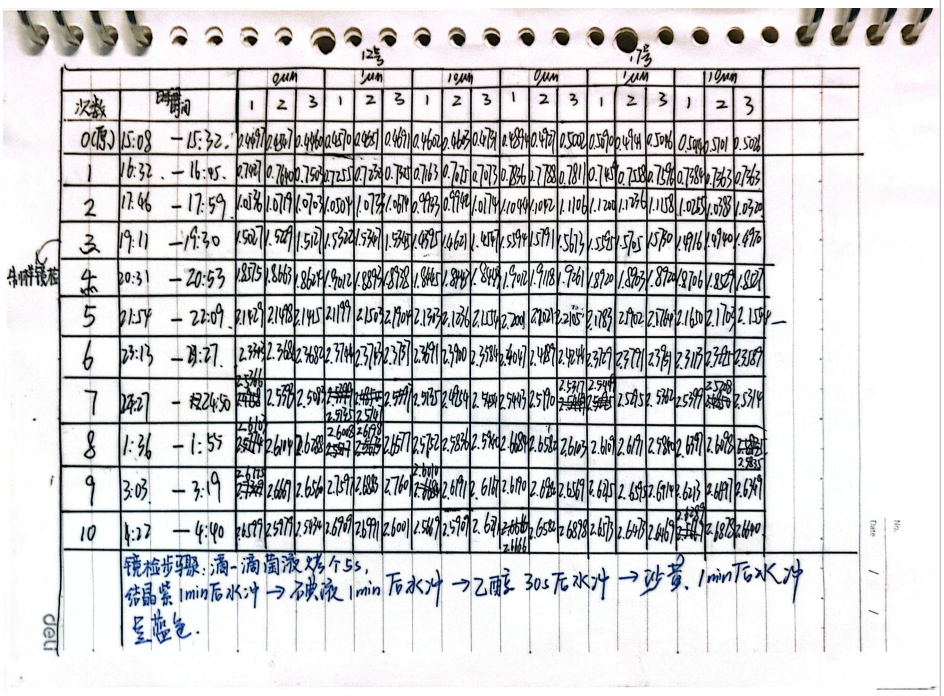
Electrophoretogram of the sixth colony PCR (the target band is 849 bp)

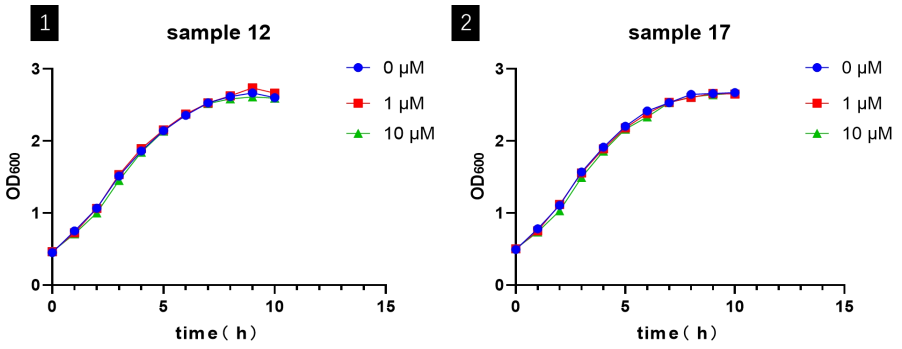
	1: 5000 bp marker; 2–18: corresponding to the picked colonies, respectively (1, 3, 5, 6, 8, 10, 11, 12, 13, 14, 17, 18, 20, 21, 22, 23, 24) 7.23 **Comparison diagrams of sequencing results (key humanized sites)** 1: alignment of sample 12 with the original sequence; 2: alignment of sample 17 with the original sequence; 3: alignment of sample 12 with the expected sequence; 4: alignment of sample 17 with the expected sequence
Explanation	SD-Ura screening only indicates that the *URA3* marker successfully entered the yeast cells; it cannot prove that the donor fragment was site-specifically integrated into the *GPA1* gene locus. In the earlier stage, multiple rounds of full-length primer amplification all failed; experimental condition issues such as marker interference and failure of the lysis solution were successively ruled out, and it was speculated that the original full-length primers had poor amplification efficiency for the recombinant strain. In the third round, the use of local fragment primers was able to amplify bands, indicating that fragment introduction had indeed occurred in some strains, but the full-length primers were difficult to detect it. Therefore, site-spanning specific primers were redesigned, with the upstream primer anchored outside the genomic homology arm and the downstream primer covering the humanized engineering region. Relying on this set of primers combined with sequencing screening, two yeast strains, 12 and 17, with completely correct genotypes for *GPA1* C-terminal humanization engineering were finally obtained.
Next plan	The reporter plasmid carrying the *yEGFP* fluorescent gene was transformed into positive strains 12 and 17.

No 12 Date：2026.07.29- 08.01 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Transformation of the *yEGFP* fluorescent plasmid into *GPA1* humanized positive strains and colony PCR verification
Note	Protocol 12（Yeast Transformation）、Protocol 13（Colony PCR for Saccharomyces cerevisia）、Protocol 5（Agarose Gel Electrophoresis）

Result	8.1  **Transformation plate images** 1: blank control; 2: transformed with the fluorescent plasmid 8.1  **Colony PCR electrophoretogram (the target bands are 1152 bp and 849 bp, respectively)** 1: 2000 bp marker; 2–5: verification of the fluorescent plasmid; 6–9: verification of the humanized sequence; 10: 2000 bp marker
Explanation	Through multiple rounds of PCR screening and sequencing identification in the earlier stage, strains 12 and 17 were confirmed as yeast engineered strains with completely correct site-specific integration for C-terminal humanization of *GPA1*. In this experiment, on that basis, the *yEGFP* fluorescent reporter plasmid was transformed in, and colony PCR was used for rapid verification of plasmid introduction. The results confirmed that the fluorescent plasmid was successfully introduced into the target strains, yielding dually verified engineered yeast possessing both the *GPA1* humanized background and the *yEGFP* fluorescent reporter system, thereby providing qualified experimental material for subsequent functional detection of fluorescent signals.
Next plan	Microscopic morphological examination, growth curve measurement, and functional verification of fluorescence signal intensity were carried out on the constructed fluorescent engineered strains.

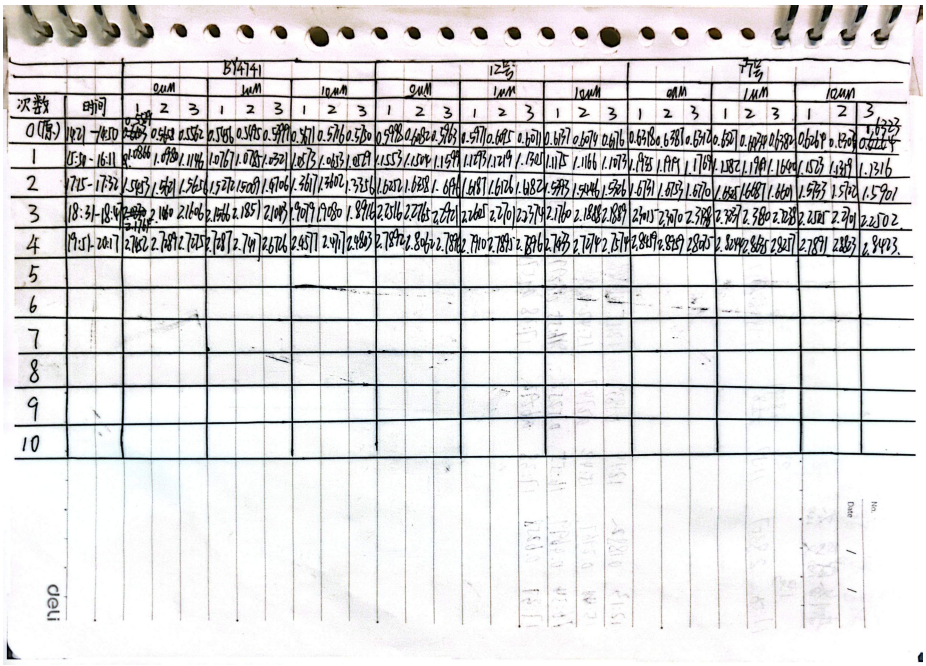
Experiment	Growth curve verification of the *GPA1* humanized yeast strains
Note	Protocol 14(Yeast Growth Curve Measurement)
Result	7.29  **Original record of the first growth curve measurement** 8.2  **Original record of the second growth curve measurement**



Growth curves obtained from the second measurement

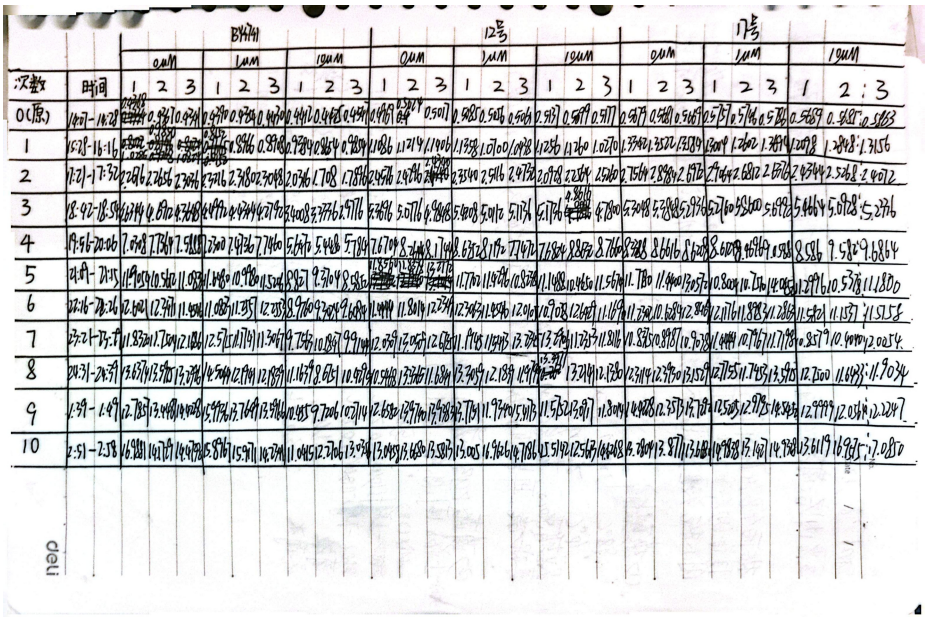
1: growth curve of sample 12; 2: growth curve of sample 17

8.8

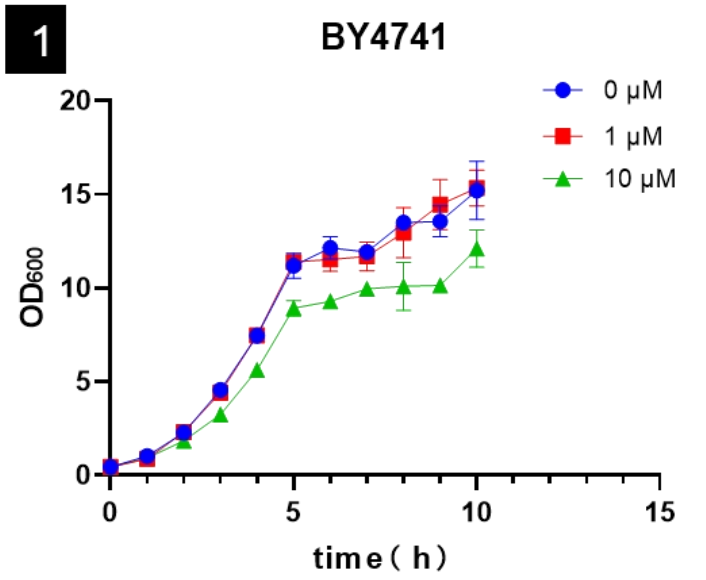


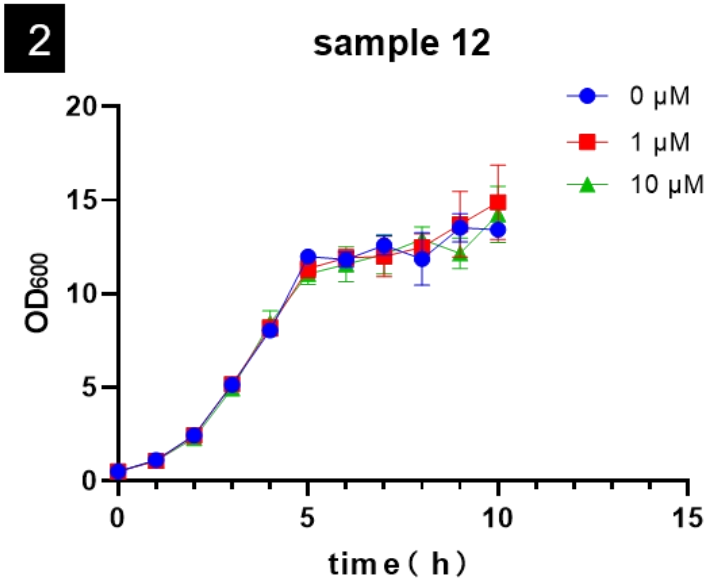
Original record of the third growth curve measurement

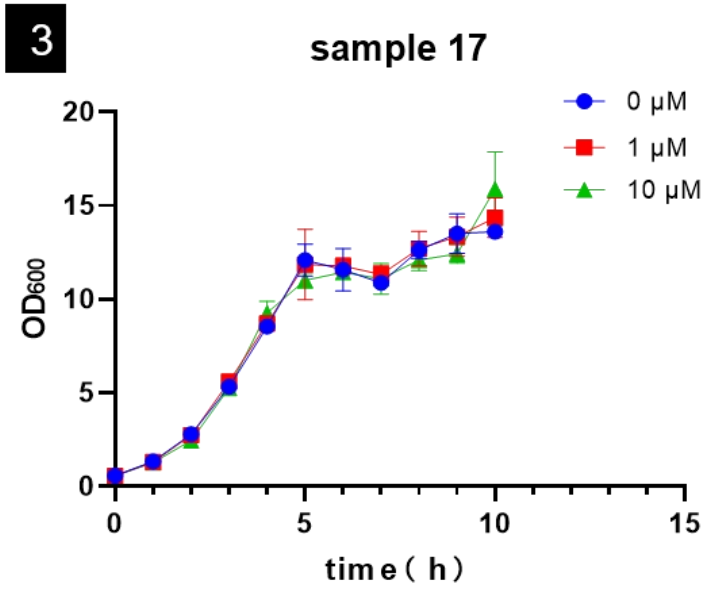
8.10



Original record of the fourth growth curve measurement


1
BY4741
0 μM
1 μM
10 μM
OD600
time (h)

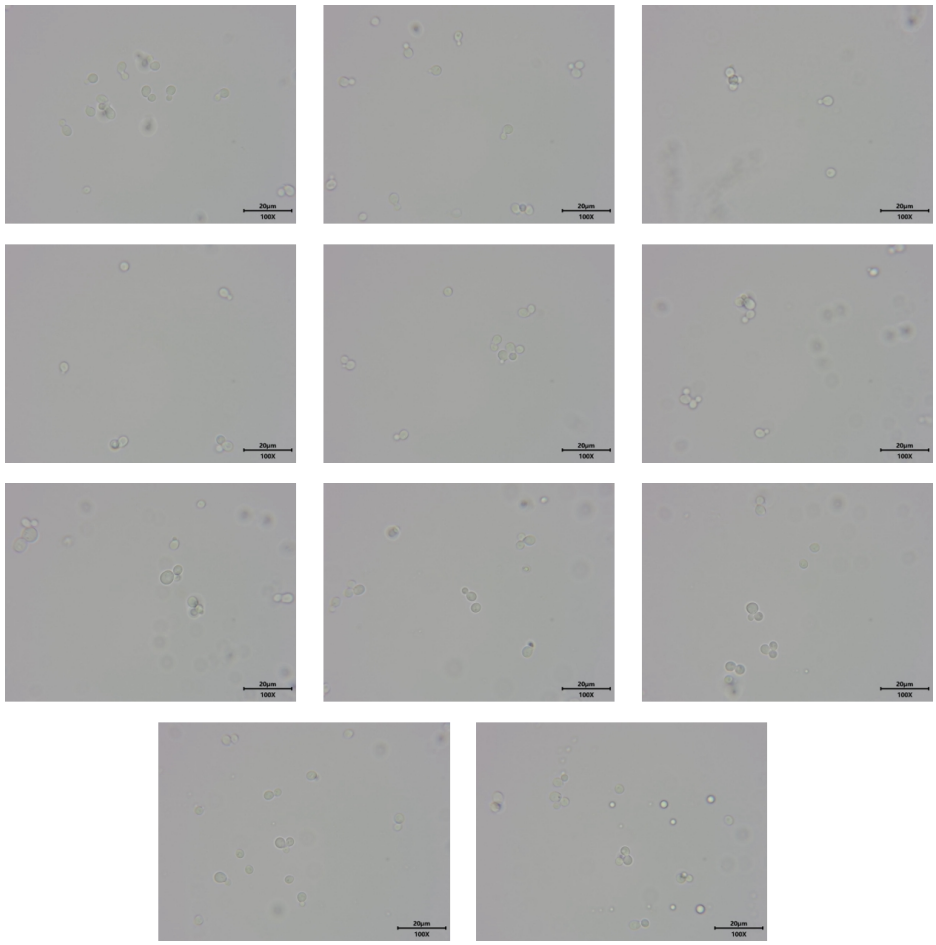

2
sample 12
0 μM
1 μM
10 μM
OD600
time (h)


3
sample 17
0 μM
1 μM
10 μM
OD600
time (h)

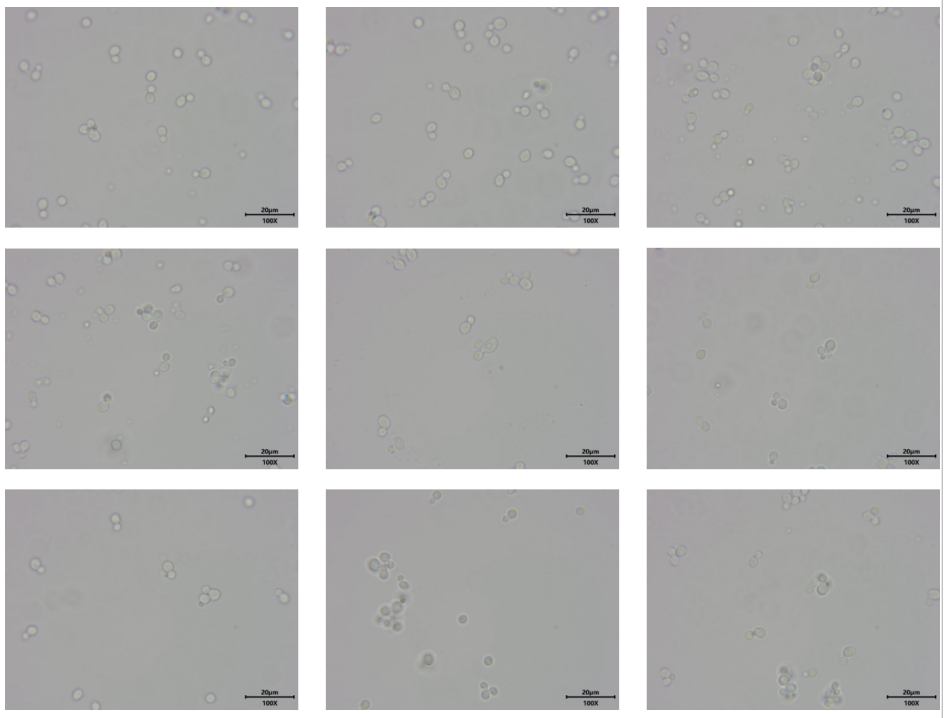
	Growth curves obtained from the fourth measurement 1: growth curve of *BY4741*; 2: growth curve of sample 12; 3: growth curve of sample 17
Explanation	This experiment used only the yeast strain itself, without introducing the *yEGFP* fluorescent plasmid, to evaluate the effects of C-terminal humanization editing of *GPA1* on yeast proliferation and on growth phenotypes under induction conditions. A total of four exploratory experiments were carried out: the first was interrupted due to contamination under YPD culture conditions; the second lacked a wild-type *BY4741* control, resulting in insufficient experimental completeness; in the third, the strains grew too rapidly, and the OD exceeded the linear detection range of the instrument, rendering the data invalid. By summarizing the problems from the first three experiments, dilution was added in the fourth experiment: when OD_{600} was greater than 0.6, the samples were diluted before measurement to ensure that the readings fell within the reliable range of the spectrophotometer, while wild-type *BY4741* was retained as a control. Phenotypic results showed that after high-concentration induction, the growth of wild-type *BY4741* was significantly inhibited, whereas the growth of the humanized strains 12 and 17 was unaffected, consistent with expectations. By comparing the growth rates and maximum OD of the wild-type strain with those of the humanized strains 12 and 17, it was confirmed that the gene-edited strains possessed the expected phenotype, providing a foundational background for subsequent microscopic morphological observation and fluorescence function experiments.
Next plan	Microscopic morphological observation of yeast cells was carried out.

No 14 Date：2026.08.01-08.02 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

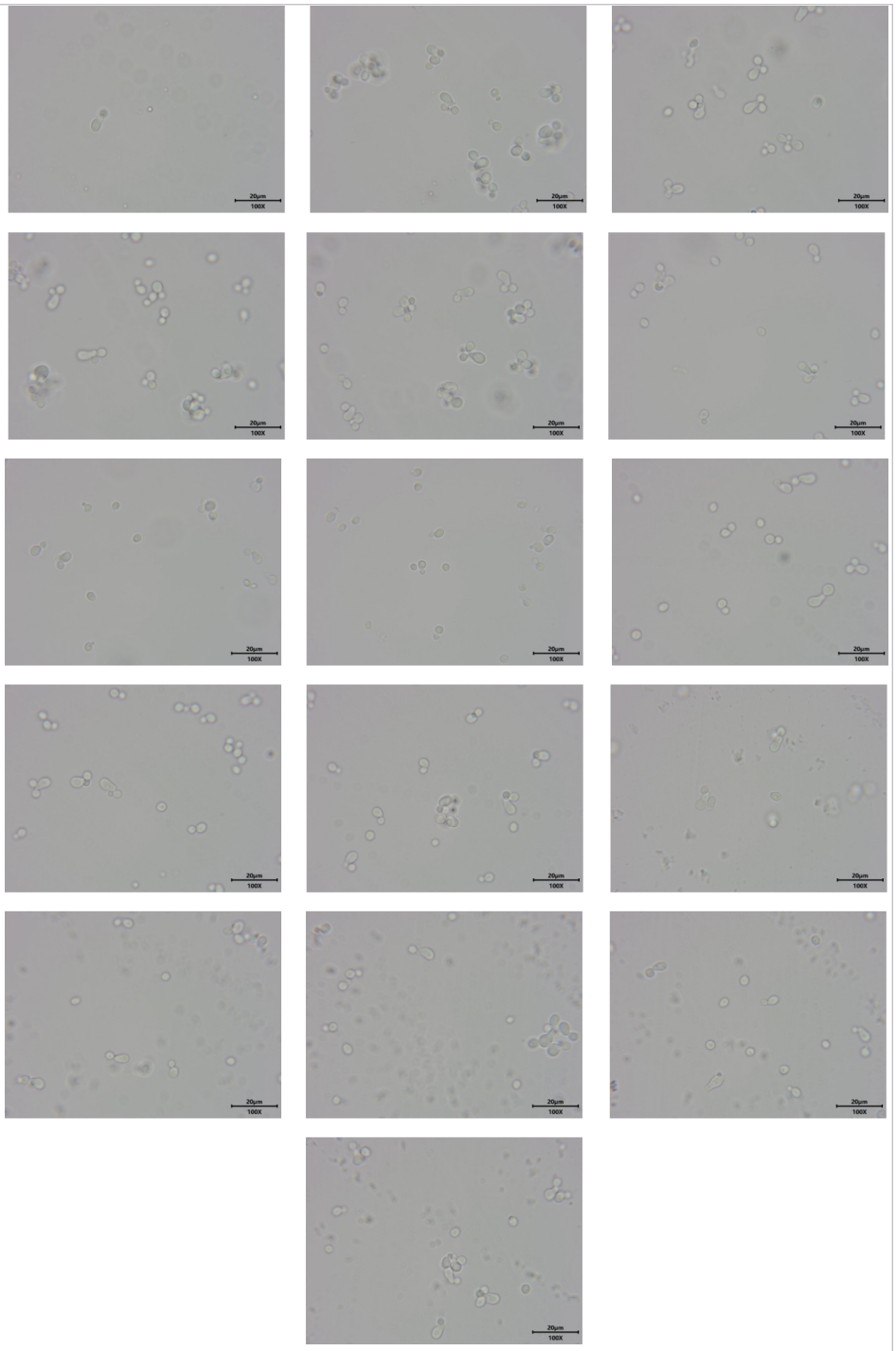
Experiment	Microscopic morphological observation of the *GPA1* humanized yeast strains
Note	Protocol 15(Microscopic Observation of Yeast Morphology)
Result	8.2 .



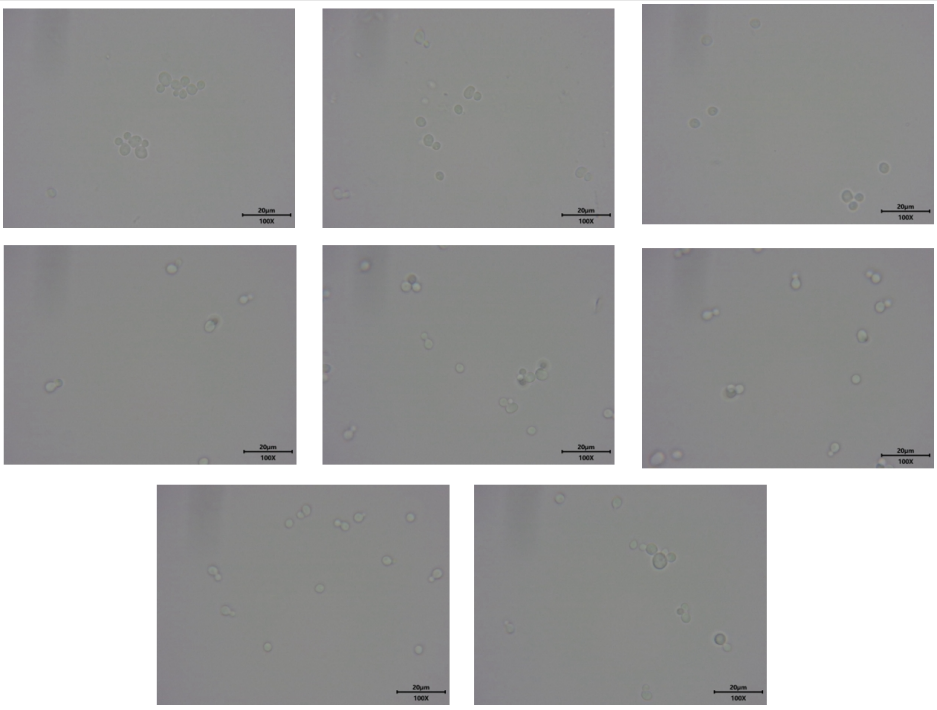
Microscopic image of sample 12 at OD600 of 0.4 under 0 µM α-factor



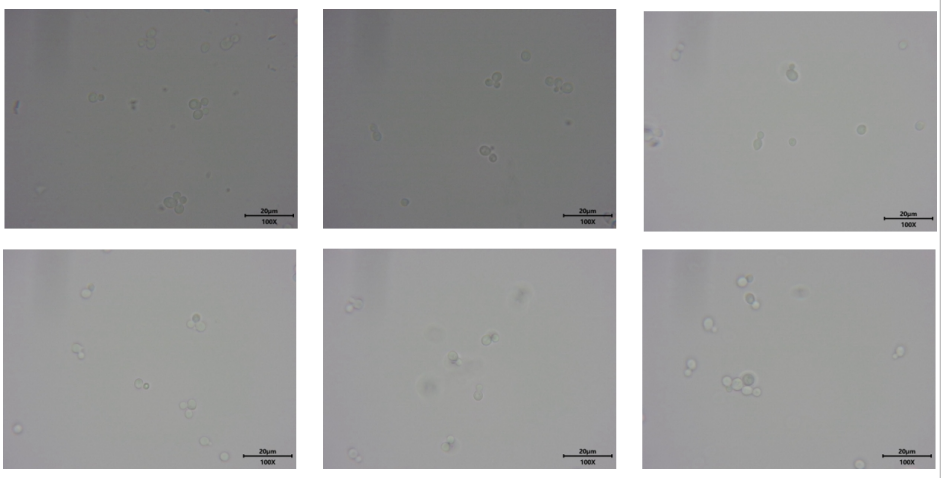
Microscopic image of sample 12 at OD600 of 0.4 under 1 µM α-factor



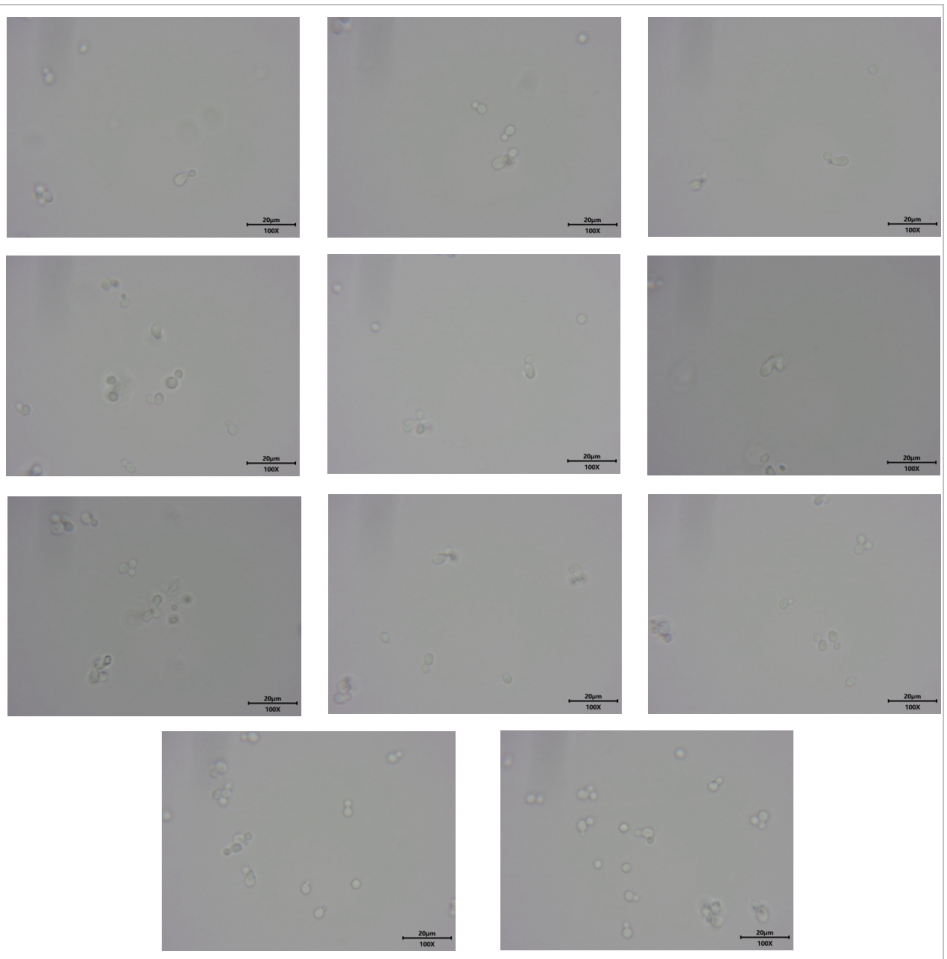
Microscopic image of sample 12 at OD600 of 0.4 under 10 µM α-factor



Microscopic image of sample 17 at OD600 of 0.4 under 0 µM α-factor

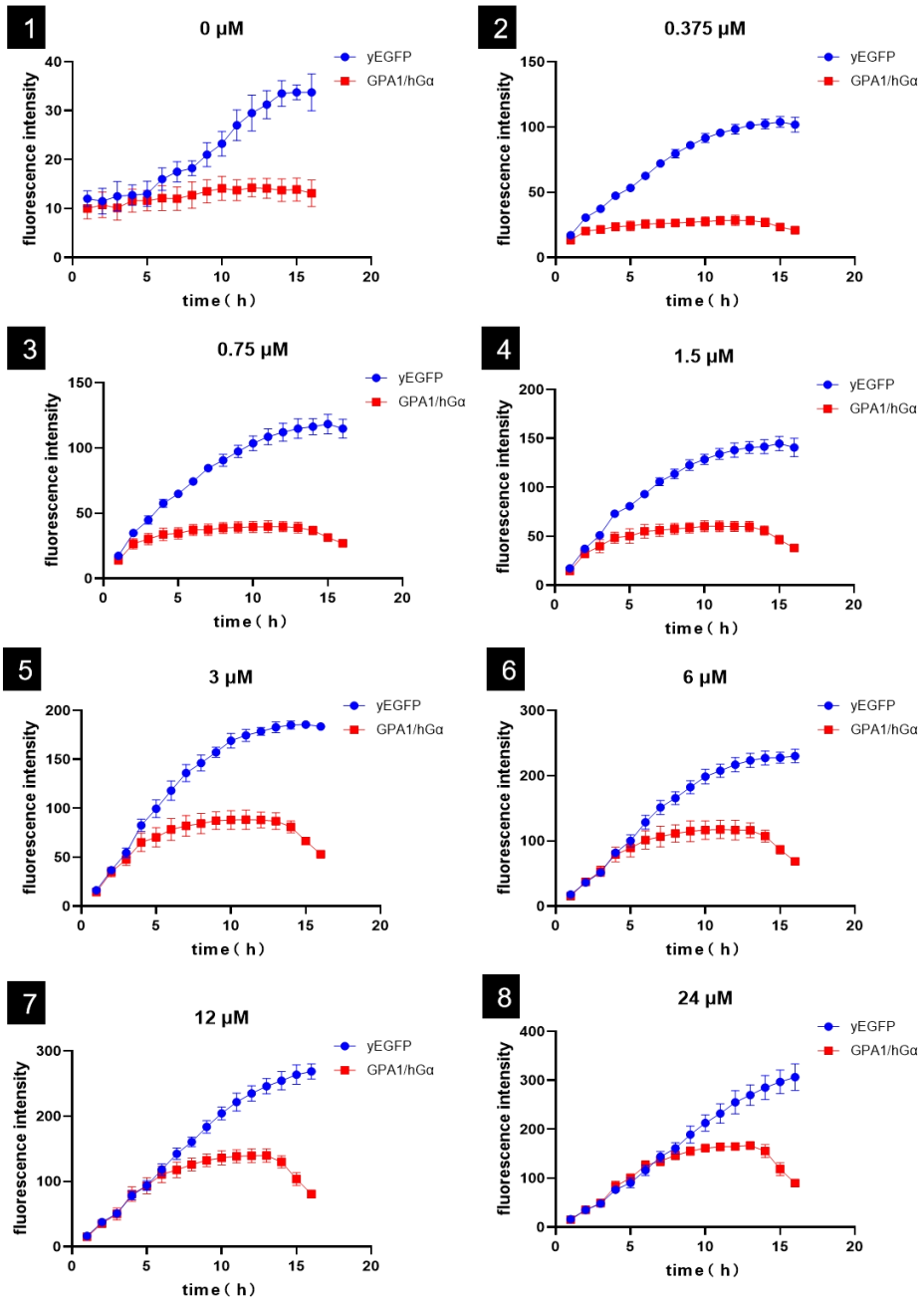


Microscopic image of sample 17 at OD600 of 0.4 under 1 µM α-factor

	 **Microscopic image of sample 17 at OD600 of 0.4 under 10 μM α-factor**
Explanation	On the basis of completing the growth curve measurement, the individual cell morphology was visually observed under a microscope to evaluate whether C-terminal humanization engineering of *GPA1* would cause cell morphology defects. Shmoo is a polarized protrusion phenotype produced by yeast in response to mating pheromone. Combining the observation results under treatments with different induction concentrations: high-concentration induction could cause the wild-type *BY4741* to produce the shmoo phenotype, whereas the humanized strains 12 and 17 showed no significant decrease in cell number and did not exhibit obvious shmoo mating protrusions. The microscopic morphological results and the growth phenotypes obtained from the growth curves corroborate each other, further demonstrating that the engineered strains were in good physiological condition and could be used for subsequent fluorescence signal detection experiments.
Next plan	Fluorescence signal intensity detection of yeast was carried out.

No 14 Date：2026.08.04- 08.05 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Fluorescence signal intensity detection of *GPA1* humanized yeast carrying the *yEGFP* reporter plasmid

Note	Protocol 16(Detection of Yeast Fluorescence Signal After Treatment)
Result	8.5  **Fluorescence signal intensity curves of *GPA1* humanized yeast carrying the *yEGFP* reporter plasmid** 1: fluorescence intensity change curve under 0 μM α-factor induction; 2: fluorescence intensity change curve under 0.375 μM α-factor induction; 3: fluorescence intensity change curve under 0.75 μM α-factor induction; 4: fluorescence intensity change curve under 1.5 μM α-factor induction; 5: fluorescence intensity change curve under 3 μM α-factor induction; 6: fluorescence intensity change curve under 6 μM α-factor induction; 7: fluorescence intensity change curve under 12 μM α-factor induction; 8:

	fluorescence intensity change curve under 24 μM α-factor induction
Explanation	In the earlier stage, genotypic sequencing verification of the strains, growth curve measurement, and microscopic morphological observation of the cells had already been completed, confirming that the humanized strains 12 and 17 were in normal physiological condition. In this experiment, with the aid of the *yEGFP* fluorescent reporter system, the activation level of the intracellular reporter gene under different induction conditions was detected, thereby reflecting the activation response of the G protein signaling pathway. Fluorescence values were normalized to OD_{600} to eliminate interference caused by differences in cell concentration. The fluorescence kinetics results indicated that C-terminal humanization engineering of *GPA1* altered the signal output characteristics of the yeast G protein pathway: the engineered strains possessed a higher peak fluorescence response and an extended time to peak, reflecting a unique signal response kinetics mediated by the humanized Gα subunit, thereby completing the verification of the major phenotypic functions of the engineered strains.
Next plan	All experimental data from the entire set of strain construction and phenotypic testing in this study were organized and compiled, and strain preservation in 80 °C glycerol was carried out according to Protocol 2.

Based on the experimental record of *STE2* gating engineering, translate it.

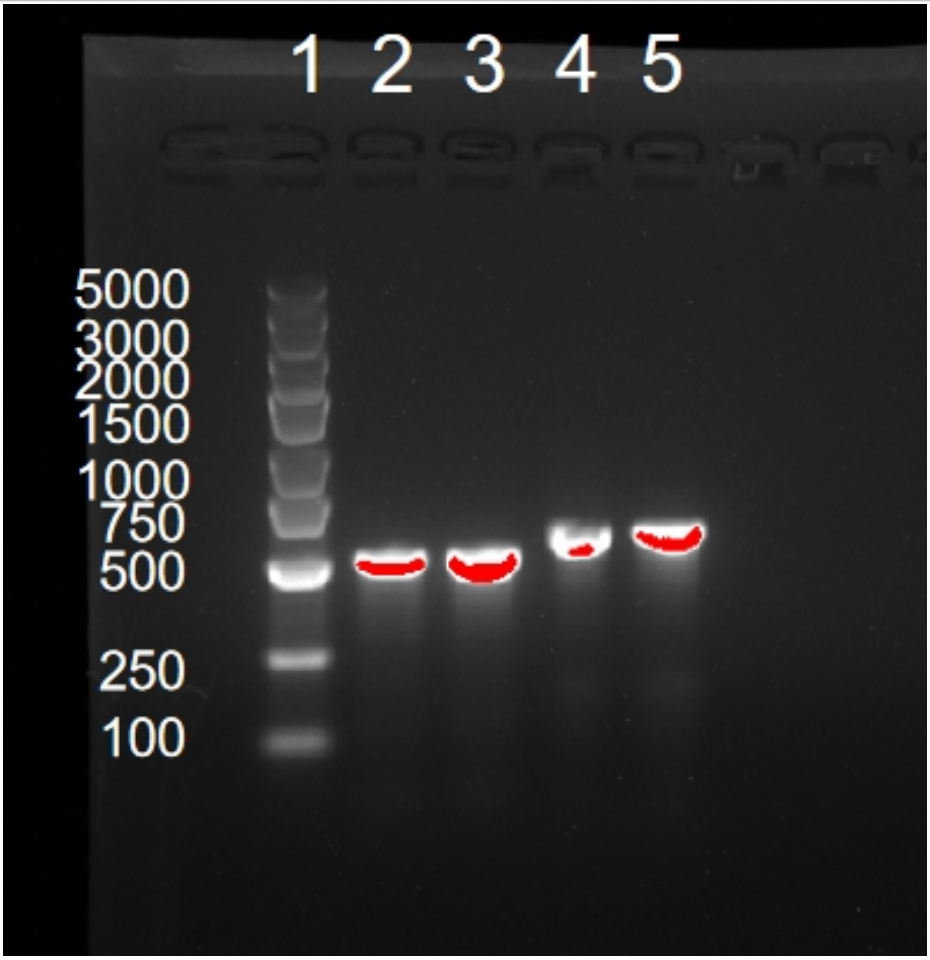
No 1 Date：2026.08.10-08.20 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Based on the preliminary investigation of the *STE2* gating engineering project, preparation for primer and target sequence synthesis
Note	YeastGenome（https://www.yeastgenome.org/） NCBI（https://www.ncbi.nlm.nih.gov/） SnapGene（https://www.snapgene.com/） Addgene（https://www.addgene.org/）
Result	Yeast signal peptide--Antagonist-- anti-H1N1_HA（114）--ALFA tag—TEVcs—LINKER The amino acid sequence is as follows: Yeast signal peptide：MRFPSIFTAVLFAASSALA Antagonist：CRGPQLKPGQPMY anti-H1N1_HA ： QVQLVESGGGLVQPGGSLRLSCAASGSFFSRYRMGWYRQAPGEQRELVA SIAYDGSTSYADPVKGRFTISRDNANTVHLQMYSLKPDDTAVYYCNLDPP GILYWGQGTQVTVSS TEVcs ： GENLYFQSGS ALFA tag ： PSRLEEELRRRLTEP LINKER：GGGSGGGSGT

	Design of the gating TTAAAAATGCACCGTTAAGAACCATATCCAAGAATCAAAAATGAGATTT CCATCTATTTTTACTGCTGTTTTGTTTGCTGCTTCTTCTGCTTTGGCTTGT AGAGGTCCACAATTGAAACCAGGTCAACCAATGTATCAAGTTCAATTG GTTGAATCTGGTGGTGGTTTGGTTCAACCAGGTGGTTCTTTGAGATTGT CTTGTGCTGCTTCTGGTTCTTTTTTTTCTAGATATAGAATGGGTTGGTATA GACAAGCTCCAGGTGAACAAAGAGAATTGGTTGCTTCTATTGCTTATGA TGGTTCTACTTCTTATGCTGATCCAGTTAAAGGTAGATTTACTATTTCTA GAGATAATGCTAATACTGTTCATTTGCAAATGTATTCTTTGAAACCAGAT GATACTGCTGTTTATTATTGTAATTTGGATCCACCAGGTATTTTGTATTGG GGTCAAGGTACTCAAGTTACTGTTTCTTCTGGTGAAAATTTGTATTTTCA ATCTGGTTCTCCATCTAGATTGGAAGAAGAATTGAGAAGAAGATTGACT GAACCAGGTGGTGGTTCTGGTGGTGGTTCTGGTACTTCTGATGCTGCTC CTTCATTGAGCAATCTATTTTATGATC **Synthetic sequence carrying homology arms**
Explanation	This part constitutes the initial preparation stage of the *STE2* gating engineering project. The background related to *STE2* receptor signaling and gating regulation was thoroughly studied to provide a theoretical basis for subsequent molecular experiments; the synthesis of primers and the engineered target sequence was completed, serving as the foundation for subsequent amplification of the *STE2* upstream and downstream homology arms, fragment assembly, and yeast editing.
Next plan	Preliminary experiments for amplification of the upstream and downstream homology arms of the *STE2* gene were carried out to explore PCR amplification conditions.

No 2 Date：2026.08.12-08.23 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

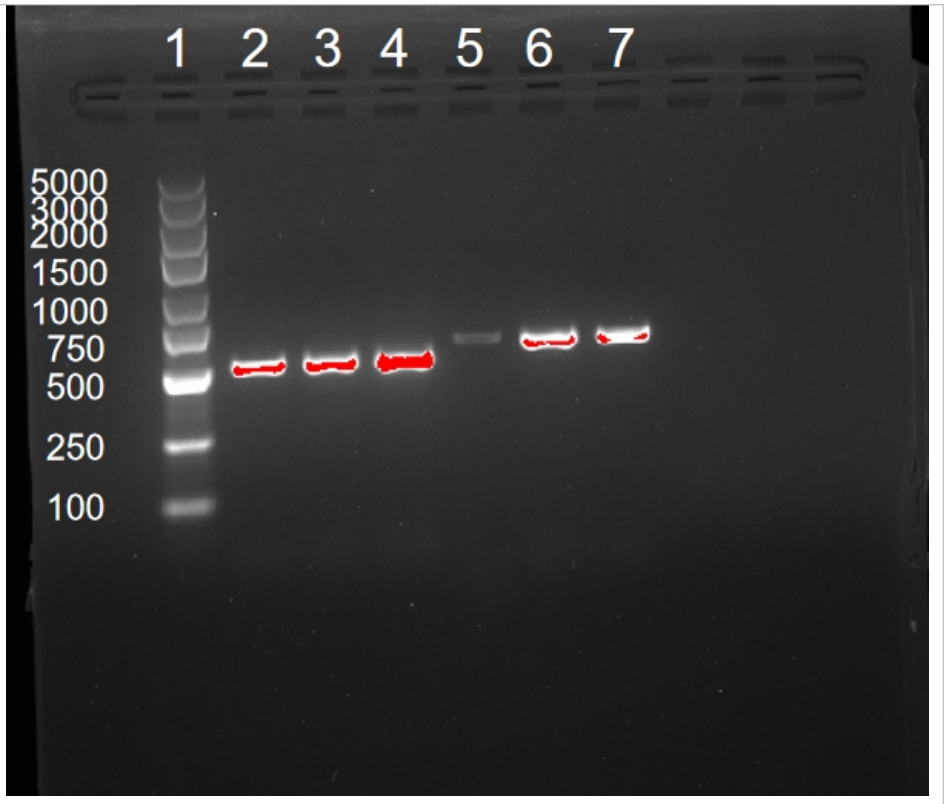
Experiment	Colony PCR amplification of the *STE2* upstream homology arm and the *STE2* downstream homology arm.
Note	Protocol 13(Colony PCR for *Saccharomyces cerevisiae*)、Protocol 5(Agarose Gel Electrophoresis)
Result	8.12



Preliminary experiment: electrophoretogram of colony PCR amplification of the *STE2* upstream homology arm and the *STE2* downstream homology arm (the target bands are 541 bp and 602 bp, respectively)

1: 5000 bp marker; 2–3: *STE2* upstream homology arm; 4–5: *STE2* downstream homology arm

8.23

	 **Formal experiment: electrophoretogram of colony PCR amplification of the *STE2* upstream homology arm and the *STE2* downstream homology arm (the target bands are 541 bp and 602 bp, respectively)** 1: 5000 bp marker; 2–4: *STE2* upstream homology arm; 5–7: *STE2* downstream homology arm
Explanation	This experiment served as a preliminary fragment amplification experiment for *STE2* gating engineering, directly using a single yeast colony as the template to obtain the *STE2* upstream and downstream homology arms by colony PCR, without the need to extract genomic DNA in advance, making the procedure simple, convenient, and rapid. Electrophoresis results showed good amplification specificity, and stable PCR reaction conditions were established, requiring no further optimization of parameters such as annealing temperature and cycle number. The qualified amplification products obtained, after recovery and purification of the PCR products, could be used for subsequent fragment assembly, thereby providing high-quality homology arm fragments for constructing the *STE2* gating engineering editing cassette.
Next plan	Plasmid extraction of *PML104*, vector restriction digestion–ligation, and *Top10* transformation experiments were carried out.

No 3 Date：2026.08.22-08.25 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

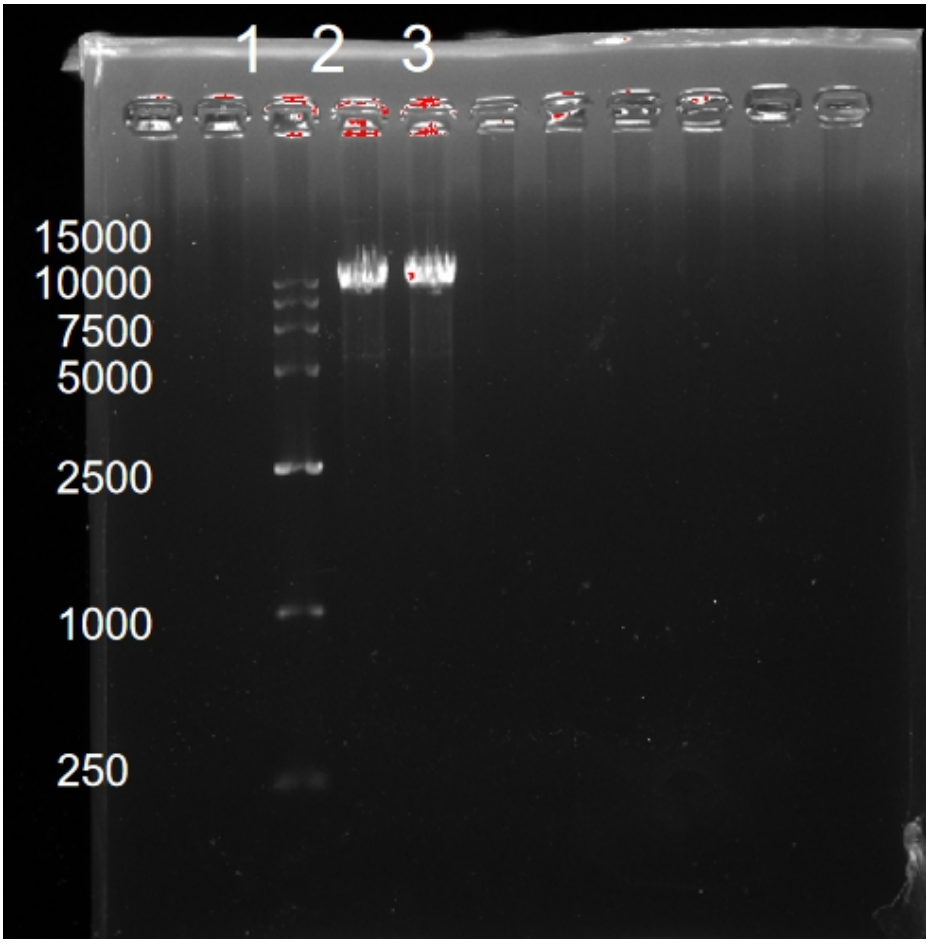
Experiment	Plasmid extraction of *PML104*, restriction digestion for vector modification, ligation, and transformation into *Top10* competent cells.
Note	Protocol 3(Plasmid Extraction from E. coli)、Protocol 4(Restriction Enzyme Digestion)、Protocol 1(E. coli Transformation and Culture Expansion)、Protocol 5(Agarose Gel Electrophoresis)

Result

8.22

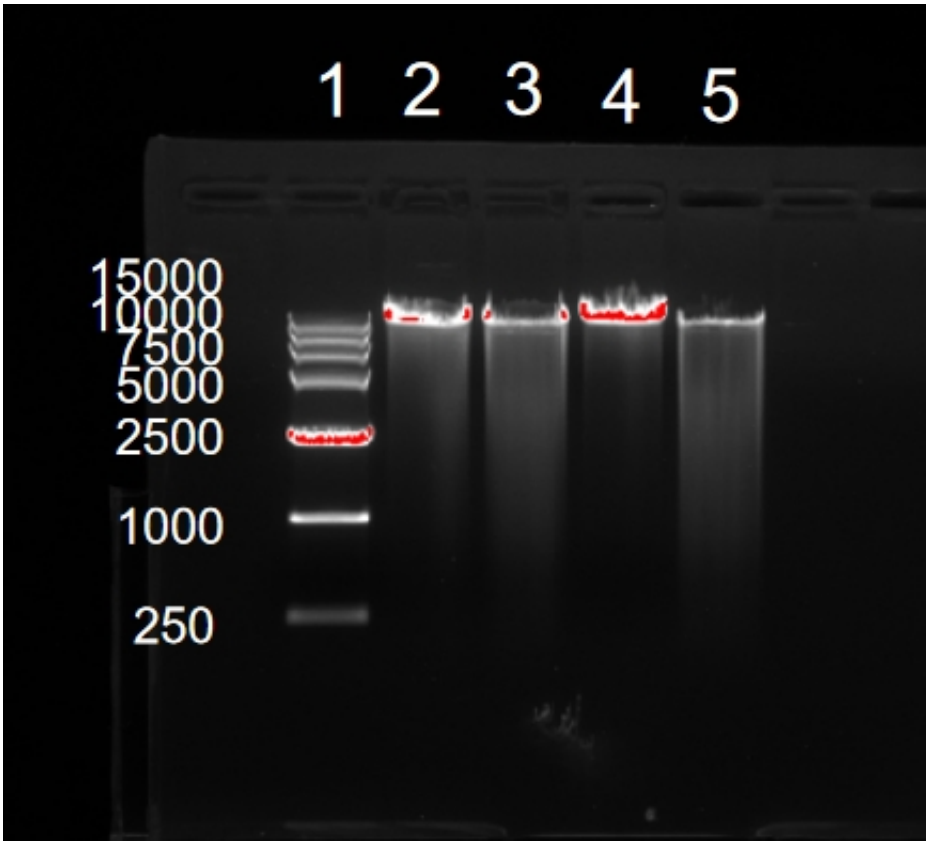
Sample	Concentration (ng/μL)	A260/A280
1	345.60	1.882
2	379.00	1.872
3	258.60	1.873
4	309.10	1.863
5	351.90	1.878

Concentration and purity of the extracted *PML104* plasmid



Electrophoretogram after verification by digestion with SwaI (the target band is 11258 bp)

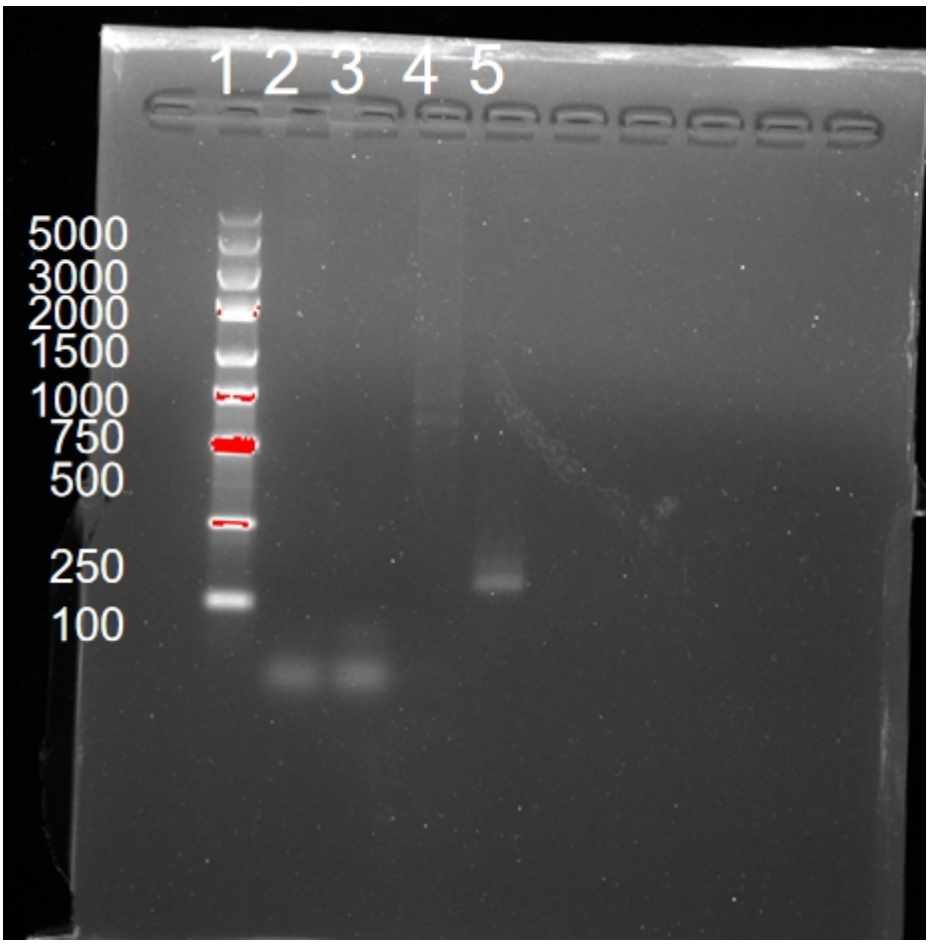
1: 15000 bp marker; 2–3: initial digestion products



Electrophoretogram after verification by digestion with BclI (the target band is 11234 bp)

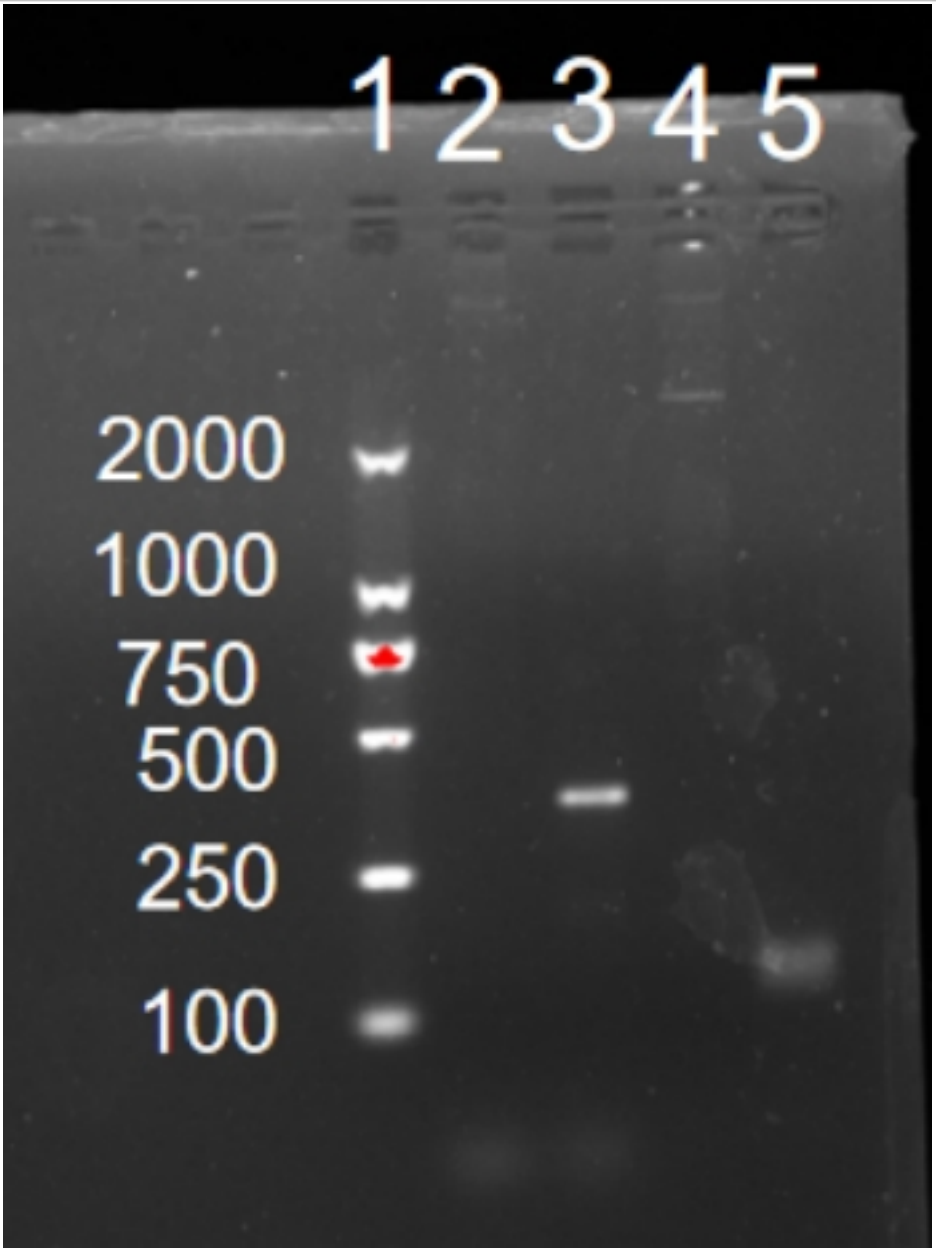
1: 15000 bp marker; 2–5: secondary digestion products

8.24



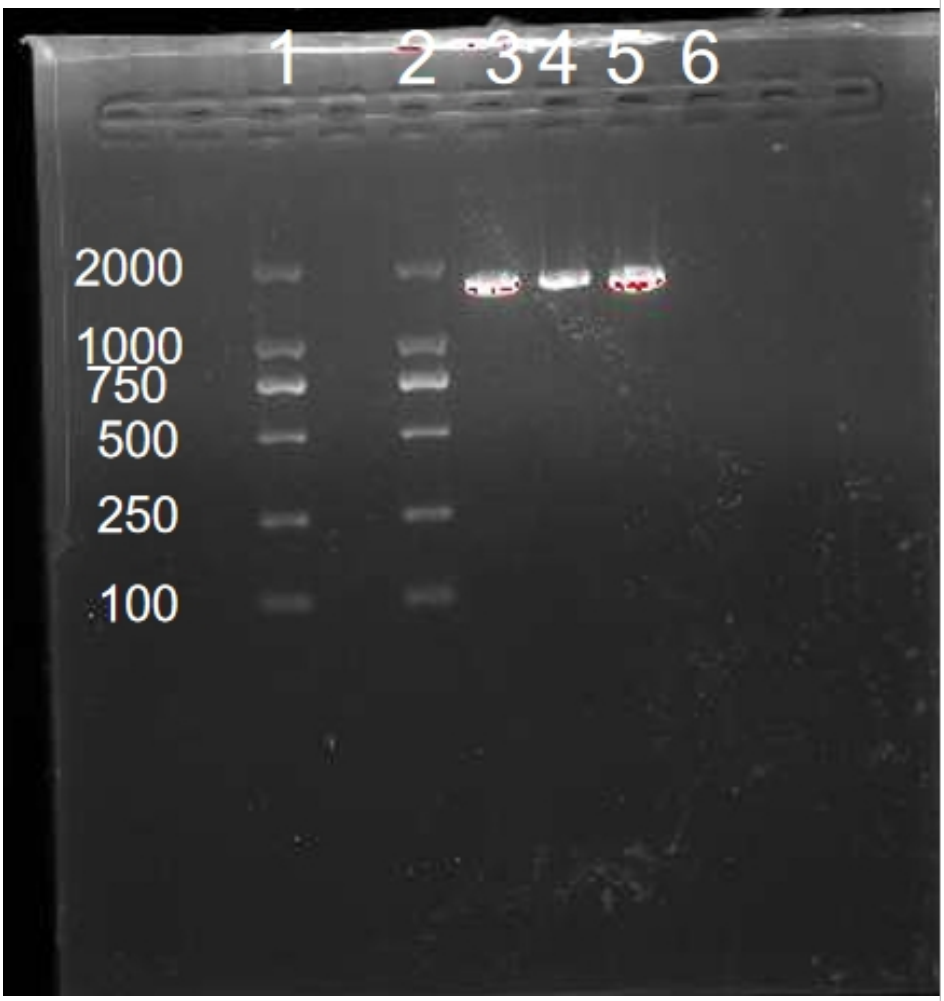
Electrophoretogram of the initial verification of the ligation products using high-fidelity enzyme (the target bands are 111 bp and 332 bp, respectively)

1: 5000 bp marker; 2: *PML104* verification 1; 3: modified *PML104* verification 1; 4: *PML104* verification 2; 5: modified *PML104* verification 2

	 **Electrophoretogram of the initial verification of the ligation products using standard enzyme (the target bands are 111 bp and 332 bp, respectively)** 1: 5000 bp marker; 2: *PML104* verification 1; 3: modified *PML104* verification 1; 4: *PML104* verification 2; 5: modified *PML104* verification 2
Explanation	This experiment aimed to insert the target sgRNA sequence into the sgRNA expression cassette of the *PML104* vector to construct a CRISPR-Cas9 editing plasmid for *STE2* gating engineering. The quality of the sgRNA annealing recovery product was acceptable, and the electrophoretic bands from the initial double restriction digestion of the vector were correct, proving that the vector digestion step itself was successful. However, the subsequent PCR identification bands of the ligation products were abnormal; regardless of whether high-fidelity enzyme or standard enzyme was used for amplification, it was impossible to distinguish the recombinant plasmid from the empty vector, and the positions of

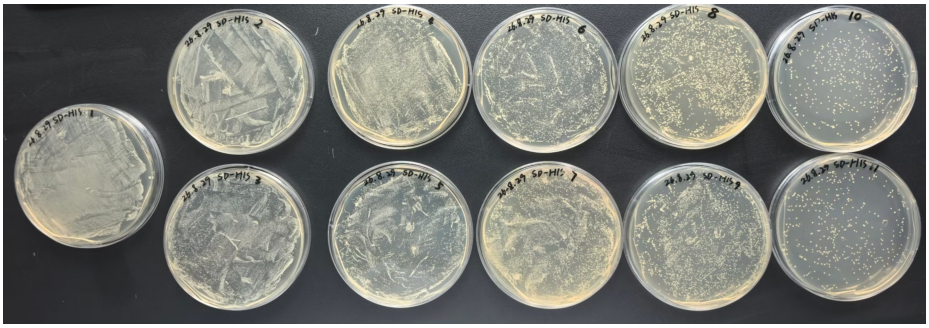
	the amplified fragments were incorrect; after transformation into *Escherichia coli*, no positive colonies were obtained. It was speculated that the ligation reaction failed to successfully insert the sgRNA sequence into the vector, and the target recombinant plasmid was not produced. Since the existing vector construction scheme was ineffective, the experimental strategy was changed, and no further attempts were made to clone sgRNA into *PML104*.
Next plan	Wait for the arrival of the synthesized fragments; prepare the *STE2* repair template by fusion PCR; subsequently transform only the repair template (without introducing the CRISPR plasmid), set up multiple concentration dilution gradients for extensive screening; pool and lyse 10 strain samples, and carry out mixed-template PCR verification.

No 4 Date：2026.08.29-09.14 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Fusion PCR for preparation of the *STE2* repair template, yeast transformation, and colony PCR.
Note	Protocol 7(Conventional PCR)、Protocol 9(PCR Purification)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 8(Overlap Extension PCR)、Protocol 14(Yeast Growth Curve Measurement)
Result	8.29  **Fusion PCR electrophoretogram (the target band is 1763 bp)**

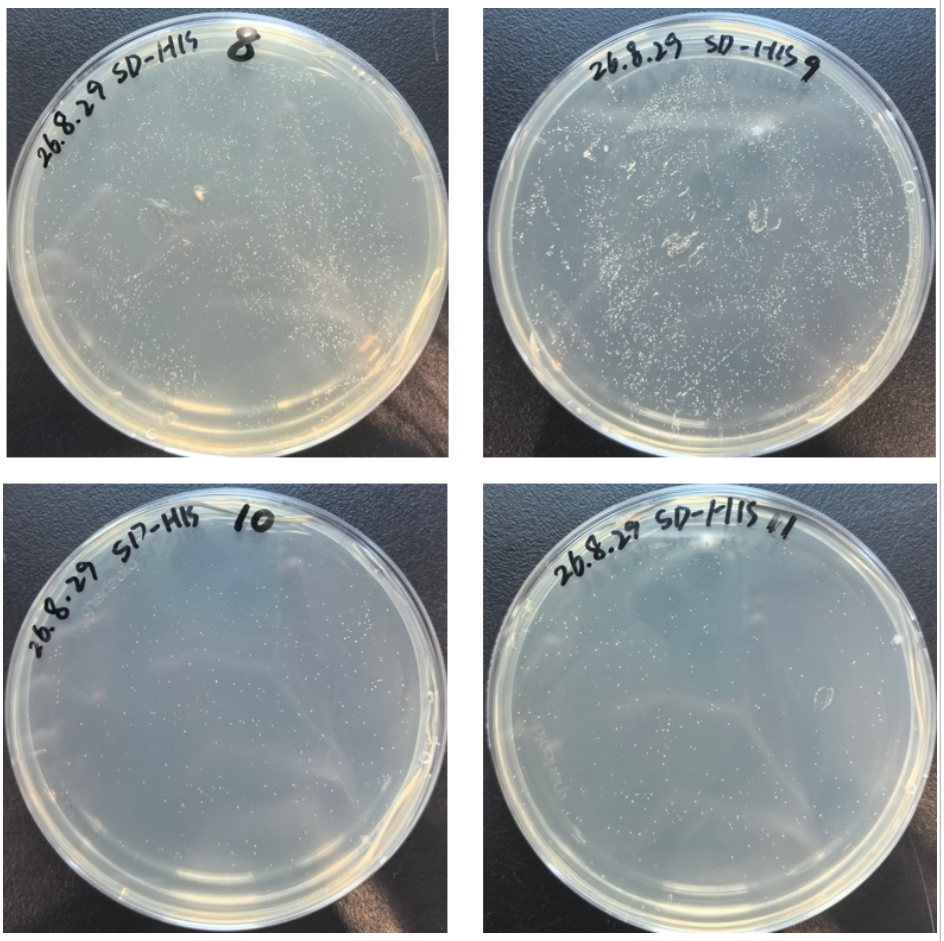
1–2: 2000 bp marker; 3–5: fusion PCR products; 6: blank control

9.5



Transformation plate images

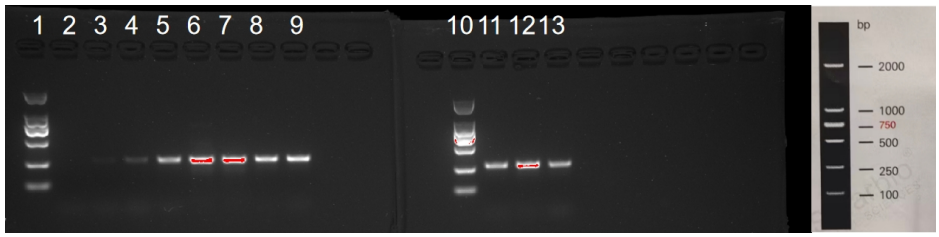
1: blank control; 2–3: original concentration; 4–5: plated after dilution to 75%; 6–7: plated after dilution to 50%; 8–9: plated after dilution to 10%; 10–11: plated after dilution to 1%



Transformation plate images

8–9: plated after dilution to 10%; 10–11: plated after dilution to 1%

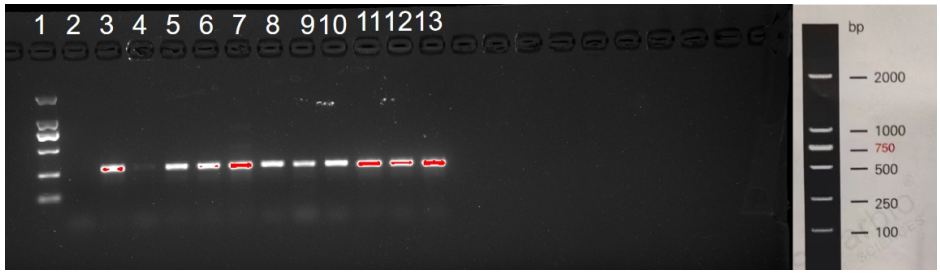
9.5



First-round colony PCR electrophoretogram (the target band is 814 bp)

1: 2000 bp marker; 2–13: PCR performed on bacterial lysates (each well contains the total DNA of 10 colonies)

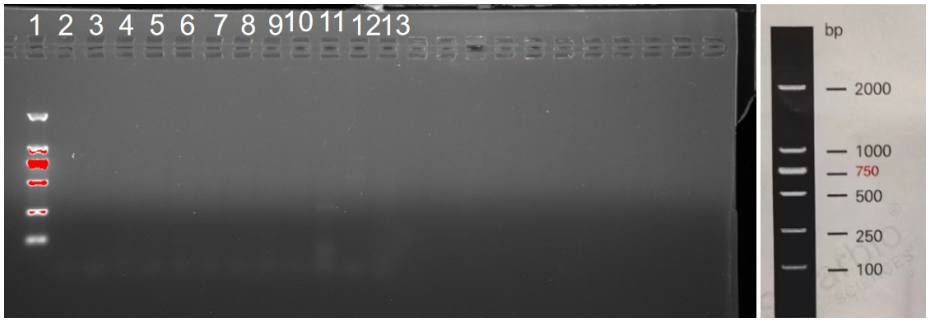
9.6



Second-round colony PCR electrophoretogram (the target band is 814 bp)

1: 2000 bp marker; 2–13: PCR performed on bacterial lysates (each well contains the total DNA of 10 colonies)

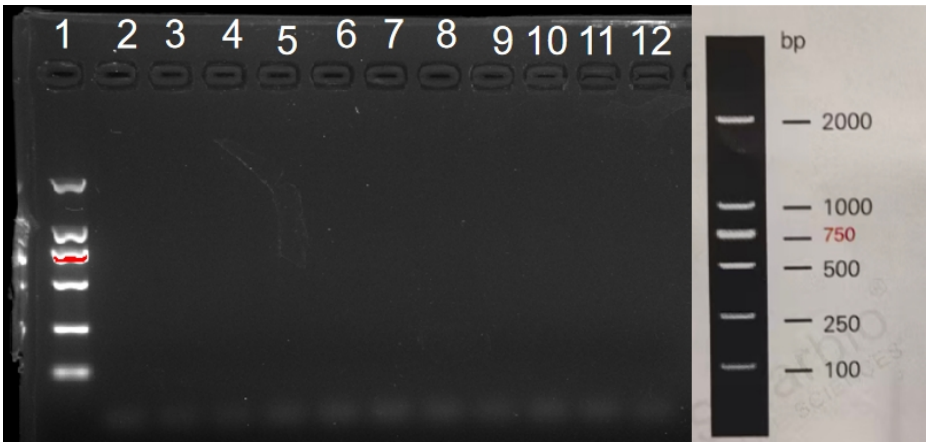
9.8



Third-round colony PCR electrophoretogram (the target band is 814 bp)

1: 2000 bp marker; 2–13: PCR performed on bacterial lysates (each well contains the total DNA of 10 colonies)

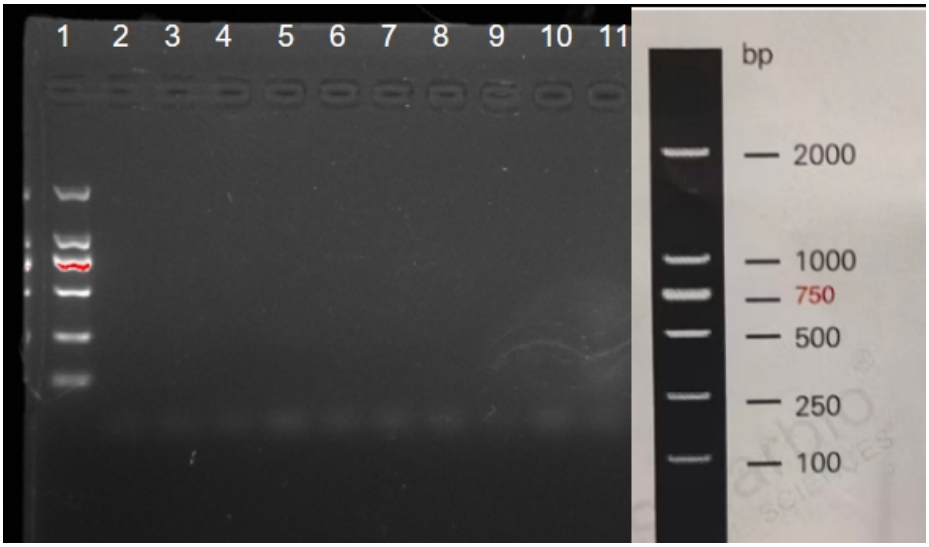
9.9



Fourth-round colony PCR electrophoretogram (the target band is 814 bp)

1: 2000 bp marker; 2–12: PCR performed on bacterial lysates (each well contains the total DNA of 10 colonies)

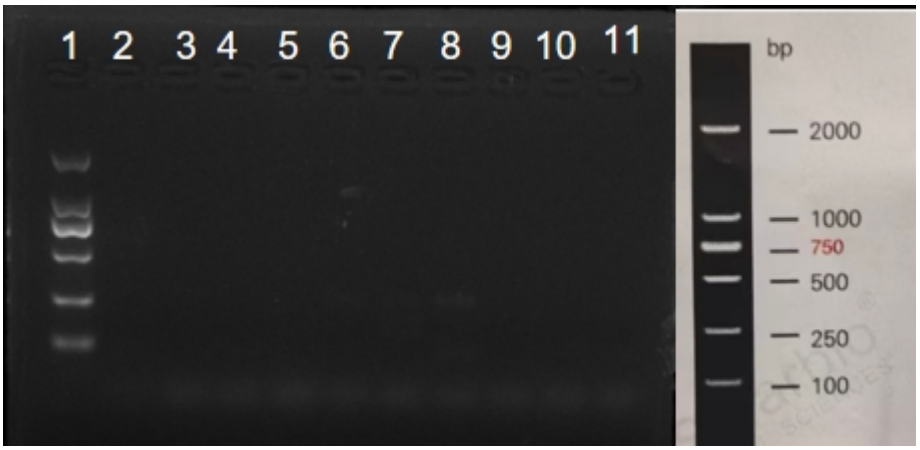
9.11



Fifth-round colony PCR electrophoretogram (the target band is 814 bp)

1: 2000 bp marker; 2–11: PCR performed on bacterial lysates (each well contains the total DNA of 10 colonies)

9.14

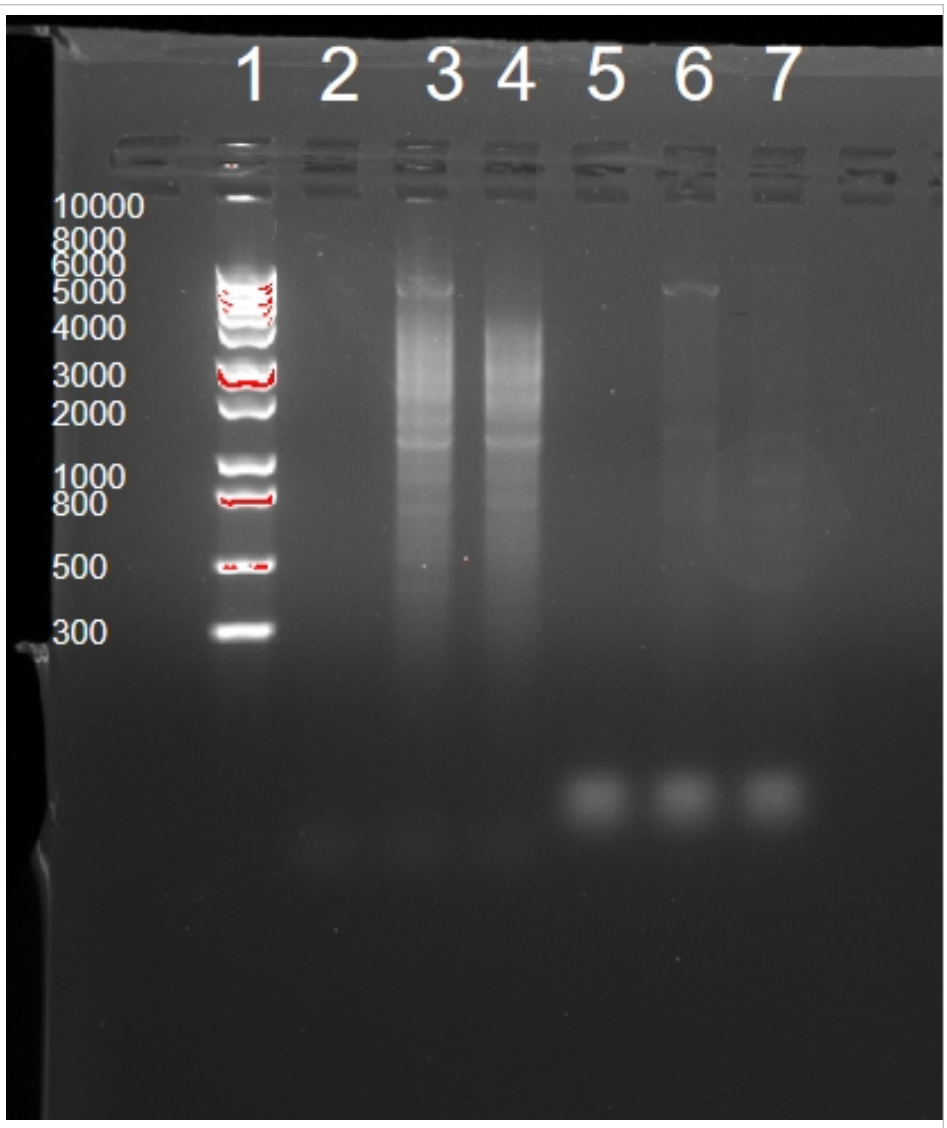


Sixth-round colony PCR electrophoretogram (the target band is 814 bp)

	1: 2000 bp marker; 2–11: PCR performed on bacterial lysates (each well contains the total DNA of 10 colonies)
Explanation	The homologous recombination efficiency of introducing only the repair template was relatively low. Although single yeast colonies were obtained in the 10% and 1% dilution groups, all of them were background colonies without the target editing, and large-scale colony PCR failed to detect positive clones. Since this repair-template-only approach made positive clone detection difficult, the experimental approach was adjusted: the strategy of transforming the repair template alone was abandoned, and the *PML104* CRISPR plasmid was instead reconstructed by PCR. Subsequently, the recombinant *PML104* plasmid and the *STE2* repair template will be co-transformed into yeast, using CRISPR to enhance genome editing efficiency.
Next plan	The *PML104* plasmid was reconstructed by PCR. After plasmid verification was completed, the *PML104* plasmid and the *STE2* repair template were co-transformed into *Saccharomyces cerevisiae* to carry out a new round of editing experiments.

No 5 Date：2026.09.08-09.16 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

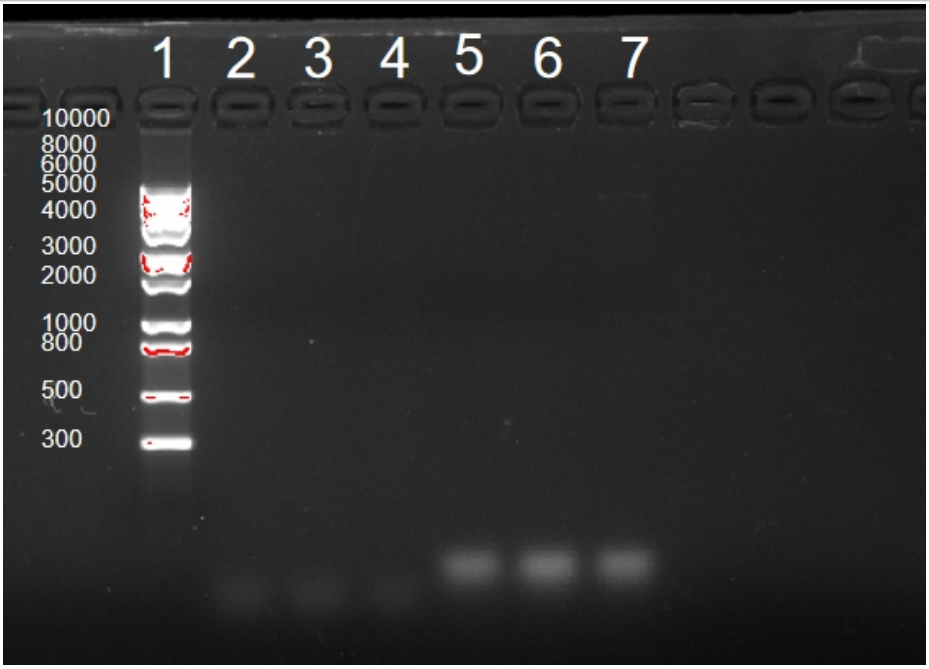
Experiment	Two-round PCR amplification of the *PML104* vector, one-step cloning assembly, and electrophoresis verification.
Note	Protocol 7(Conventional PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 10(Gel Extraction of PCR Products)、Protocol 11(Gibson Assembly)
Result	9.8



Electrophoretogram of the first two-round PCR of *PML104* (the target bands are 5930 bp and 5359 bp, respectively)

1: 10000 bp marker; 2: *STE2*-sgRNA large (blank); 3–4: *STE2*-sgRNA large (experimental); 5: *STE2*-sgRNA small (blank); 6–7: *STE2*-sgRNA small (experimental)

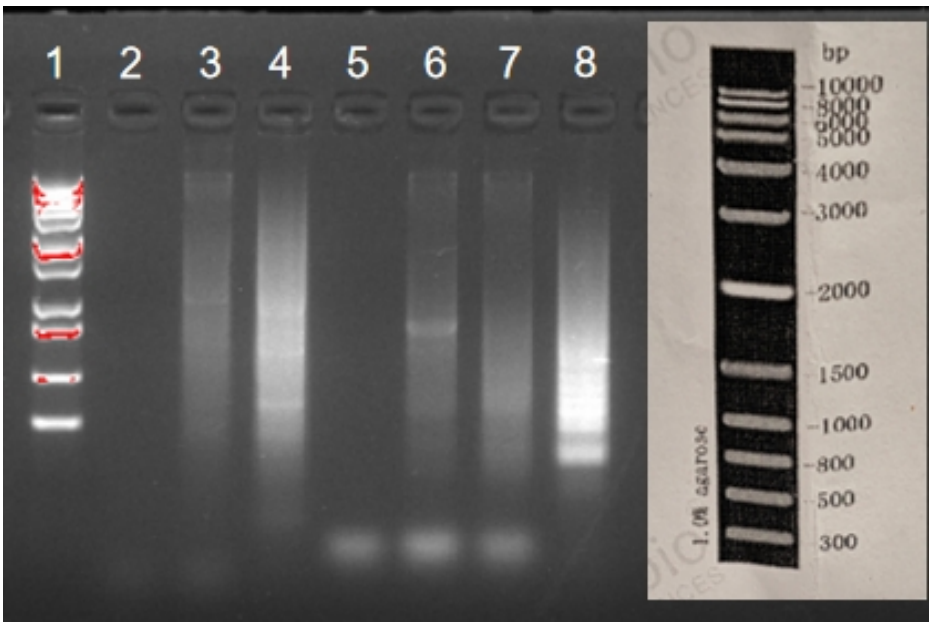
9.9



Electrophoretogram of the second two-round PCR of *PML104* (the target bands are 5930 bp and 5359 bp, respectively)

1: 10000 bp marker; 2: *STE2*-sgRNA large (blank); 3–4: *STE2*-sgRNA large (experimental); 5: *STE2*-sgRNA small (blank); 6–7: *STE2*-sgRNA small (experimental)

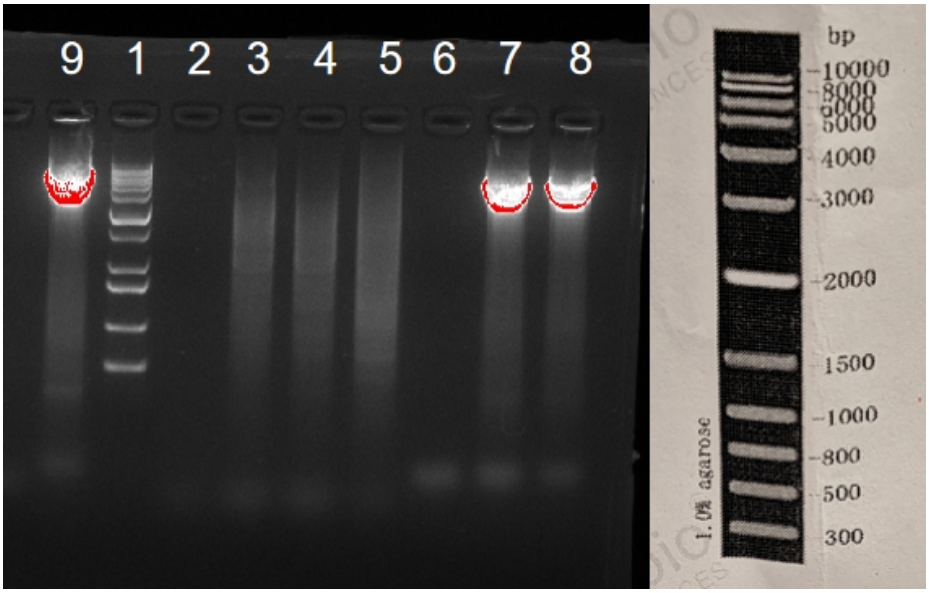
9.11



Electrophoretogram of the third two-round PCR of *PML104* (the target bands are 5930 bp and 5359 bp, respectively)

1: 10000 bp marker; 2: *STE2*-sgRNA large (blank); 3–4: *STE2*-sgRNA large (experimental); 5: *STE2*-sgRNA small (blank); 6–8: *STE2*-sgRNA small (experimental)

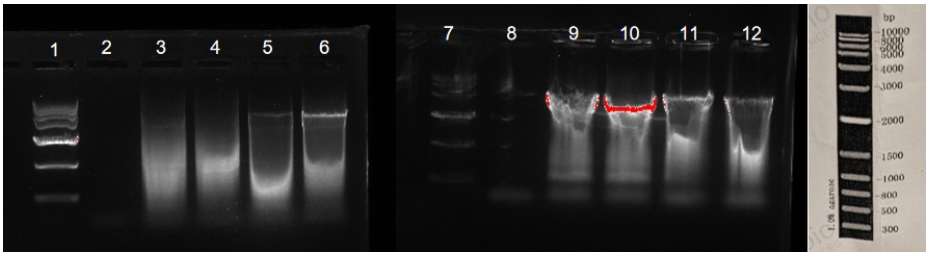
9.14



Electrophoretogram of the fourth two-round PCR of *PML104* (the target bands are 5930 bp and 5359 bp, respectively)

1: 10000 bp marker; 2: *STE2*-sgRNA large (blank); 3–5: *STE2*-sgRNA large (experimental); 6: *STE2*-sgRNA small (blank); 7–9: *STE2*-sgRNA small (experimental)

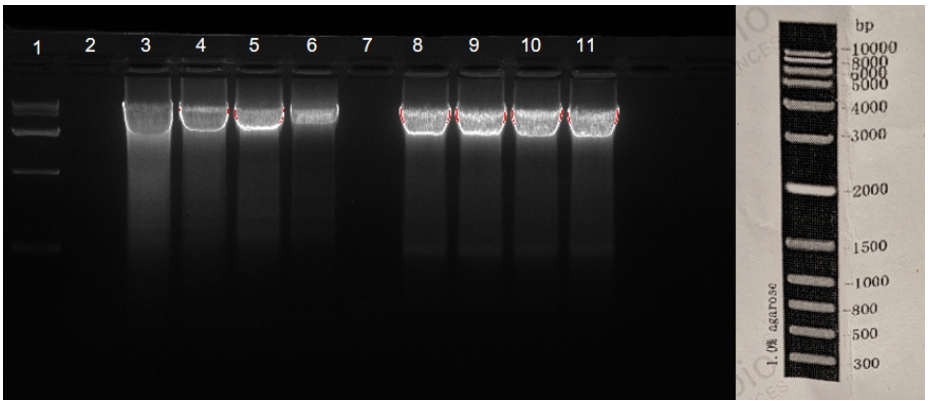
9.15



Electrophoretogram of the fifth two-round PCR of *PML104* (the target bands are 5930 bp and 5359 bp, respectively)

1: 10000 bp marker; 2: *STE2*-sgRNA large (blank); 3–6: *STE2*-sgRNA large (experimental); 7: 10000 bp marker; 8: *STE2*-sgRNA small (blank); 9–12: *STE2*-sgRNA small (experimental)

9.16

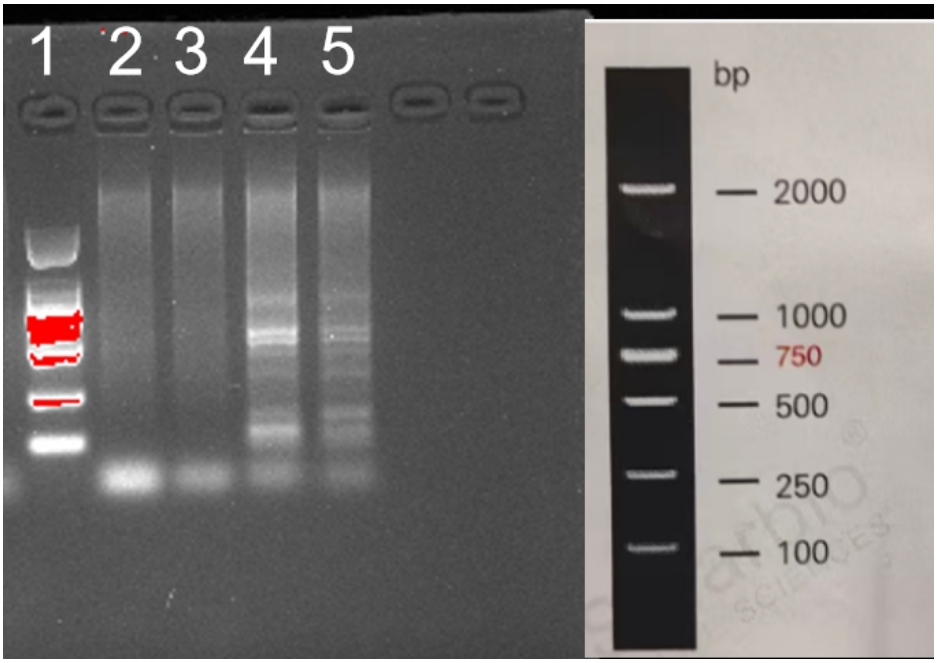


Electrophoretogram of the sixth two-round PCR of *PML104* (the target bands are 5930 bp and 5359 bp, respectively)

1: 10000 bp marker; 2: *STE2*-sgRNA large (blank); 3–6: *STE2*-sgRNA large (experimental); 7: 10000 bp marker; 8: *STE2*-sgRNA small (blank); 9–12: *STE2*-sgRNA small (experimental)

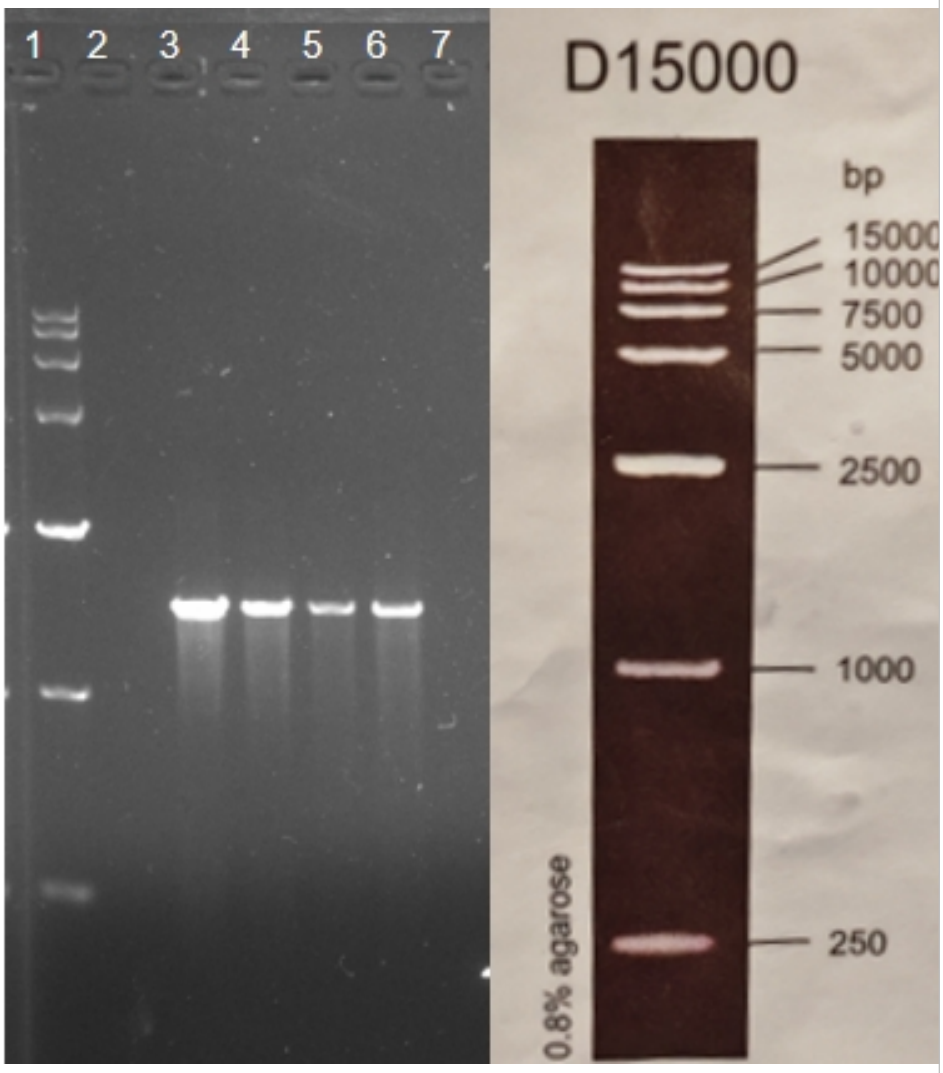
Electrophoretogram after gel extraction recovery for verification (the target bands are 5930 bp and 5359 bp, respectively)

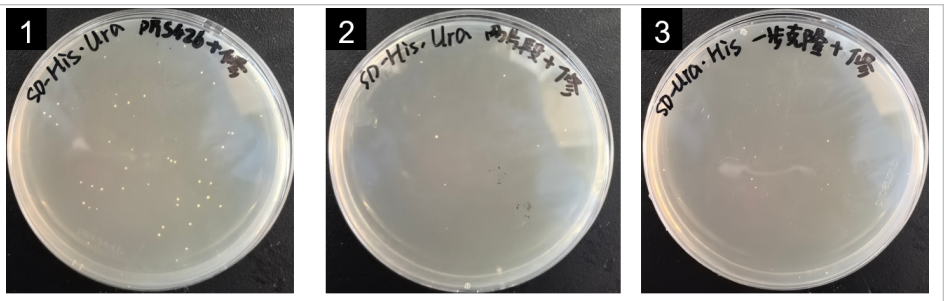
1: 10000 bp marker; 2: *STE2*-sgRNA large; 3: *STE2*-sgRNA small

	 **Electrophoretogram for one-step cloning verification (the target bands are 111 bp and 332 bp, respectively)** 1: 10000 bp marker; 2: *STE2*-sgRNA large; 3: *STE2*-sgRNA small
Explanation	The earlier traditional BclI/SwaI double restriction digestion scheme for *PML104* failed, making it impossible to construct the sgRNA recombinant plasmid; therefore, the strategy was changed to PCR amplification combined with one-step cloning assembly. Through multiple rounds of preliminary experiments, the *STE2*-sgRNA amplification system was successfully optimized, and high-quality purified fragments were obtained, but the *in vitro* one-step cloning assembly performed poorly. *Saccharomyces cerevisiae* possesses extremely strong endogenous homologous recombination capacity, so fragments that failed to assemble *in vitro* can still undergo self-ligation and recombination assembly within yeast cells. Combined with findings from earlier experiments: the editing efficiency of transforming the repair template alone was extremely low, with almost no positives, which was speculated to result from the absence of plasmid selection pressure preventing enrichment of recombination events. Therefore, the *pRS426* empty vector plasmid carrying the same selection marker was introduced as a screening backbone, and multiple control systems were set up to rule out false-negative problems caused by insufficient selection pressure.
Next plan	The one-step cloning products that failed to assemble *in vitro* were directly used for *Saccharomyces cerevisiae* transformation. At the same time, multiple yeast transformation experimental groups with different combinations of vectors and fragments were set up for efficiency comparison, respectively co-transforming the recombinant assembly products with the *STE2* repair template, the linear *PML104* vector plus sgRNA fragment with the *STE2* repair template, and the *pRS426* empty vector plasmid plus the *STE2* repair template. Through uniform selection pressure and multi-system comparison, the problem of extremely low transformation efficiency when transforming the repair template alone in the

	earlier stage was investigated. Subsequently, single colonies grown from each transformation group were picked for batch colony PCR verification, comparing the positive editing situations among the different experimental groups.

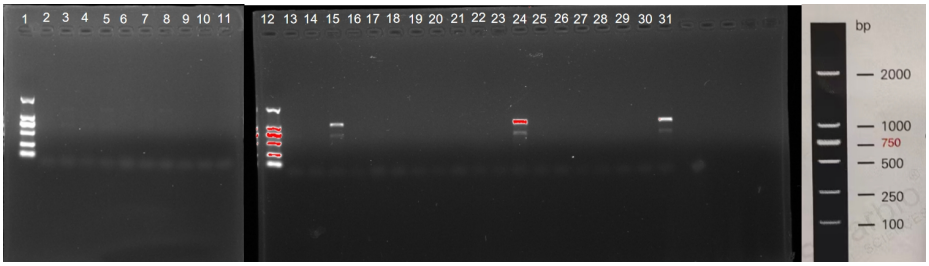
No 6 Date：2026.09.16-09.23 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Large-scale preparation of the *STE2* repair template by fusion PCR, *Saccharomyces cerevisiae* transformation with multiple systems, and batch colony PCR verification.
Note	Protocol 8(Overlap Extension PCR)、Protocol 14(Yeast Growth Curve Measurement)、Protocol 13(Colony PCR for *Saccharomyces cerevisiae*)、Protocol 5(Agarose Gel Electrophoresis)
Result	9.18  **Fusion PCR electrophoretogram (the target band is 1763 bp)** 1: 15000 bp marker; 2: blank control; 3–7: fusion PCR products 9.19



Transformation plate images

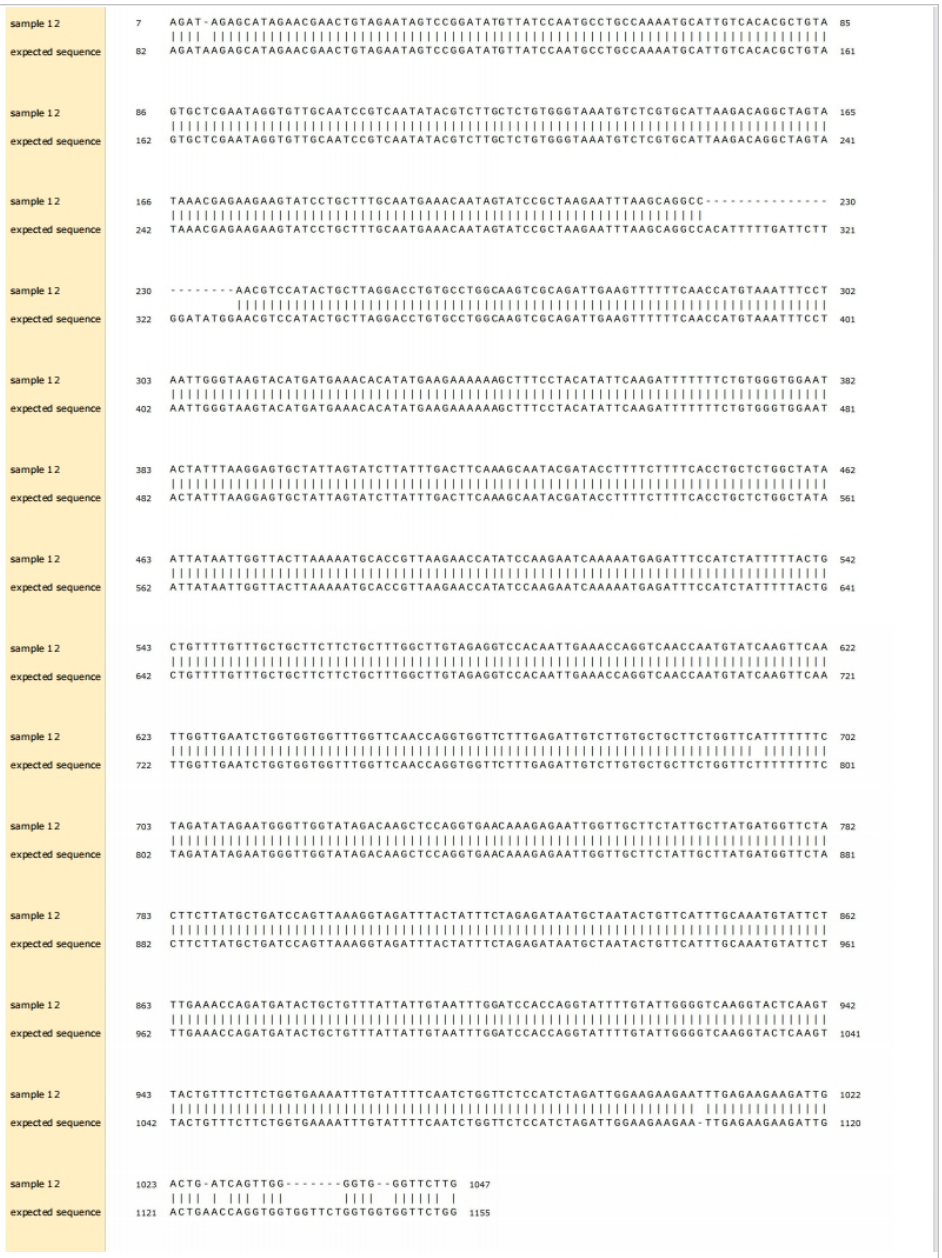
1: *pRS426* + repair template; 2: *PML104* two fragments + repair template; 3: one-step cloning product + repair template



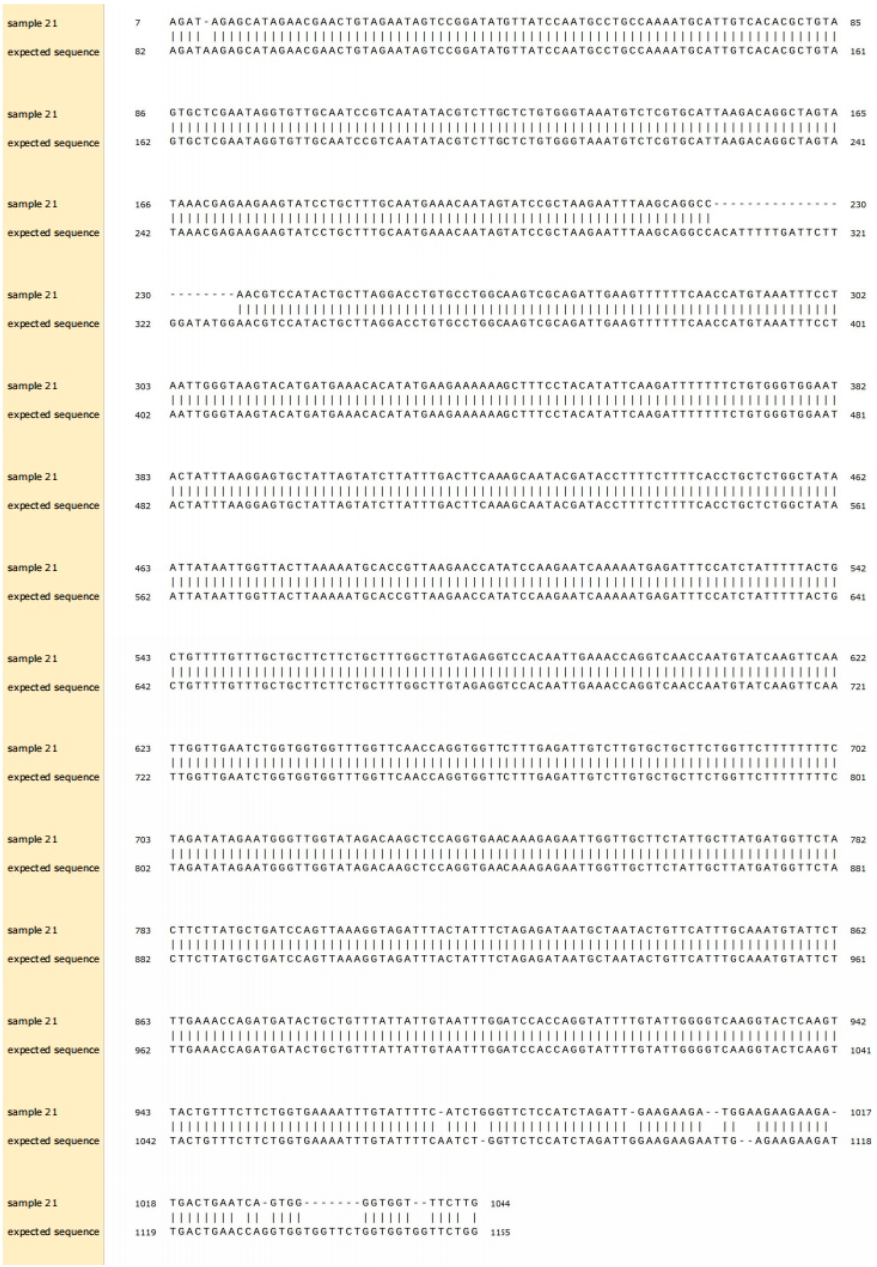
Colony PCR electrophoretogram (the target band is 1098 bp)

1: 2000 bp marker; 2: blank control; 3–11: PCR performed on nine colonies randomly picked from plate 1 (corresponding to colonies 1–9); 12: 2000 bp marker; 13: PCR performed on one colony randomly picked from plate 1 (corresponding to colony 10); 14–23: PCR performed on all ten colonies grown on plate 2 (corresponding to colonies 10–20); 24–31: PCR performed on all eight colonies grown on plate 3 (corresponding to colonies 21–28)

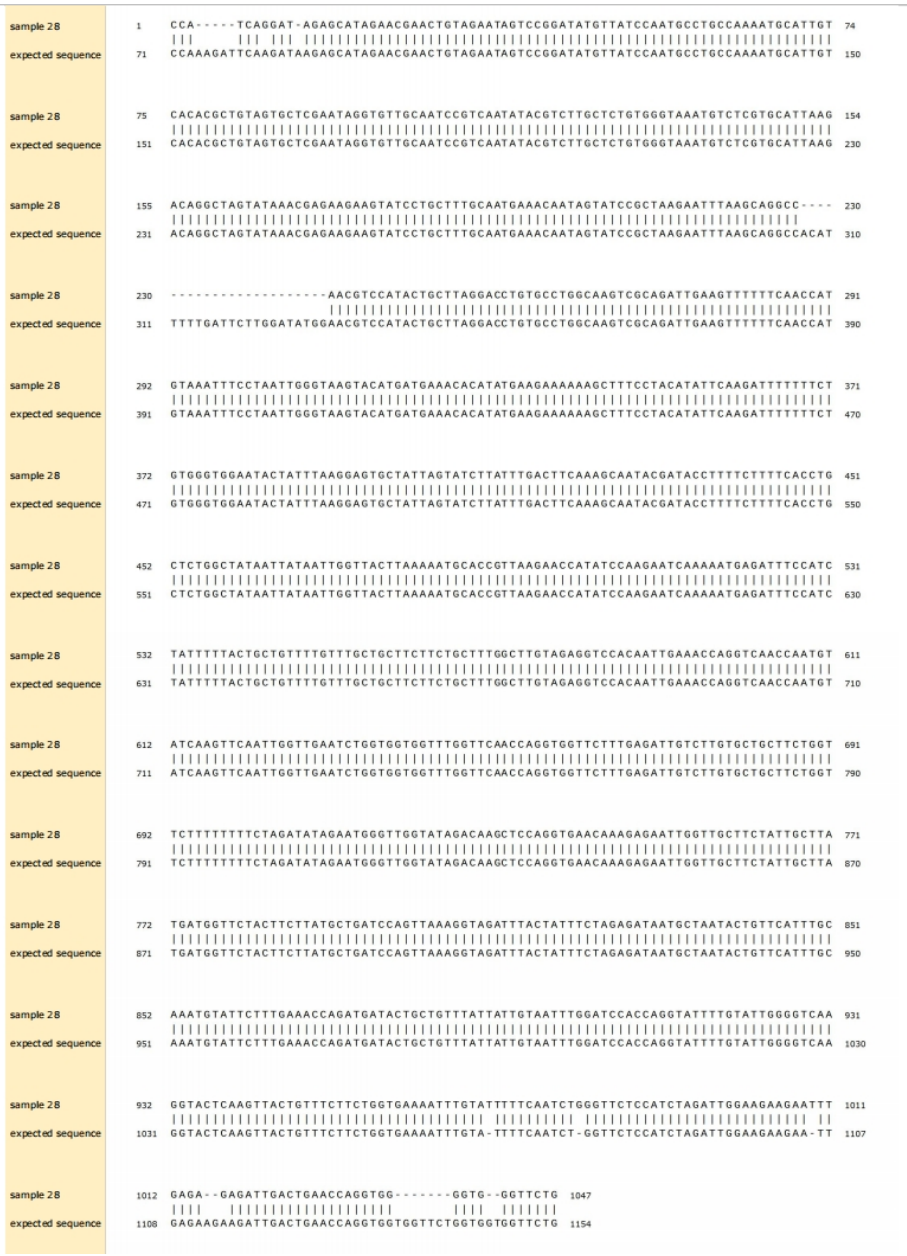
9.23



Alignment of sample 12 with the expected sequence



Alignment of sample 21 with the expected sequence

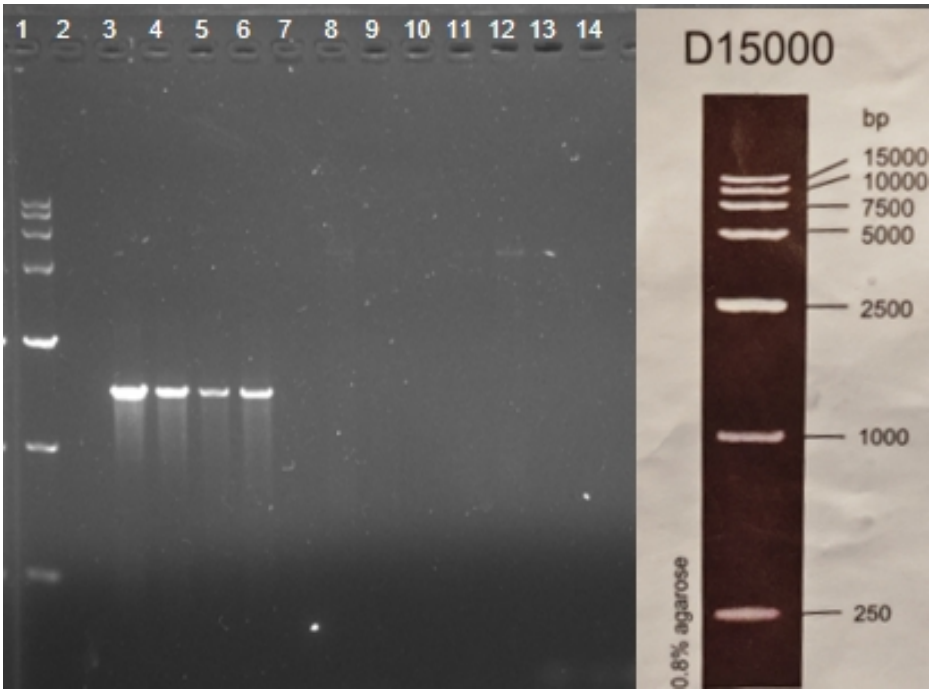


Alignment of sample 28 with the expected sequence

Explanation	In the earlier experimental group in which only the repair template was transformed, there were almost no positive strains, demonstrating that genomic homologous recombination alone has extremely low efficiency and requires plasmid selection pressure to assist in enriching positive strains. The positive strains obtained in this screening came respectively from the two *PML104* systems: transformation of the *in vitro* assembly products and *in vivo* recombination transformation of pure fragments; whereas the *pRS426* empty vector control group obtained no positive edited strains, indicating that when relying solely on the URA selection marker to provide selection pressure and lacking CRISPR-Cas9 cleavage, the efficiency of site-specific genome editing is insufficient. At the same time, the two-fragment co-transformation group without *in vitro* assembly also successfully obtained positive strains, further confirming that *Saccharomyces cerevisiae* possesses extremely strong endogenous

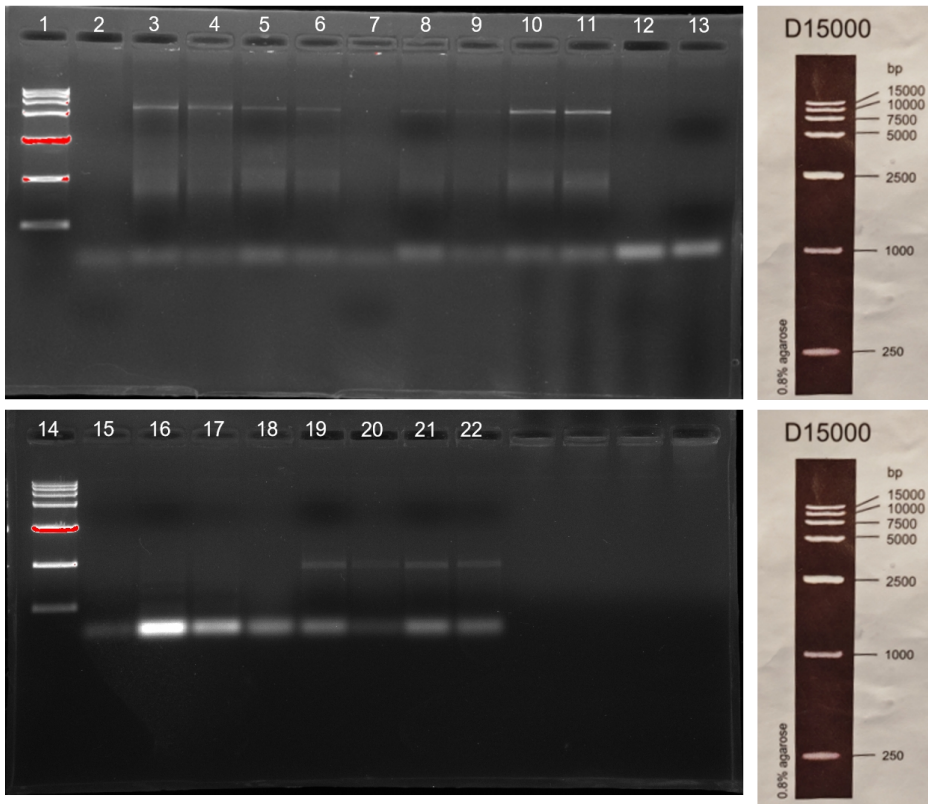
	homologous recombination capacity and can efficiently accomplish autonomous *in vivo* splicing and assembly of vectors and fragments. Sequencing verification confirmed that the key site sequences of the three positive strains were completely correct, yielding reliable *STE2* gating-engineered positive strains that can be used for subsequent functional experimental analysis.
Next plan	On the basis of successfully obtaining *STE2* engineered strains with correct sequence verification, new primers were designed to amplify the *PML104* vector, and a novel repair template was constructed by fusion PCR. This time, the signal peptide coding segment within the synthetic sequence will be removed; subsequently, yeast transformation will be carried out to construct the corresponding engineered strains, so as to comparatively investigate the effects of this signal peptide sequence on the extracellular membrane expression, protein localization, and gating function of the *STE2* gating protein.

No 7 Date：2026.09.18-09.22 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Two-round PCR amplification of *PML104* using new and old primers, segmented PCR for modification of the fragment, and fusion PCR for constructing the novel *STE2* repair template with the signal peptide of the synthetic sequence removed.
Note	Protocol 7(Conventional PCR)、Protocol 8(Overlap Extension PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 9(Gel Extraction of PCR Products)
Result	9.18  **Electrophoretogram of the two-round PCR of *PML104* (the target bands are 5930 bp, 5857 bp, 5359 bp, and 5431 bp, respectively)** 1: 15000 bp marker; 2–6: electrophoresis results of other experiments; 7: *STE2*_sgRNA large 1 (blank); 8: *STE2*_sgRNA large 1 (experimental); 9: *STE2*_sgRNA large 2 (blank); 10: *STE2*_sgRNA large 2 (experimental); 11:

*STE2*_sgRNA small 1 (blank); 12: *STE2*_sgRNA small 1 (experimental); 13: *STE2*_sgRNA small 2 (blank); 14: *STE2*_sgRNA small 2 (experimental)

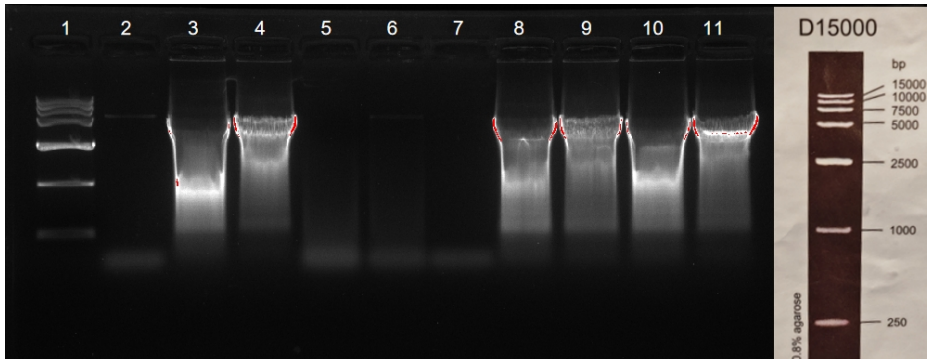
9.20



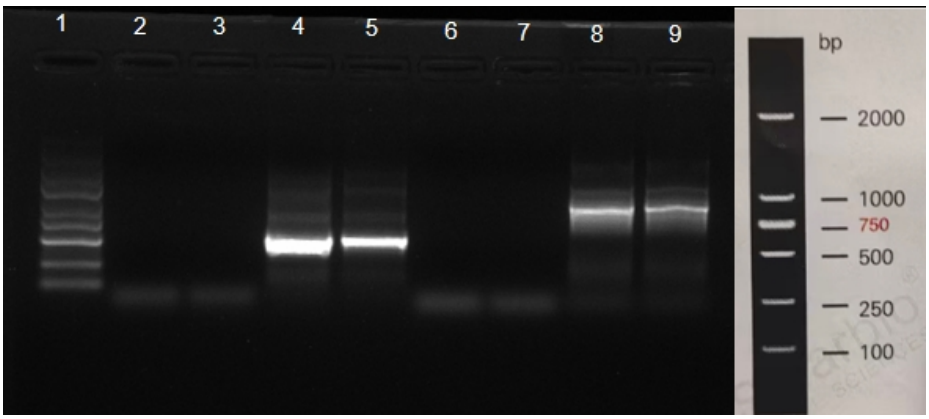
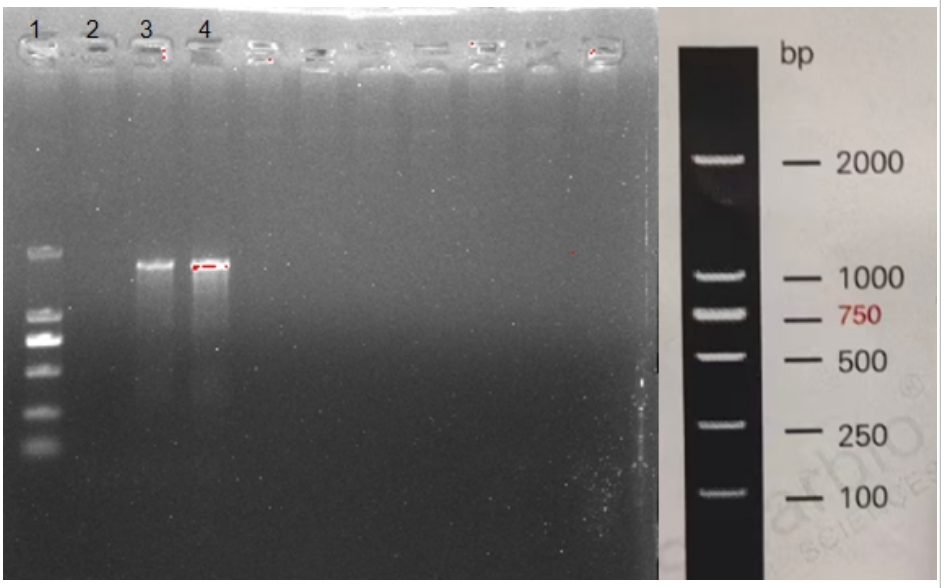
Electrophoretogram of the two-round PCR of *PML104* (the target bands are 5930 bp, 5857 bp, 5359 bp, and 5431 bp, respectively)

1: 15000 bp marker; 2: *STE2*_sgRNA large 1 (blank); 3–6: *STE2*_sgRNA large 1 (experimental); 7: *STE2*_sgRNA large 2 (blank); 8–11: *STE2*_sgRNA large 2 (experimental); 12: *STE2*_sgRNA small 1 (blank); 13: *STE2*_sgRNA small 1 (experimental); 14: 15000 bp marker; 15: *STE2*_sgRNA small 1 (blank); 15–17: *STE2*_sgRNA small 1 (experimental); 18: *STE2*_sgRNA small 2 (blank); 19–22: *STE2*_sgRNA small 2 (experimental)

9.21

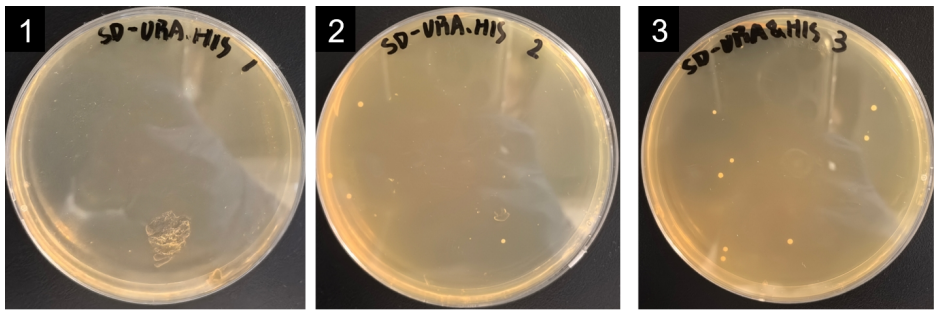
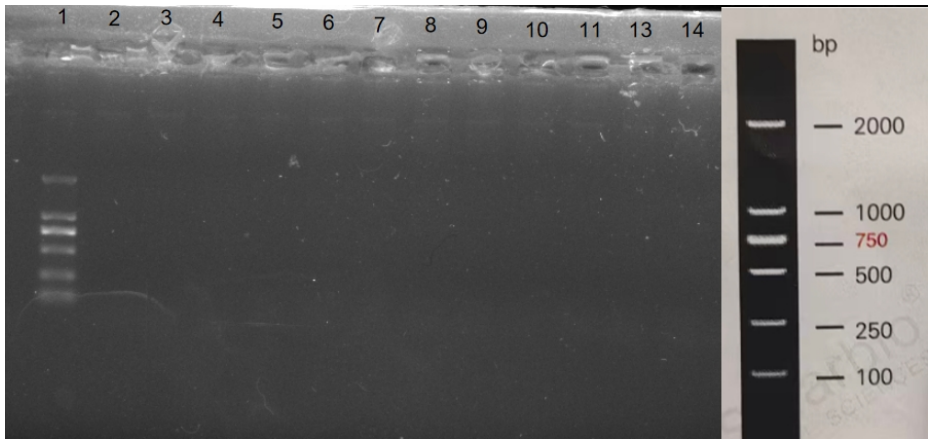


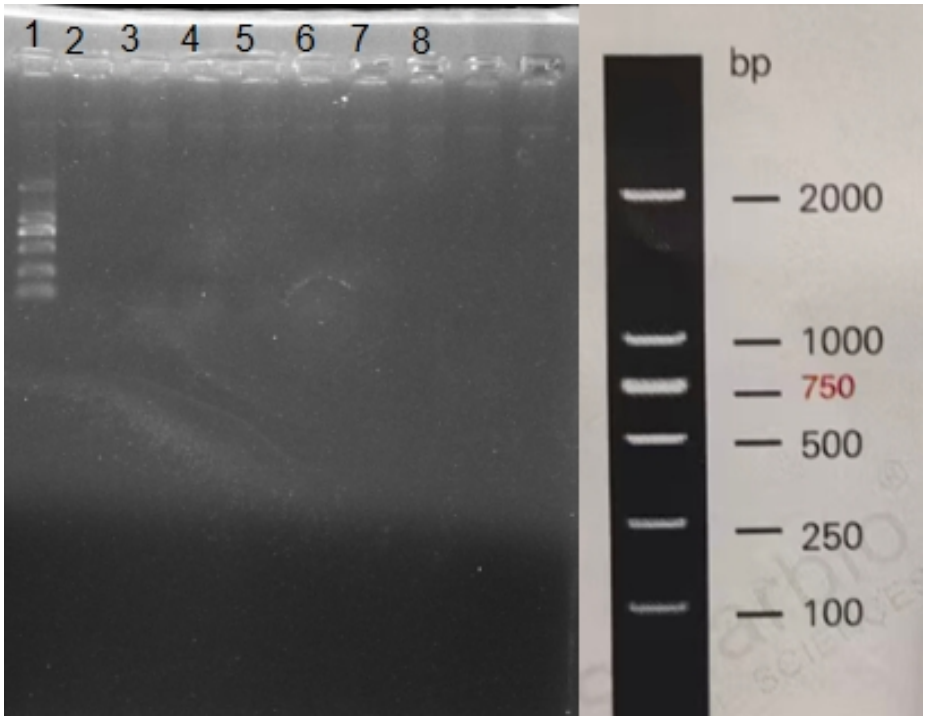
Electrophoretogram of the *PML104* PCR (the target bands are 5359 bp and

	5431 bp, respectively) 1: 15000 bp marker; 2: *STE2*_sgRNA small 1 (blank); 3–6: *STE2*_sgRNA small 1 (experimental); 7: *STE2*_sgRNA small 2 (blank); 8–11: *STE2*_sgRNA small 2 (experimental)  **Electrophoretogram for signal peptide deletion (the target bands are 562 bp and 1088 bp, respectively)** 1: 2000 bp marker; 2–3: upper segment blank control; 4–5: upper segment (experimental); 6–7: lower segment blank control; 8–9: lower segment (experimental) 9.22  **Fusion PCR electrophoretogram for signal peptide deletion (the target band is 1632 bp)** 1: 12000 bp marker; 2: blank control; 3–4: experimental group
Explanation	To investigate the effects of the signal peptide within the synthetic sequence on the extracellular membrane expression, protein localization, and gating function of the *STE2* gating protein, this experiment constructed a control repair template with the signal peptide removed through primer redesign and segmented excision of the signal peptide sequence. After multiple rounds of debugging and optimization of the earlier amplification system, the two-round PCR of *PML104*

	with new and old primers, fragment modification, and fusion PCR were all successfully achieved, yielding a novel repair template with precise sequence excision and reliable structure, thereby providing key material for subsequent strain construction and signal peptide function control experiments.
Next plan	The novel *STE2* repair template with the signal peptide removed, constructed in this round, was transformed into *Saccharomyces cerevisiae*. After single colonies grew on the plates, batch colony PCR identification was carried out to screen positive engineered strains.

No 8 Date：2026.09.23-10.7 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

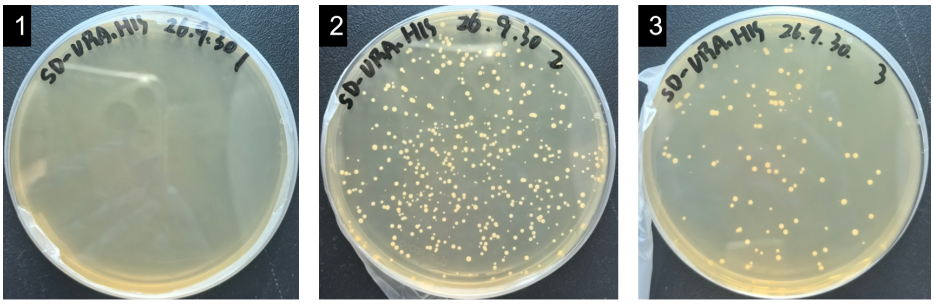
Experiment	*Saccharomyces cerevisiae* transformation with the novel *STE2* repair template lacking the signal peptide and colony PCR verification.
Note	Protocol 14(Yeast Growth Curve Measurement)、Protocol 13(Colony PCR for *Saccharomyces cerevisiae*)、Protocol 5(Agarose Gel Electrophoresis)
Result	9.26  **Transformation plates** 1: blank control; 2–3: experimental group  **Colony PCR electrophoretogram (the target band is 1041 bp)** 1: 2000 bp marker; 2: blank control; 3–14: colony PCR 9.28



Second-round colony PCR electrophoretogram (the target band is 1041 bp)

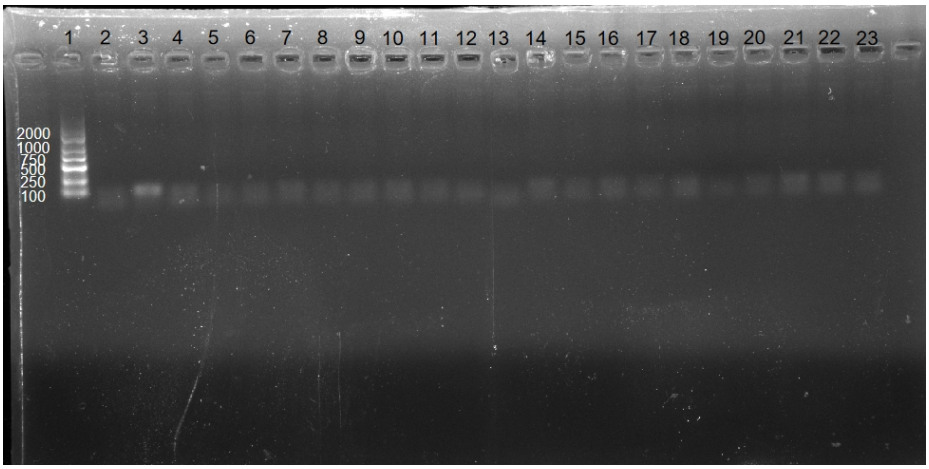
1: 2000 bp marker; 2: blank control; 3–8: colony PCR

10.7



Second-round transformation plates

1: blank control; 2–3: experimental group

	 **Colony PCR electrophoretogram (the target band is 1041 bp)** 1: 2000 bp marker; 2: blank control; 3–23: colony PCR
Explanation	This transformation with the signal peptide-removed repair template yielded no positive results. The core reason was speculated to be that this novel repair template had undergone multiple rounds and numerous instances of PCR amplification and fusion splicing, and repeated amplification cycles readily introduced cryptic base mutations, base shifts, or terminal sequence loss, resulting in an abnormal repair template sequence that was unable to complete homologous recombination editing in yeast normally. Combined with the preliminary results of subsequent functional verification, it could be confirmed that the original gating-engineered strain retaining the signal peptide was already able to achieve normal extracellular membrane expression of the protein, so the necessity of this signal peptide knockout verification experiment was somewhat reduced. Therefore, this experiment was temporarily set aside, and no further blind optimization of amplification conditions was continued.
Next plan	Subsequently, priority will be given to conducting preliminary functional verification experiments on the *STE2* engineered strains to clarify the membrane expression and functional characteristics of the gating protein; after the functional experimental results are complete, a comprehensive assessment will then be made as to whether to optimize the amplification scheme, reduce the number of PCR iterations, and re-initiate the signal peptide knockout control experiment, or to directly terminate this part of the control experiment.

No 9 Date：2026.09.28-10.01 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

Experiment	Preliminary experiment for verification of membrane localization and gating response function of *STE2* gating-engineered positive strains
Note	Protocol 20(Functional Validation of the Yeast Sensor)
Result	9.29

	1	2	3	4	5	6	7	8	9	10	11	12
A	1 μM α-factor+TEVp engineered BY4741 strain(1)	10μM α-factor+TEVp engineered BY4741 strain(1)	20μM α-factor+TEVp engineered BY4741 strain(1)	0 μM α-factor+TEVp engineered BY4741 strain(1)		20μM α-factor engineered BY4741 strain(1)		10μM α-factor+TEVp BY4741 strain(1)	10μM α-factor BY4741 strain(1)	TEVp BY4741 strain(1)		
B	1 μM α-factor+TEVp engineered BY4741 strain(2)	10μM α-factor+TEVp engineered BY4741 strain(2)	20μM α-factor+TEVp engineered BY4741 strain(2)	0 μM α-factor+TEVp engineered BY4741 strain(2)		20μM α-factor engineered BY4741 strain(2)		10μM α-factor+TEVp BY4741 strain(2)	10μM α-factor BY4741 strain(2)	TEVp BY4741 strain(2)		
C	1 μM α-factor+TEVp engineered BY4741 strain(3)	10μM α-factor+TEVp engineered BY4741 strain(3)	20μM α-facto+TEVp engineered BY4741 strain(3)	0 μM α-factor+TEVp engineered BY4741 strain(3)		20μM α-factor engineered BY4741 strain(3)		10μM α-factor+TEVp BY4741 strain(3)	10μM α-factor BY4741 strain(3)	TEVp BY4741 strain(3)		
D												
E	1 μM α-factor+HA engineered BY4741 strain(1)	10 μM α-factor+HA engineered BY4741 strain(1)	20 μM α-factor+HA engineered BY4741 strain(1)	0 μM α-factor+HA engineered BY4741 strain(1)		20 μM α-factor engineered BY4741 strain(1)		10 μM α-factor+HA BY4741 strain(1)	10μM α-factor BY4741 strain(1)	HA BY4741 strain(1)		
F	1 μM α-factor+HA engineered BY4741 strain(2)	10 μM α-factor+HA engineered BY4741 strain(2)	20 μM α-factor+HA engineered BY4741 strain(2)	0 μM α-factor+HA engineered BY4741 strain(2)		20 μM α-factor engineered BY4741 strain(2)		10 μM α-factor+HA BY4741 strain(2)	10μM α-factor BY4741 strain(2)	HA BY4741 strain(2)		
G	1 μM α-factor+HA engineered BY4741 strain(3)	10 μM α-factor+HA engineered BY4741 strain(3)	20 μM α-factor+HA engineered BY4741 strain(3)	0 μM α-factor+HA engineered BY4741 strain(3)		20 μM α-factor engineered BY4741 strain(3)		10 μM α-factor+HA BY4741 strain(3)	10μM α-factor BY4741 strain(3)	HA BY4741 strain(3)		
H												

Table of samples added to each well for fluorescence measurement

	1	2	3	4	5	6	7	8	9	10	11	12
A	14	27	59	12	1	42	5	52	57	53	0	5
B	16	20	61	12	2	32	0	59	54	55	4	0
C	11	27	69	10	5	38	5	56	56	57	0	6
D	6	1	6	2	3	5	1	6	1	4	3	2
E	12	28	56	12	6	30	5	58	56	57	1	6
F	20	23	62	11	4	59	1	60	54	56	4	1
G	14	30	48	11	6	54	3	58	56	58	2	3
H	2	7	2	4	6	2	7	3	3	6	0	6

4 h fluorescence values

	1	2	3	4	5	6	7	8	9	10	11	12
A	14	41	79	10	4	63	2	56	54	54	1	1
B	12	28	67	15	3	49	5	54	56	57	0	4
C	12	42	83	15	3	65	6	53	59	55	0	7
D	2	3	6	0	6	3	2	6	0	5	3	1
E	18	30	69	13	2	61	1	60	57	54	7	1
F	21	27	75	16	1	72	4	55	60	53	5	4
G	13	47	89	13	7	64	4	58	56	58	4	2
H	4	6	1	6	3	2	6	1	5	5	0	7

5 h fluorescence values

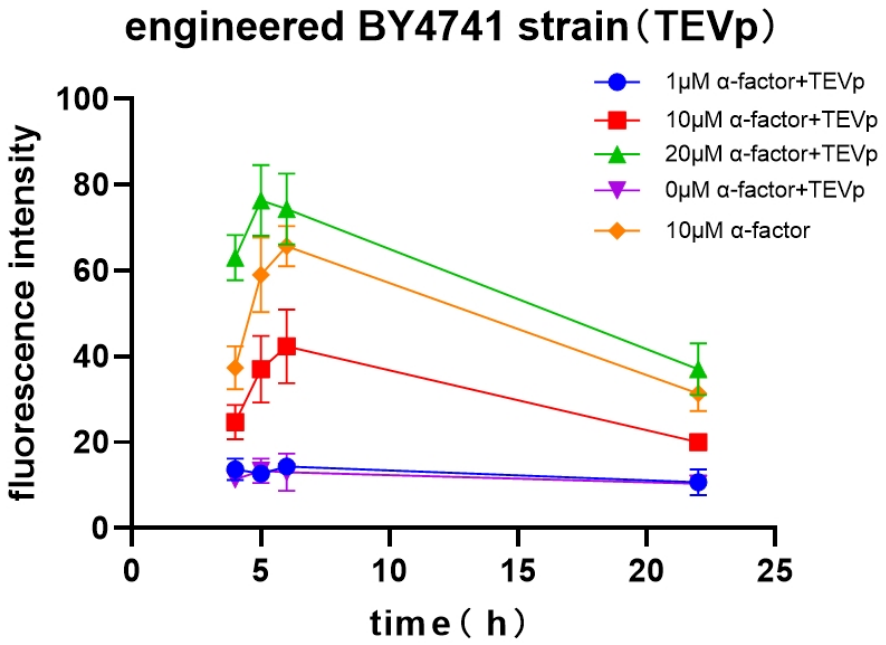
	1	2	3	4	5	6	7	8	9	10	11	12
A	15	44	77	8	3	64	1	56	52	54	2	1
B	14	33	65	15	2	62	6	53	58	56	0	5
C	14	50	81	16	2	71	5	53	60	52	2	5
D	6	2	3	4	1	6	1	5	4	1	6	1
E	14	47	68	14	5	69	6	56	58	55	3	6
F	23	28	78	16	1	73	1	58	57	53	6	0
G	15	67	104	15	4	62	6	54	61	55	3	5
H	5	4	2	6	1	6	2	4	5	1	6	0

6 h fluorescence values

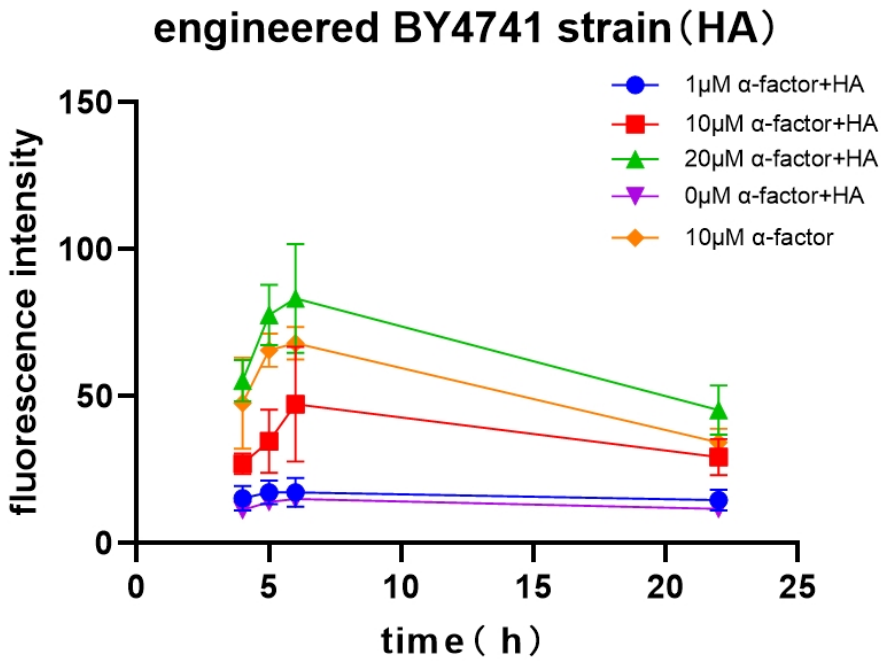
10.01

	1	2	3	4	5	6	7	8	9	10	11	12
A	8	18	31	10	0	28	2	47	54	46	0	2
B	10	21	37	9	2	30	0	57	52	52	3	0
C	14	21	43	12	0	36	2	55	60	50	4	3
D	3	1	6	1	3	4	0	5	2	1	5	0
E	15	28	40	11	2	37	0	60	57	53	6	2
F	18	24	41	14	0	37	2	56	60	53	6	4
G	11	36	55	10	6	29	4	57	55	58	1	4
H	1	5	4	2	7	2	4	4	1	6	1	5

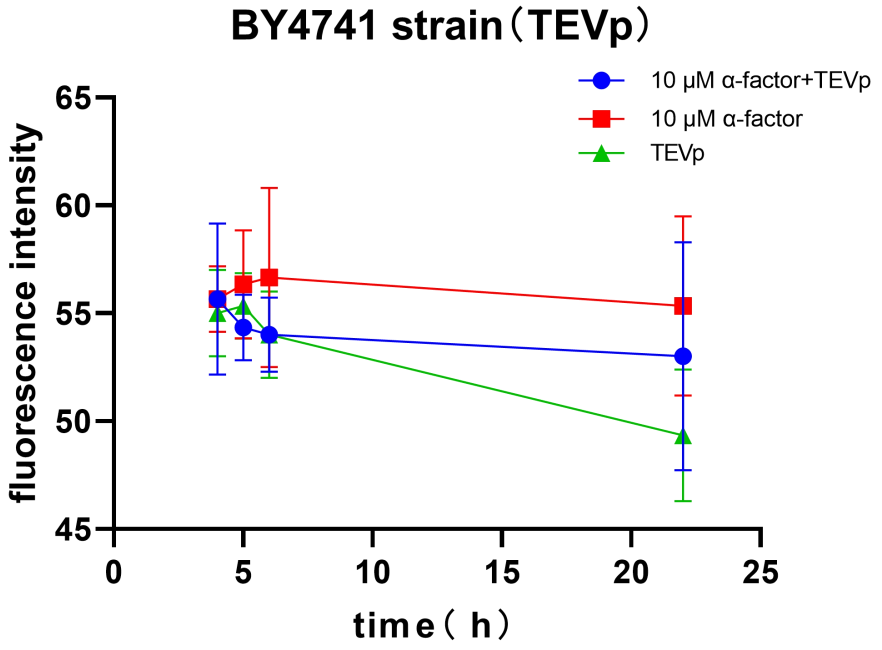
22 h fluorescence values



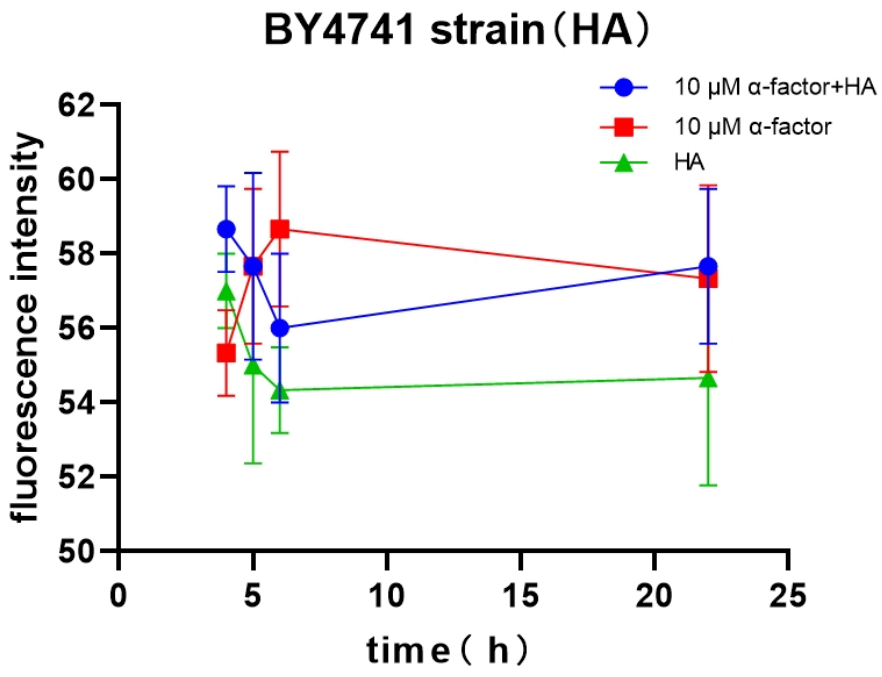
engineered *BY4741* strain (TEVp)



engineered *BY4741* strain (HA)



***BY4741* strain (TEVp)**

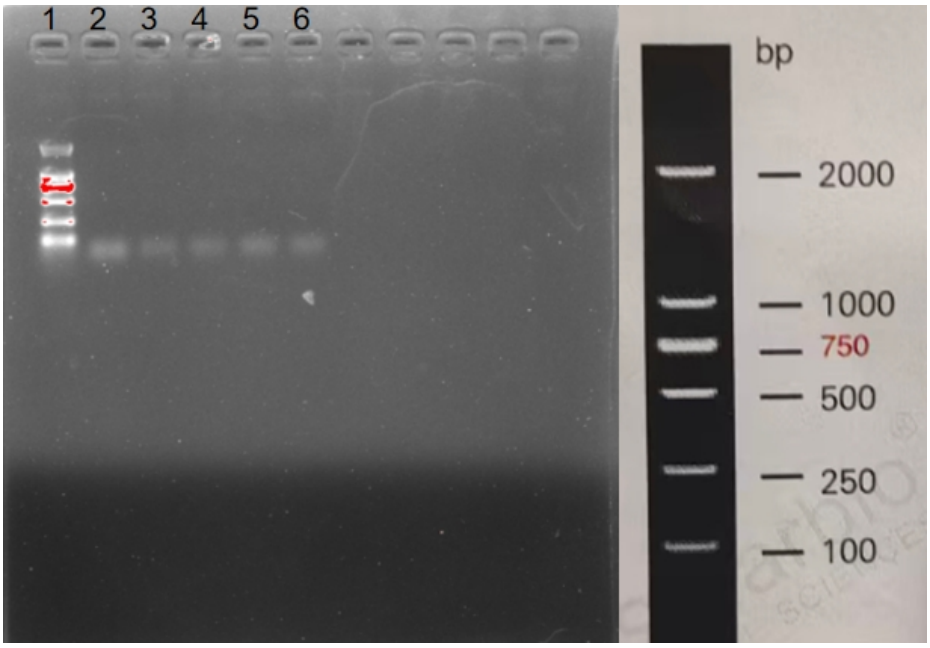


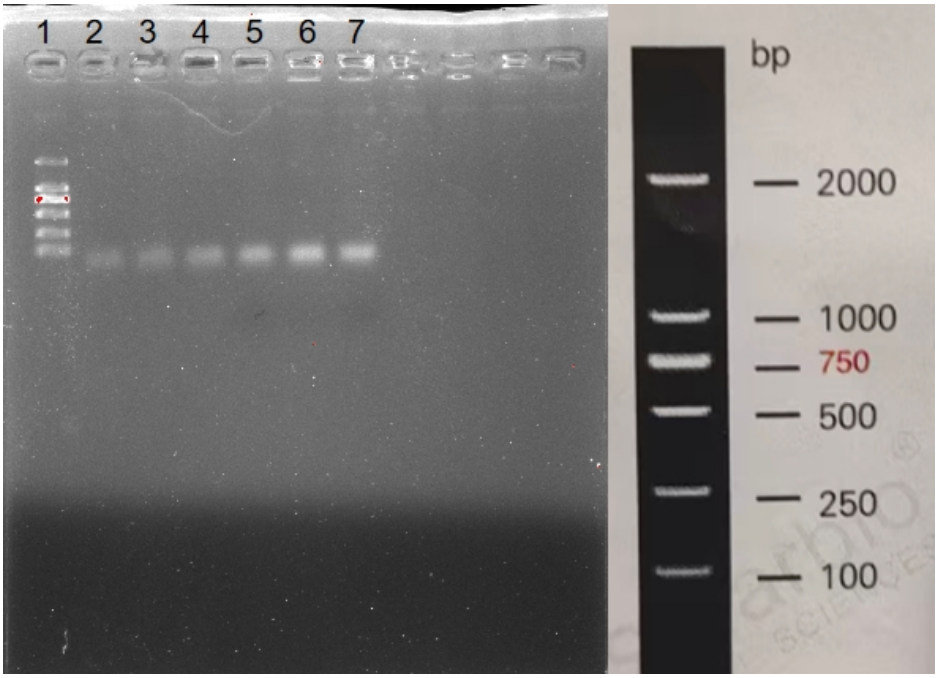
***BY4741* strain（HA）**

Explanation	Preliminary fluorescence detection results indicated that the *STE2* gating-engineered strains constructed in this study were able to transport the protein to the extracellular side of the cell membrane and could respond to α-factor signaling, with fluorescence intensity increasing as the inducer concentration rose, demonstrating that the gating system possesses basic responsiveness. However, under conditions without added α-factor and without

	regulatory elements, the control group still exhibited fluorescence signal, indicating that the interaction between the self-inhibitory domain and *STE2* is defective and cannot achieve a stable basal inhibitory effect. The fluorescence values of the wild-type *BY4741* groups were similar, and this background fluorescence originates from the strain itself, representing inherent background interference of the strain rather than gating response.
Next plan	Amplify the *yEGFP* fragment, ligate it to the extracellular side of *STE2*, replace the corresponding region of the original synthetic sequence, and construct a new repair template, so as to visually verify by fluorescence localization whether the *STE2* protein is successfully transported to the extracellular side of the cell membrane; after clarifying the protein membrane localization status, subsequently redesign an entirely new self-inhibitory peptide sequence, optimize the gating module, and once again carry out control verification experiments for this gating system.

No 10 Date：2026.09.22-10.10 Experimenters: Yuelin Zheng Recorder: Yuelin Zheng

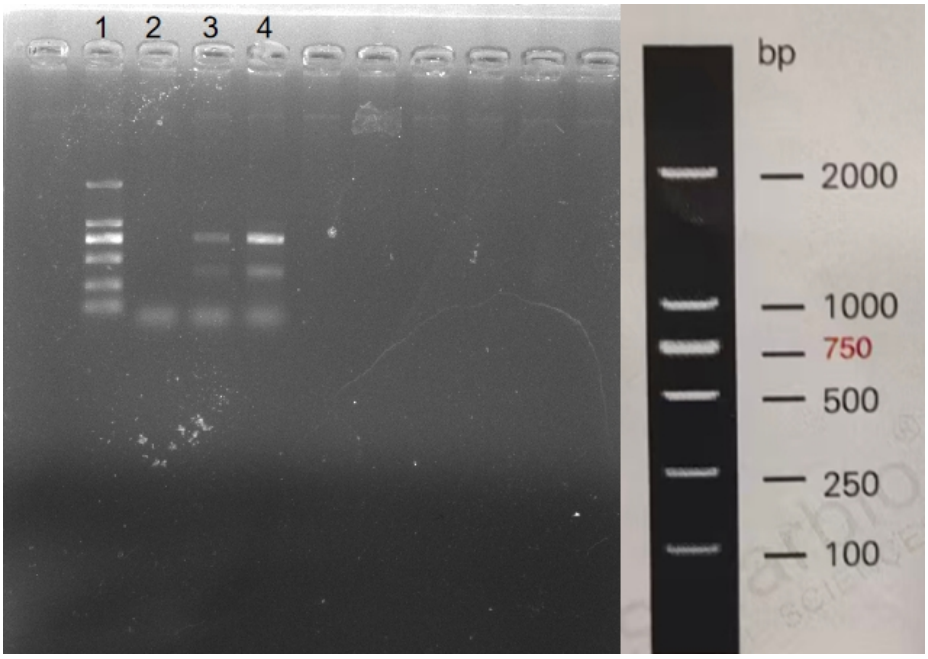
Experiment	PCR amplification of the *yEGFP* fragment, construction of the *STE2-yEGFP* fusion repair template, and *Saccharomyces cerevisiae* transformation.
Note	Protocol (Conventional PCR)、Protocol 8(Overlap Extension PCR)、Protocol 5(Agarose Gel Electrophoresis)、Protocol 14(Yeast Growth Curve Measurement)
Result	9.22  **Electrophoretogram of the first *yEGFP* amplification (the target band is 768 bp)** 1: 2000 bp marker; 2: blank control; 3 – 6: experimental group 9.23



Electrophoretogram of the second *yEGFP* amplification (the target band is 768 bp)

1: 2000 bp marker; 2: blank control; 3 – 7: experimental group

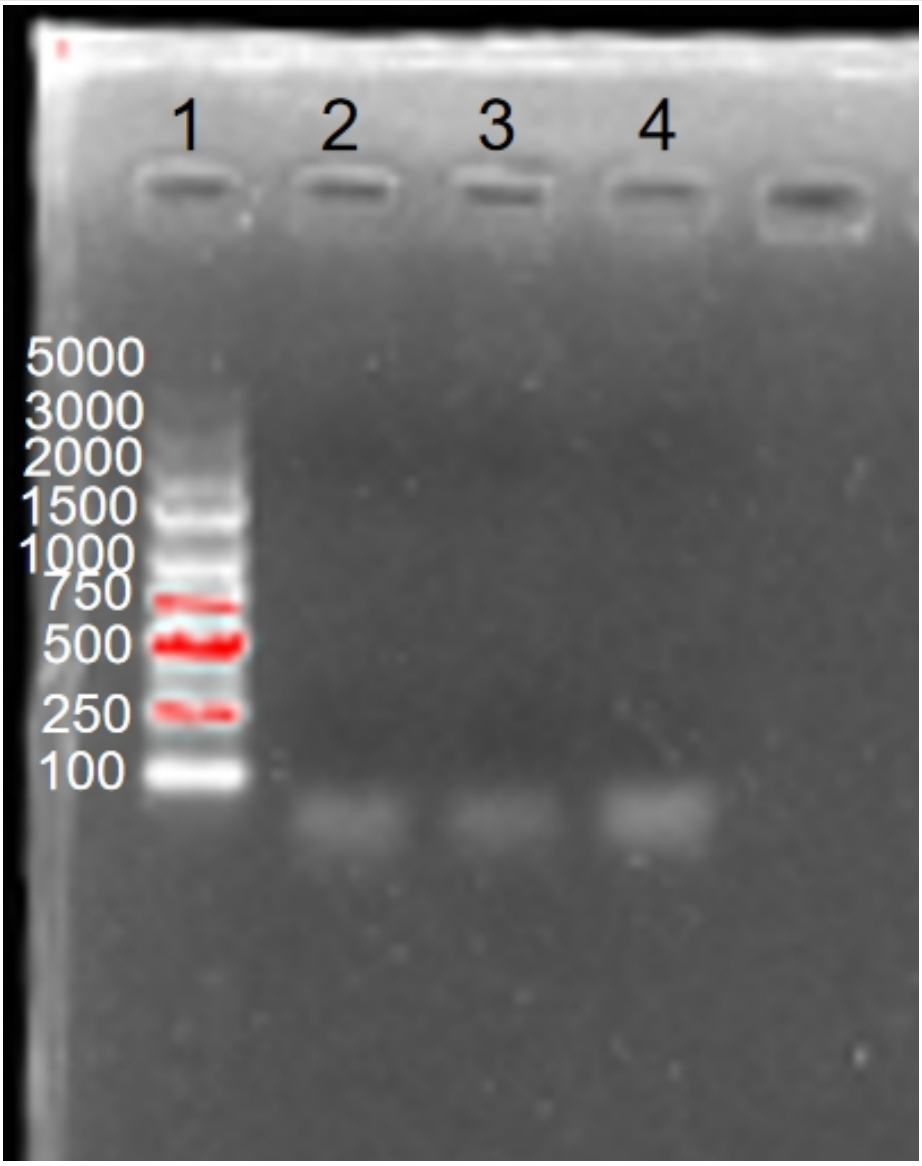
9.26



Electrophoretogram of the third *yEGFP* amplification (standard enzyme) (the target band is 768 bp)

1: 2000 bp marker; 2: blank control; 3 – 4: experimental group

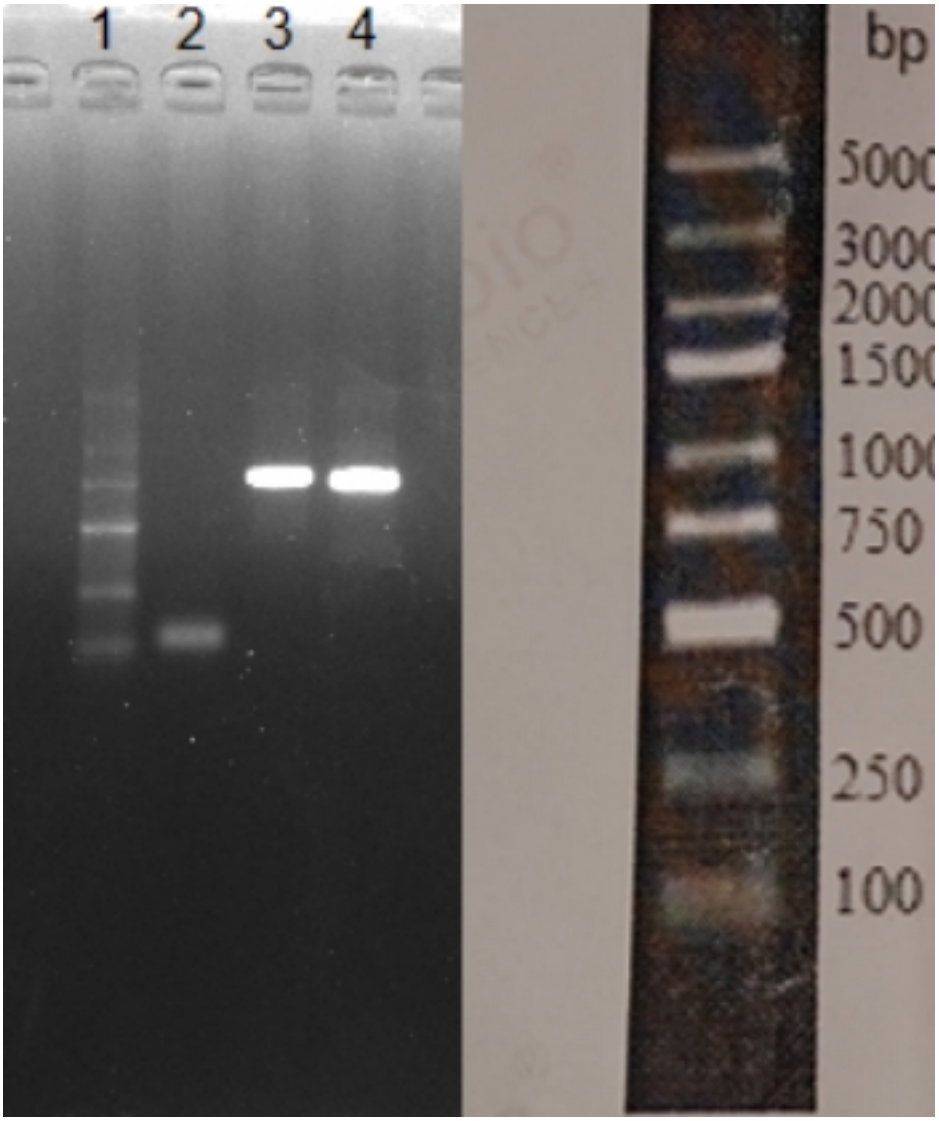
9.29



Electrophoretogram of the fourth *yEGFP* amplification (high-fidelity enzyme) (the target band is 768 bp)

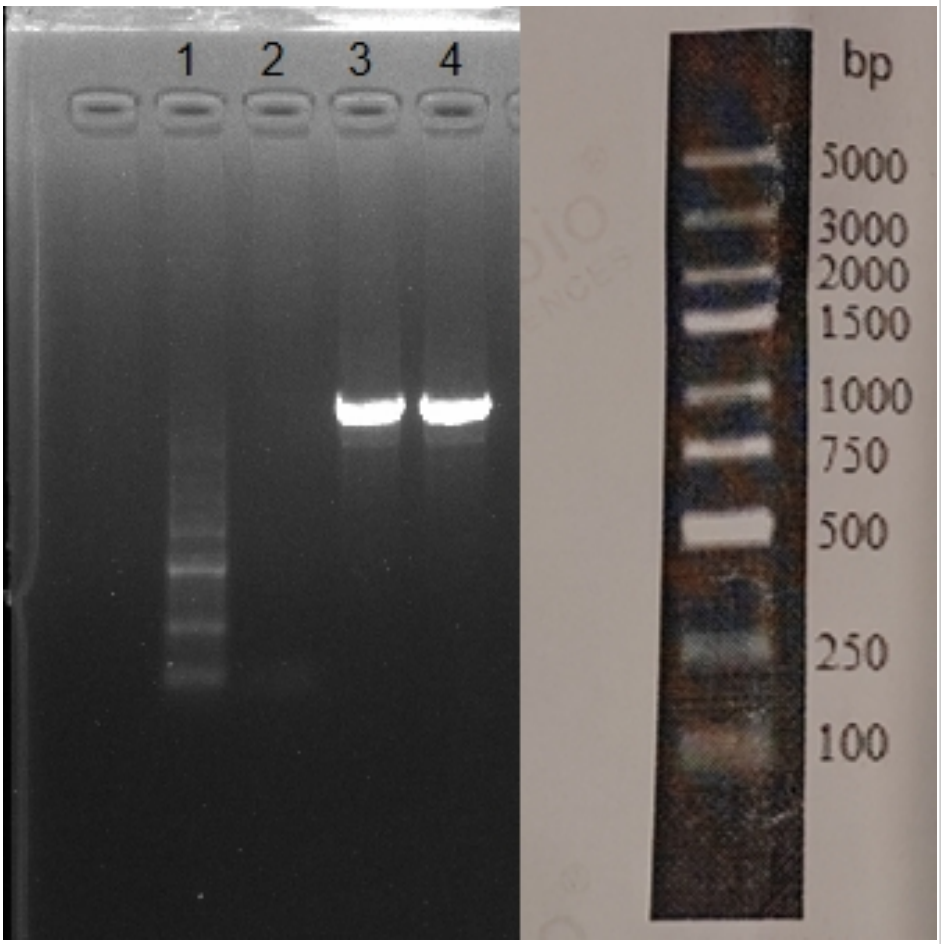
1: 2000 bp marker; 2: blank control; 3 – 4: experimental group

10.10



Electrophoretogram of the fifth *yEGFP* amplification (new high-fidelity enzyme) (the target band is 768 bp)

1: 2000 bp marker; 2: blank control; 3 – 4: experimental group

	 **Fusion PCR electrophoretogram (the target band is 1875 bp)** 1: 2000 bp marker; 2: blank control; 3 – 4: experimental group
Explanation	In this round, *yEGFP* amplification progressed from multiple failures to success. The key improvement was switching to a high-performance, brand-new high-fidelity enzyme; the new enzyme showed significantly enhanced amplification specificity, error tolerance, and amplification efficiency, resolving the core problem of no product in earlier amplifications. Through fusion PCR, the *STE2-yEGFP* fusion repair template was successfully assembled, enabling fusion expression of the *yEGFP* fluorescent tag on the extracellular side of the *STE2* protein, which allows intuitive and visualized determination of the cell membrane localization of the *STE2* protein, thereby providing reliable experimental material for subsequently clarifying extracellular membrane expression, further optimizing the self-inhibitory peptide sequence, and improving the *STE2* gating functional system.
Next plan	Batch colony PCR verification was performed on the single colonies obtained from this transformation, and *STE2-yEGFP* positive clone strains were screened and obtained for subsequent fluorescence localization observation and identification of extracellular membrane protein expression.